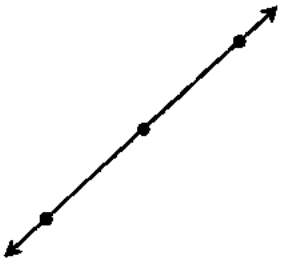
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Rate of Change	A ratio that shows how one variable changes with respect to another On a linear graph, this is called the slope of the line!
Slope 	• Slope is written as a ratio of the **vertical change** (rise) to the **horizontal change** (run) between any two points on a line. • This remains constant for any two points on the same line. • Slope is written as a fraction in simplest form. • Variable for slope: m

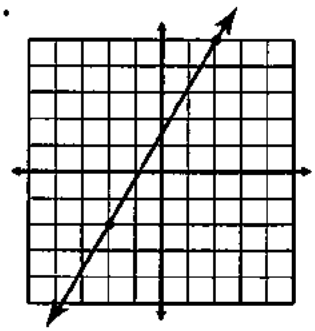
Types of Slope

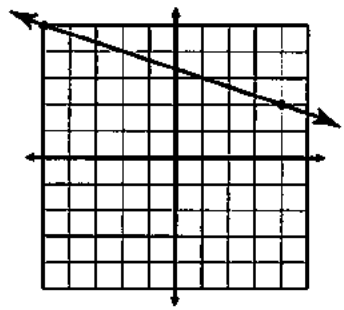
			
Positive	Negative	Zero	Undefined

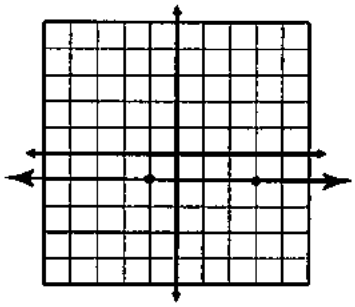
Finding Slope on a Graph

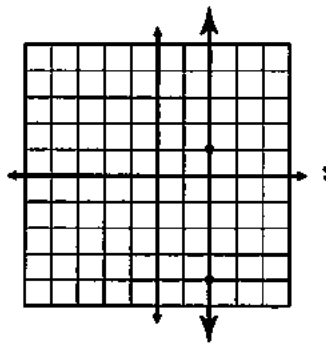
$$m = \frac{rise}{run}$$

Directions: Find the slope of each line. Write your answer in simplest form!

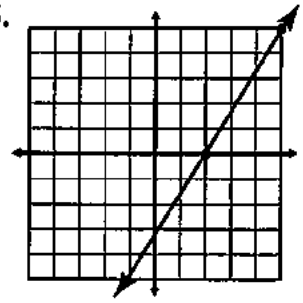
1.  $\frac{7}{4}$

2.  $\frac{-3}{9} = \frac{-1}{3}$

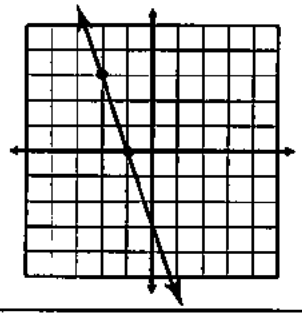
3.  $\frac{0}{4} = 0$

4.  $\frac{5}{0}$ = undef.

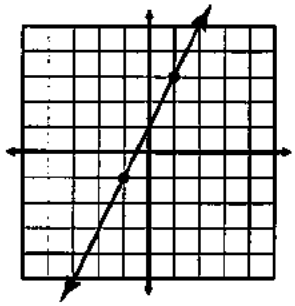
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5. 

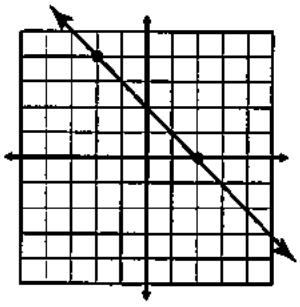
$\frac{5}{3}$

6. 

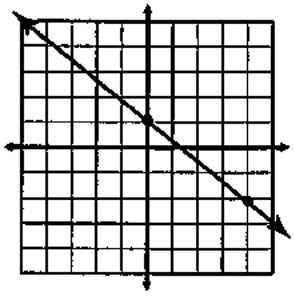
$\frac{-3}{1} = -3$

7. 

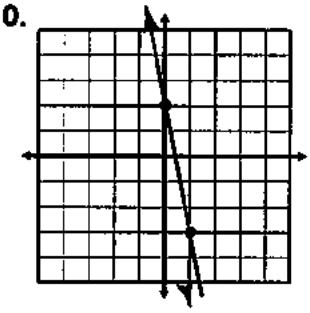
$\frac{4}{2} = 2$

8. 

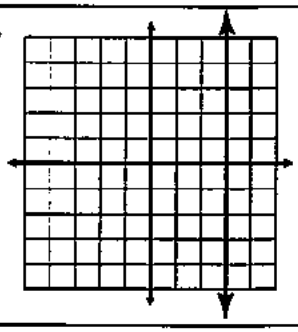
$\frac{-4}{4} = -1$

9. 

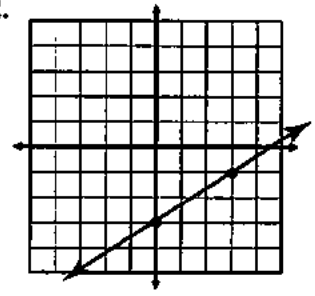
$\frac{-3}{4}$

10. 

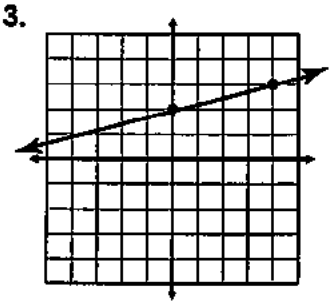
$\frac{-5}{1} = -5$

11. 

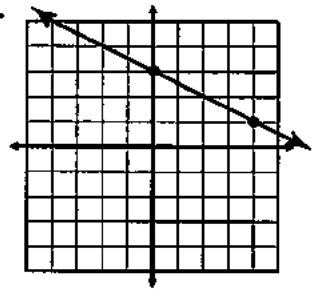
undef.

12. 

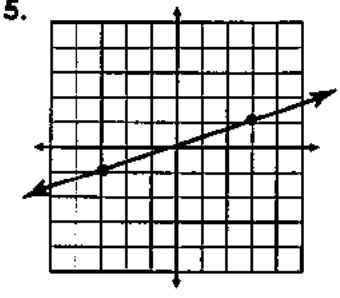
$\frac{2}{3}$

13. 

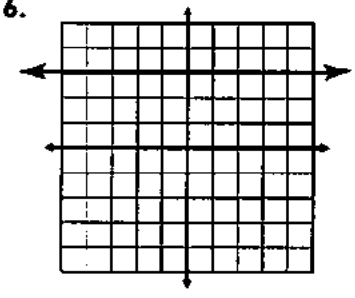
$\frac{1}{4}$

14. 

$\frac{-2}{4} = \frac{-1}{2}$

15. 

$\frac{2}{6} = \frac{1}{3}$

16. 

0

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
SLOPE FORMULA	The **slope formula** is used to find the slope between two points (x_1, y_1) and (x_2, y_2).
	Formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$
	It is important to remember to SIMPLIFY your answer!
EXAMPLES	**Directions:** Find the slope between each pair of points.

1. (1, 1) and (4, 3)

$m = \frac{3-1}{4-1} = \boxed{\frac{2}{3}}$

2. (-2, 4) and (10, -2)

$m = \frac{-2-4}{10+2} = \frac{-6}{12} = \boxed{-\frac{1}{2}}$

3. (-4, 5) and (-8, -5)

$m = \frac{-5-5}{-8+4} = \frac{-10}{-4} = \boxed{\frac{5}{2}}$

4. (10, 0) and (-2, 4)

$m = \frac{4-0}{-2-10} = \frac{4}{-12} = \boxed{-\frac{1}{3}}$

5. (5, 9) and (3, 9)

$m = \frac{9-9}{3-5} = \frac{0}{-2} = \boxed{0}$

6. (-7, 8) and (-7, 5)

$m = \frac{5-8}{-7+7} = \frac{-3}{0} = \boxed{\text{undef.}}$

7. (-1, 9) and (2, 3)

$m = \frac{3-9}{2+1} = \frac{-6}{3} = \boxed{-2}$

8. (-4, 13) and (6, -2)

$m = \frac{-2-13}{6+4} = \frac{-15}{10} = \boxed{-\frac{3}{2}}$

9. (5, 6) and (6, 5)

$m = \frac{5-6}{6-5} = \frac{-1}{1} = \boxed{-1}$

10. (9, -4) and (1, -4)

$m = \frac{-4+4}{1-9} = \frac{0}{-8} = \boxed{0}$

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11. (5, -9) and (3, -2)

$m = \frac{-2+9}{3-5} = \boxed{\frac{7}{-2}}$

12. (4, 6) and (4, 8)

$m = \frac{8-6}{4-4} = \frac{2}{0} = \boxed{\text{undef.}}$

GOING BACKWARDS

Sometimes you must complete the ordered pairs using a given slope.

Example: If the slope of the line passing through the points (-5, 6) and $(5, y)$ is $-\frac{4}{5}$, find y.

$\frac{-4}{5} = \frac{y-6}{5+5}$ $\quad \frac{-4}{5} = \frac{y-6}{10}$ $\quad -40 = 5y - 30$

$-10 = 5y$

$\boxed{-2 = y}$

MORE EXAMPLES

Directions: Find the missing value using the given slope.

13. (-3, -2) and $(x, 6)$; $m = 2$

$2 = \frac{6+2}{x+3}$ $\quad 2x+6 = 8$

$2x = 2$

$\boxed{x = 1}$

14. (0, -4) and $(x, -7)$; $m = \frac{3}{2}$

$\frac{3}{2} = \frac{-7+4}{x-0}$ $\quad \frac{3}{2} = \frac{-3}{x}$

$3x = -6$

$\boxed{x = -2}$

15. (-3, -4) and $(-5, y)$; $m = -\frac{9}{2}$

$\frac{-9}{2} = \frac{y+4}{-5+3}$ $\quad \frac{-9}{2} = \frac{y+4}{-2}$

$2y + 8 = 18$

$2y = 10$

$\boxed{y = 5}$

16. $(x, 2)$ and (6, 3); $m = -\frac{1}{2}$

$-\frac{1}{2} = \frac{3-2}{6-x}$ $\quad -\frac{1}{2} = \frac{1}{6-x}$

$-6 + x = 2$

$\boxed{x = 8}$

17. $(-3, y)$ and (1, -7); $m = -4$

$-4 = \frac{-7-y}{1+3}$ $\quad -7 - y = -16$

$-y = -9$

$\boxed{y = 9}$

18. $(x, 7)$ and (11, 8); $m = -\frac{1}{5}$

$-\frac{1}{5} = \frac{8-7}{11-x}$ $\quad -\frac{1}{5} = \frac{1}{11-x}$

$5 = -11 + x$

$\boxed{16 = x}$

19. $(4, y)$ and (0, 5); $m = \frac{3}{4}$

$\frac{3}{4} = \frac{5-y}{0-4}$ $\quad \frac{3}{4} = \frac{5-y}{-4}$

$20 - 4y = -12$

$-4y = -32$

$\boxed{y = 8}$

20. $(-3, y)$ and (9, -2); $m = \frac{1}{3}$

$\frac{1}{3} = \frac{-2-y}{9+3}$ $\quad \frac{1}{3} = \frac{-2-y}{12}$

$12 = -6 - 3y$

$18 = -3y$

$\boxed{-6 = y}$

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slope formula tic-tac-toe!

Directions: With your partner, decide who is X and who is O. Each pick a problem below, find the slope, then find your answer in the grid below. Mark your answer with your X and O. Cross out the two problems you just did, then pick two more. Continue until someone gets four in a row, column, or diagonal.

(10, -6) and (13, -7) $m = -\frac{1}{3}$

(5, 5) and (15, -7) $m = -\frac{12}{5}$

(0, -9) and (-3, -3) $m = -2$

(-2, 9) and (-2, -5) m= undef.

(-3, 4) and (-10, 6) $m = -\frac{2}{7}$

(2, -8) and (4, -6) $m = 1$

(4, 19) and (2, 11) $m = 4$

(-2, -9) and (-6, -6) $m = -\frac{3}{4}$

(-17, -13) and (-9, 7) $m = \frac{5}{2}$

(-8, 7) and (-11, 3) $m = \frac{4}{3}$

(-1, 7) and (-4, 7) $m = 0$

(11, 5) and (2, 1) $m = \frac{4}{9}$

(3, 10) and (-13, 2) $m = \frac{1}{2}$

(2, -12) and (18, -8) $m = \frac{1}{4}$

(-7, -4) and (-14, 3) $m = -1$

(-1, -4) and (-6, -2) $m = -\frac{2}{5}$

$m = 0$	$m = \frac{5}{2}$	$m = 4$	$m = -1$
$m = -\frac{3}{4}$	$m = \frac{1}{4}$	$m = -2$	$m = -\frac{2}{7}$
$m = 1$	$m = \frac{4}{9}$	$m = -\frac{2}{5}$	$m = -\frac{1}{3}$
$m = -\frac{6}{5}$	$m = \frac{1}{2}$	$m =$ undefined	$m = \frac{4}{3}$

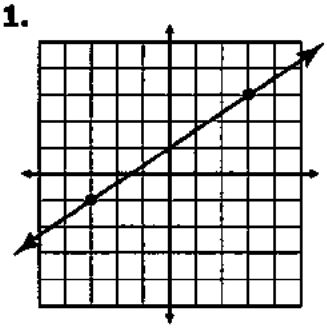
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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 1:** Slope

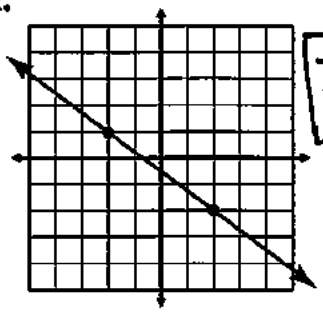
Directions: Find the slope of the line show on the graph.

1.



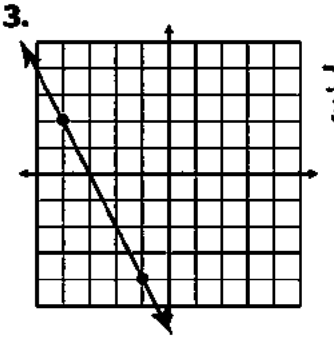
$\frac{4}{6} = \boxed{\frac{2}{3}}$

2.



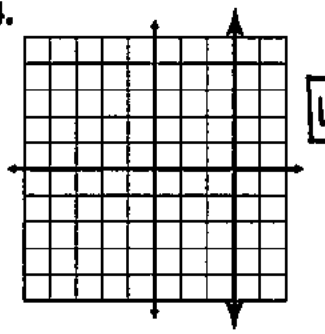
$\boxed{\frac{-3}{4}}$

3.



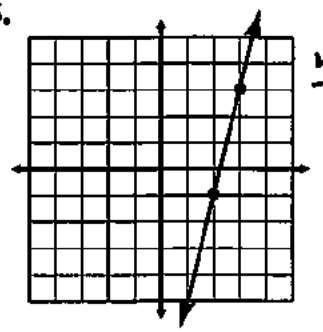
$\frac{-6}{3} = \boxed{-2}$

4.



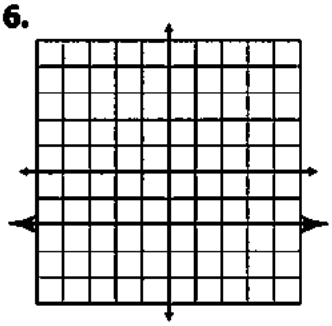
$\boxed{\text{undef.}}$

5.



$\frac{4}{1} = \boxed{4}$

6.



$\boxed{0}$

Directions: Use the slope formula to find the slope between the given two points.

7. (4, 9) and (1, 6)

$\frac{6-9}{1-4} = \frac{-3}{-3} = \boxed{1}$

8. (-4, -1) and (-2, -5)

$\frac{-5+1}{-2+4} = \frac{-4}{2} = \boxed{-2}$

9. (5, 3) and (5, -9)

$\frac{-9-3}{5-5} = \frac{-12}{0}$

$= \boxed{\text{undef.}}$

10. (2, 1) and (8, 9)

$\frac{9-1}{8-2} = \frac{8}{6} = \boxed{\frac{4}{3}}$

11. (14, -8) and (7, -6)

$\frac{-6+8}{7-14} = \boxed{\frac{2}{-7}}$

12. (4, -3) and (8, -3)

$\frac{-3+3}{8-4} = \frac{0}{4} = \boxed{0}$

Directions: Find the missing value so that the line passing through the points has the given slope.

13. $(x, -4)$ and $(2, 8)$; $m = -3$

$-3 = \frac{8+4}{2-x}$

$-6+3x = 12$

$3x = 18$

$\boxed{x = 6}$

14. $(10, y)$ and $(3, 4)$; $m = -\frac{2}{7}$

$\frac{-2}{7} = \frac{4-y}{3-10}$

$28 - 7y = 14$

$-7y = -14$

$\boxed{y = 2}$

15. $(7, 1)$ and $(x, 4)$; $m = \frac{3}{4}$

$\frac{3}{4} = \frac{4-1}{x-7}$

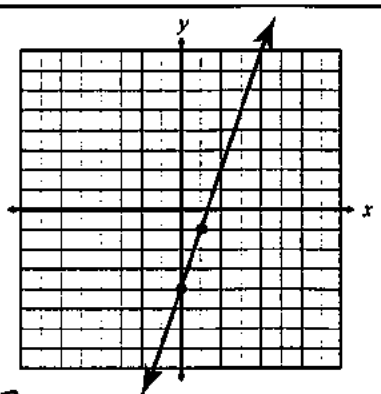
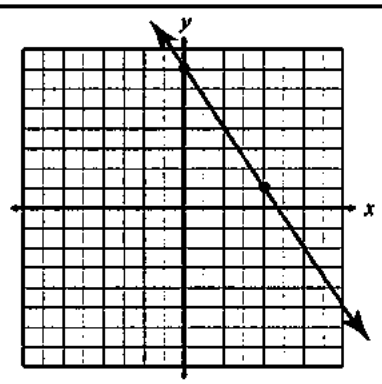
$3x - 21 = 12$

$3x = 33$

$\boxed{x = 11}$

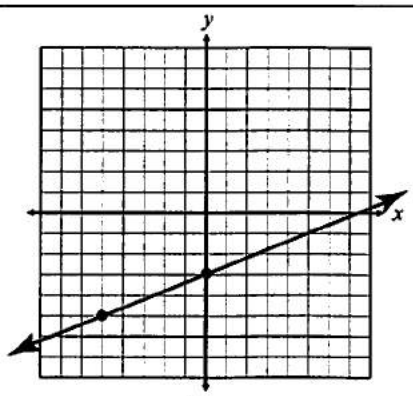
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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Slope-Intercept Form	Linear equations are frequently written in **slope-intercept form:** $y = mx + b$ m is the slope and b is the y-intercept
Examples	**Directions:** Given the slope and y-intercept of the line, write the equation in slope-intercept form. **1.** slope = 2; y-intercept = -1 — $y = 2x - 1$ **2.** slope = $-\frac{3}{5}$; y-intercept = 4 — $y = -\frac{3}{5}x + 4$ **3.** slope = -3; y-intercept = 2 — $y = -3x + 2$ **4.** slope = -1; y-intercept = 7 — $y = -x + 7$ **5.** slope = $\frac{1}{4}$; y-intercept = 0 — $y = \frac{1}{4}x$ **6.** slope = $-\frac{5}{2}$; y-intercept = -3 — $y = -\frac{5}{2}x - 3$
Given a Graph	**Directions:** Identify the slope and y-intercept of the line on the graph. Then, write the equation of the line in slope-intercept form. **7.**  $m = \frac{3}{1} = 3$ $b = -4$ Equation: $y = 3x - 4$ **8.**  $m = \frac{-6}{4} = -\frac{3}{2}$ $b = 7$ Equation: $y = -\frac{3}{2}x + 7$

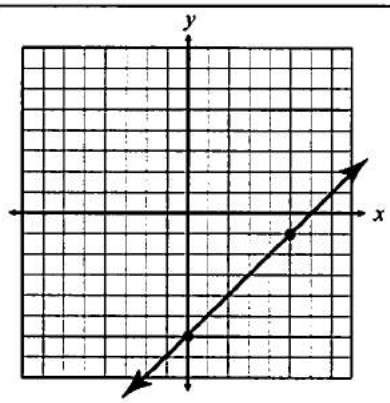
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9.



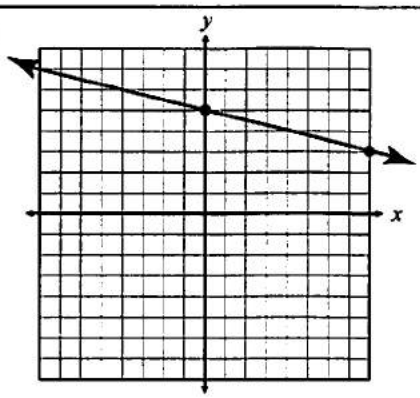
Equation: $y = \frac{2}{5}x - 3$

10.



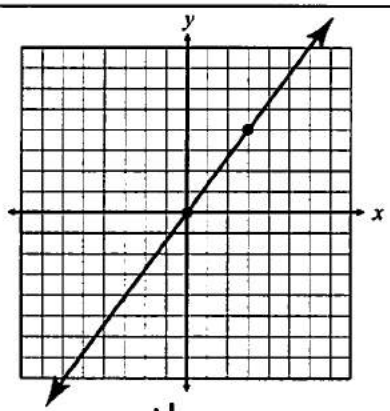
Equation: $y = x - 6$

11.



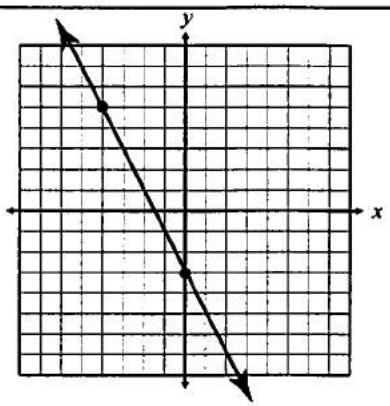
Equation: $y = -\frac{1}{4}x + 5$

12.



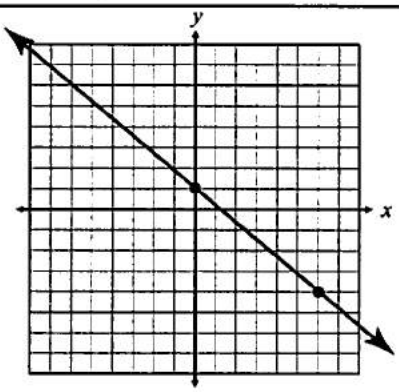
Equation: $y = \frac{4}{3}x$

13.



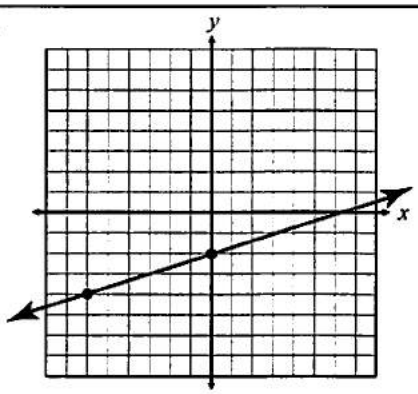
Equation: $y = -2x - 3$

14.



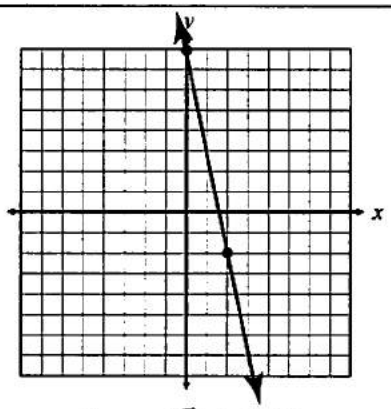
Equation: $y = -\frac{5}{6}x + 1$

15.



Equation: $y = \frac{1}{3}x - 2$

16.



Equation: $y = -5x + 8$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
STANDARD FORM	Linear equations are also frequently written in **standard form:** $\boxed{Ax + By = C}$
CONVERTING Standard Form ➔ Slope-Intercept Form	**Equations written in standard form can be converted to slope-intercept form by solving for y!**

Directions: Write each equation in slope-intercept form. Identify the slope and y-intercept.

1. $2x + y = 3$

$-2x \quad -2x$

$\boxed{y = -2x + 3}$

$m = -2;\ b = 3$

2. $4x + 5y = -30$

$-4x \quad -4x$

$\frac{5y}{5} = \frac{-4x}{5} - \frac{30}{5}$

$\boxed{y = -\frac{4}{5}x - 6}$

$m = -4/5;\ b = -6$

3. $x - 3y = 12$

$-x \quad -x$

$\frac{-3y}{-3} = \frac{-x}{-3} + \frac{12}{-3}$

$\boxed{y = \frac{1}{3}x - 4}$

$m = 1/3;\ b = -4$

4. $x - y = -8$

$-x \quad -x$

$\frac{-y}{-1} = \frac{-x}{-1} - \frac{8}{-1}$

$\boxed{y = x + 8}$

$m = 1;\ b = 8$

5. $4x - y = 0$

$-4x \quad -4x$

$\frac{-y}{-1} = \frac{-4x}{-1}$

$\boxed{y = 4x}$

$m = 4;\ b = 0$

6. $3x - 2y = -14$

$-3x \quad -3x$

$\frac{-2y}{-2} = \frac{-3x}{-2} - \frac{14}{-2}$

$\boxed{y = \frac{3}{2}x + 7}$

$m = 3/2;\ b = 7$

7. $-6x + 4y = 20$

$+6x \quad +6x$

$\frac{4y}{4} = \frac{6x}{4} + \frac{20}{4}$

$\boxed{y = \frac{3}{2}x + 5}$

$m = 3/2;\ b = 5$

8. $x + 5y = -15$

$-x \quad -x$

$\frac{5y}{5} = \frac{-x}{5} - \frac{15}{5}$

$\boxed{y = -\frac{1}{5}x - 3}$

$m = -1/5;\ b = -3$

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9. $3x + 3y = -12$

$-3x \quad -3x$

$\frac{3y}{3} = \frac{-3x}{3} - \frac{12}{3}$

$\boxed{y = -x - 4}$

$m = -1$; $b = -4$

10. $-2x - 4y = 36$

$+2x \quad +2x$

$\frac{-4y}{-4} = \frac{2x}{-4} + \frac{36}{-4}$

$\boxed{y = -\frac{1}{2}x - 9}$

$m = -1/2$; $b = -9$

11. $12x - 9y = -9$

$-12x \quad -12x$

$\frac{-9y}{-9} = \frac{-12x}{-9} - \frac{9}{-9}$

$\boxed{y = \frac{4}{3}x + 1}$

$m = \frac{4}{3}$; $b = 1$

12. $4x + 10y = 10$

$-4x \quad -4x$

$\frac{10y}{10} = \frac{-4x}{10} + \frac{10}{10}$

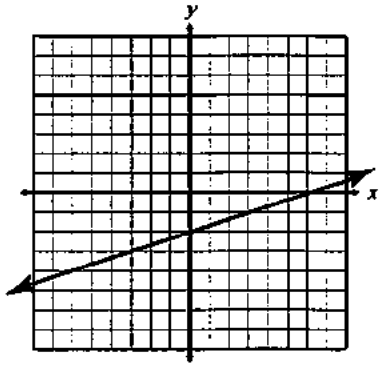
$\boxed{y = -\frac{2}{5}x + 1}$

$m = -2/5$; $b = 1$

GIVEN GRAPHS

Directions: Determine the equation that best represents the line shown on the graph. ***Hint: Convert them to slope-intercept form!**

13.



$m = 1/3$

$b = -2$

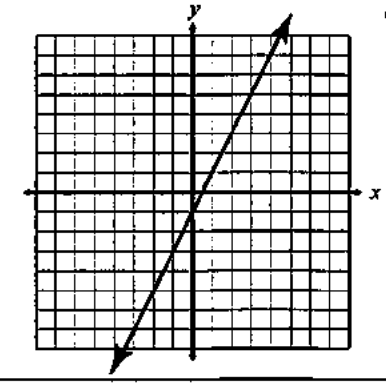
A. $x + 3y = -6$ → $y = -1/3x - 2$

B. $x - 3y = 6$ → $y = 1/3x - 2$

C. $3x + y = -2$ → $y = -3x - 2$

D. $3x - y = 2$ → $y = 3x - 2$

14.



$m = 2$

$b = -1$

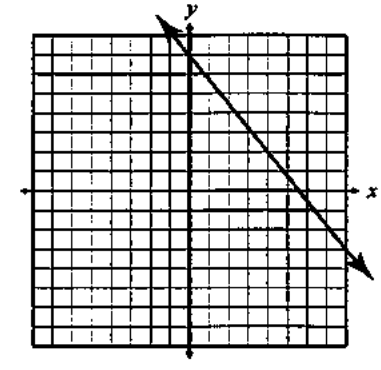
A. $x + 2y = -2$ → $y = -1/2x - 1$

B. $x - 2y = 2$ → $y = 1/2x - 1$

C. $2x + y = -1$ → $y = -2x - 1$

D. $2x - y = 1$ → $y = 2x - 1$

15.



$m = -5/4$

$b = 7$

A. $4x + 5y = 35$ → $y = -\frac{4}{5}x + 7$

B. $4x - 5y = -35$ → $y = \frac{4}{5}x + 7$

C. $5x + 4y = 28$ → $y = -\frac{5}{4}x + 7$

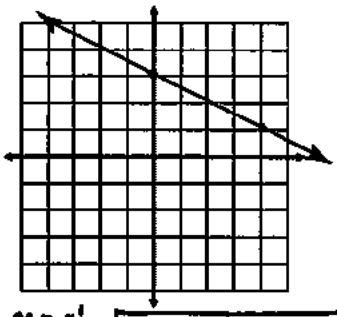
D. $5x - 4y = -28$ → $y = \frac{5}{4}x + 7$

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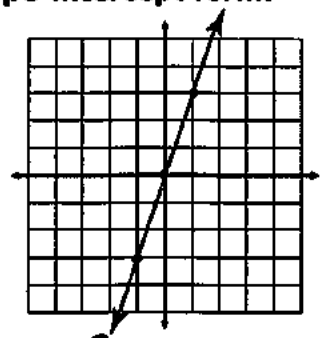
entrance ticket

Name: _____

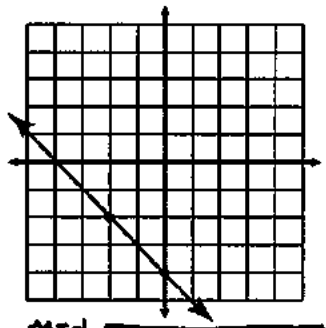
Write the equation of the line in slope-intercept form:

1) 

$m = -\frac{1}{2}$, $b = 3$ → $y = -\frac{1}{2}x + 3$

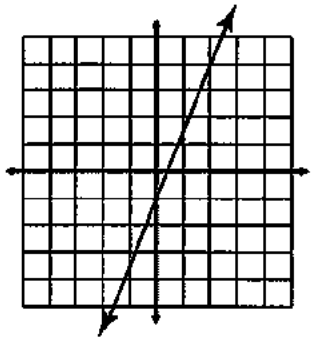
2) 

$m = 3$, $b = 0$ → $y = 3x$

3) 

$m = -1$, $b = -4$ → $y = -x - 4$

Identify the equation of the line graphed.

4) 

A. $2x + 5y = -5$

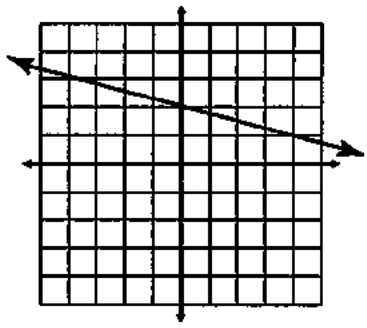
B. $2x - 5y = 5$

C. $5x + 2y = -2$

D. $5x - 2y = 2$

$-2y = -5x + 2$

$y = \frac{5}{2}x - 1$

5) 

A. $4x + y = 2$

B. $x + 4y = 8$

C. $4x - y = -2$

D. $x - 4y = -8$

$4y = -x + 8$

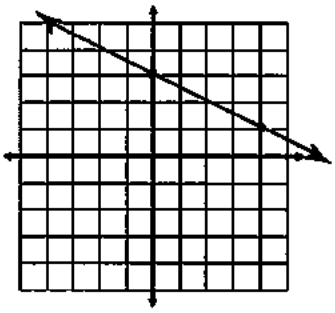
$y = -\frac{1}{4}x + 2$

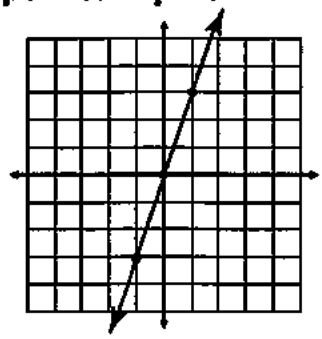
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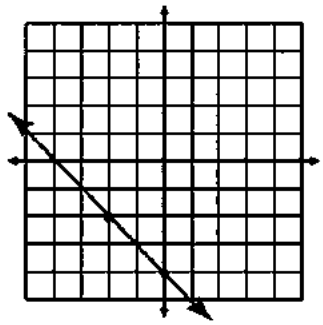
entrance ticket

Name: _____

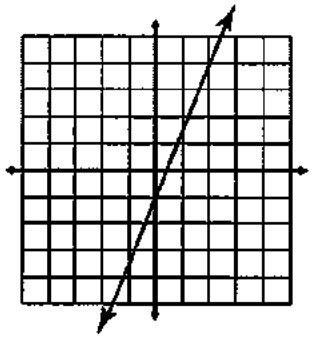
Write the equation of the line in slope-intercept form:

1) 

2) 

3) 

Identify the equation of the line graphed.

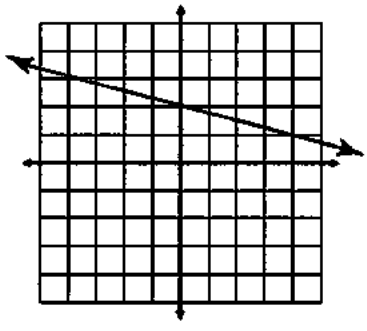
4) 

A. $2x + 5y = -5$

B. $2x - 5y = 5$

C. $5x + 2y = -2$

D. $5x - 2y = 2$

5) 

A. $4x + y = 2$

B. $x + 4y = 8$

C. $4x - y = -2$

D. $x - 4y = -8$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 2:** Slope-Intercept & Standard Form

**** This is a 2-page document! ****

Directions: Give the equation for each form below.

Slope-Intercept Form	**Standard Form**
$y = mx + b$	$Ax + By = C$

Directions: Write the equation of the line with the given slope and y-intercept in slope-intercept form.

1. slope = $\frac{1}{3}$; y-intercept = -1

$y = \frac{1}{3}x - 1$

2. slope = $-\frac{2}{5}$; y-intercept = 0

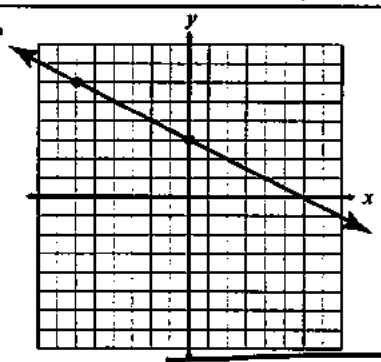
$y = \frac{-2}{5}x$

3. slope = 1; y-intercept = 8

$y = x + 8$

Directions: Write the equation of the line shown on the graph in slope-intercept form.

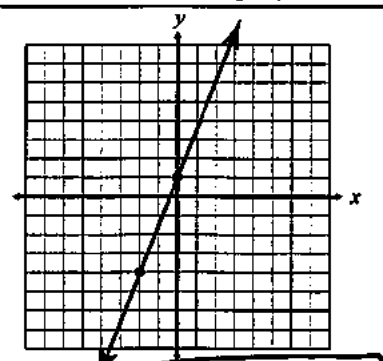
4.



$m = -1/2$, $b = 3$

$y = -\frac{1}{2}x + 3$

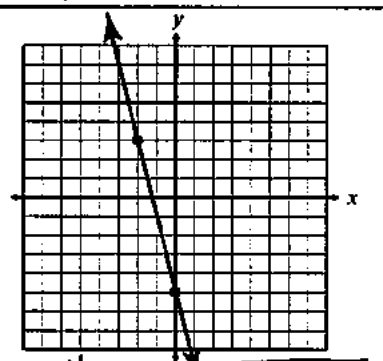
5.



$m = \frac{5}{2}$, $b = 1$

$y = \frac{5}{2}x + 1$

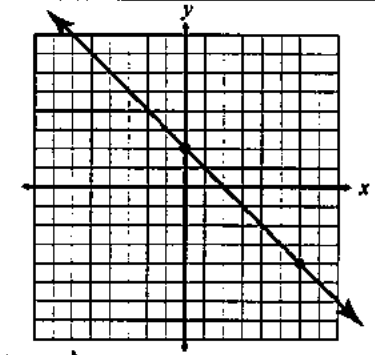
6.



$m = -4$, $b = -5$

$y = -4x - 5$

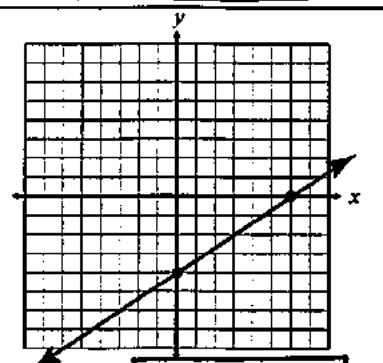
7.



$m = -1$, $b = 2$

$y = -x + 2$

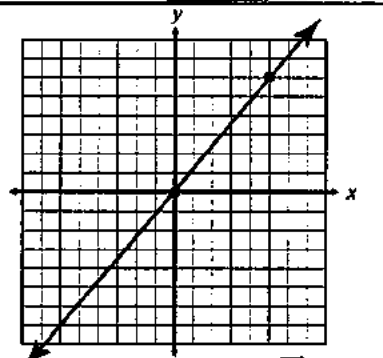
8.



$m = \frac{2}{3}$, $b = -4$

$y = \frac{2}{3}x - 4$

9.



$m = \frac{6}{5}$, $b = 0$

$y = \frac{6}{5}x$

Directions: Convert the following equations from standard form to slope-intercept form.

10. $6x + 2y = 10$

$-6x \quad -6x$

$\frac{2y}{2} = \frac{-6x}{2} + \frac{10}{2}$

$y = -3x + 5$

11. $x + 2y = 4$

$-x \quad -x$

$\frac{2y}{2} = \frac{-x}{2} + \frac{4}{2}$

$y = -\frac{1}{2}x + 2$

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12. $x + y = -3$

$-x \quad -x$

$y = -x - 3$

13. $-2x + 3y = -6$

$+2x \quad +2x$

$\frac{3y}{3} = \frac{2x}{3} - \frac{6}{3}$

$y = \frac{2}{3}x - 2$

14. $4x + y = -1$

$-4x \quad -4x$

$y = -4x - 1$

15. $5x - 2y = 14$

$-5x \quad -5x$

$\frac{-2y}{-2} = \frac{-5x}{-2} + \frac{14}{-2}$

$y = \frac{5}{2}x - 7$

16. $10x - 2y = -6$

$-10x \quad -10x$

$\frac{-2y}{-2} = \frac{-10x}{-2} - \frac{6}{-2}$

$y = 5x + 3$

17. $8x + 6y = 12$

$-8x \quad -8x$

$\frac{6y}{6} = \frac{-8x}{6} + \frac{12}{6}$

$y = -\frac{4}{3}x + 2$

18. $6x - 10y = 20$

$-6x \quad -6x$

$\frac{-10y}{-10} = \frac{-6x}{-10} + \frac{20}{-10}$

$y = \frac{3}{5}x - 2$

19. $3x + 9y = -36$

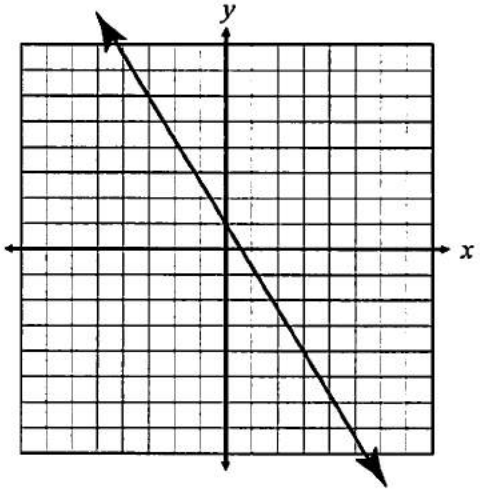
$-3x \quad -3x$

$\frac{9y}{9} = \frac{-3x}{9} - \frac{36}{9}$

$y = -\frac{1}{3}x - 4$

Directions: Determine the equation that best represents the line shown on the graph.

20.



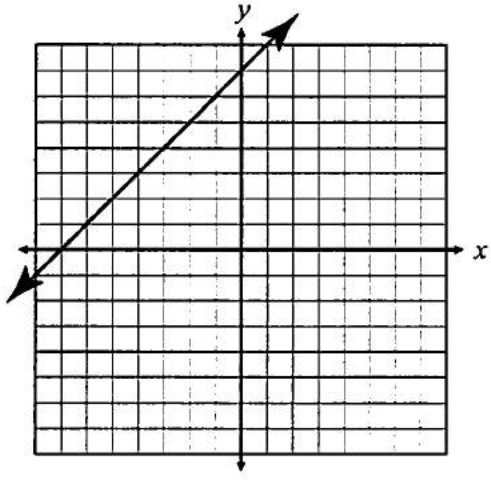
A. $-5x + 3y = 3$

(C.) $5x + 3y = 3$

B. $5x + 3y = -3$

D. $5x - 3y = -3$

21.



(A.) $x - y = -7$

C. $x - y = 7$

B. $x + y = -7$

D. $x + y = 7$

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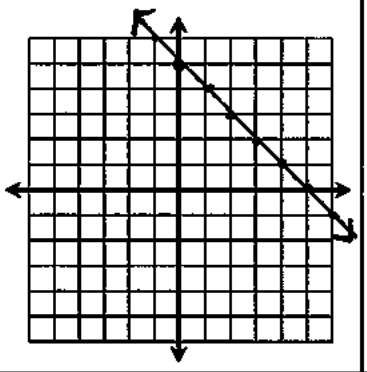
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
GRAPHING LINEAR EQUATIONS (By Slope-Intercept)	**Use the steps below to graph an equation using slope-intercept form:**

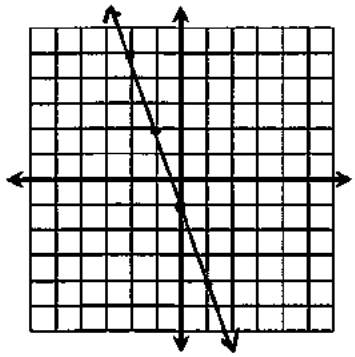
❶	Write the equation in **slope-intercept form**.
❷	Graph the ***y*-intercept.** This is always point $(0, b)$.
❸	Use the **slope** of the line to create more points. Remember slope is rise/run!
❹	Use a ruler to draw a line that extends through the points, placing an arrow on both ends.

Directions: Graph each equation using the slope-intercept method.

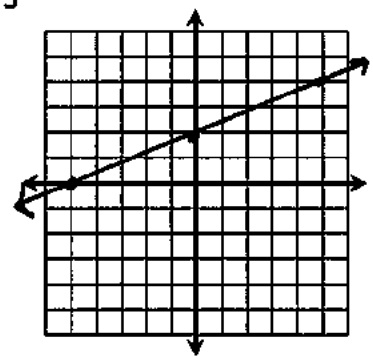
1. $y = -x + 5$



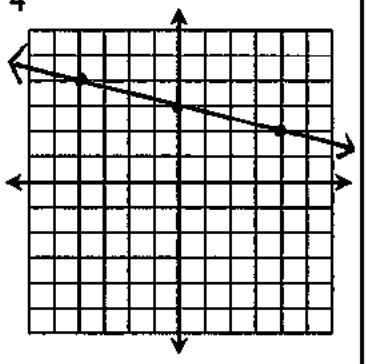
2. $y = -3x - 1$



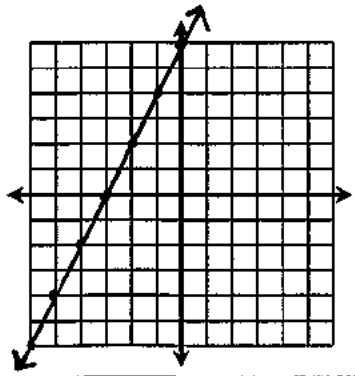
3. $y = \frac{2}{5}x + 2$



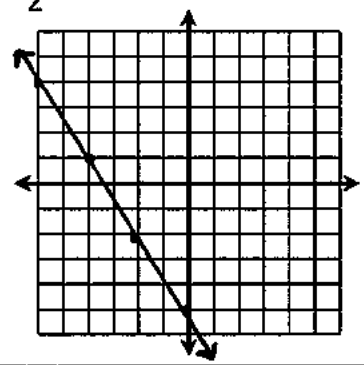
4. $y = -\frac{1}{4}x + 3$



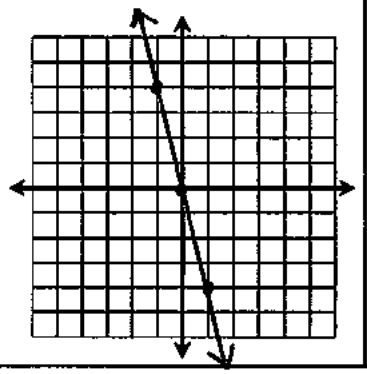
5. $y = 2x + 6$



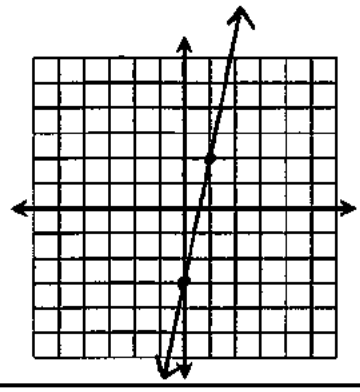
6. $y = -\frac{3}{2}x - 5$



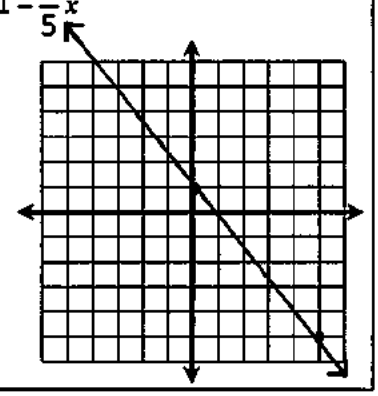
8. $y = -4x$



7. $y = -3 + 5x$



9. $y = 1 - \frac{6}{5}x$



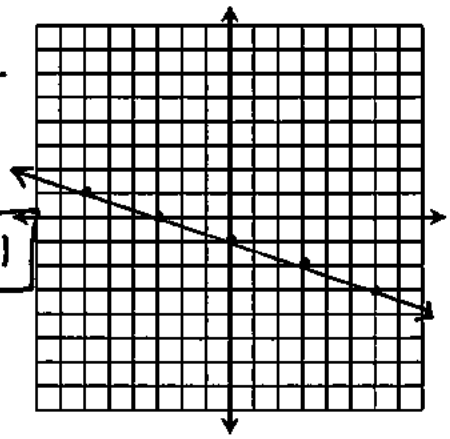
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10. $x + 3y = -3$

$-x \quad -x$

$\frac{3y}{3} = \frac{-x}{3} - \frac{3}{3}$

$y = -\frac{1}{3}x - 1$

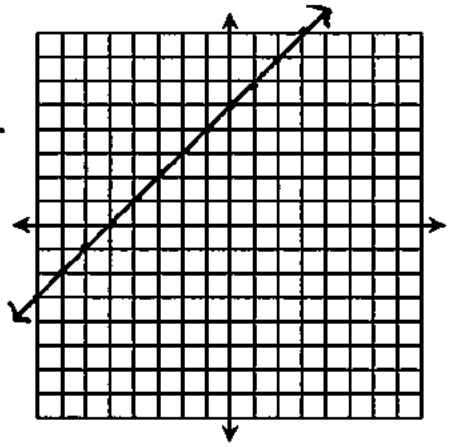


11. $x - y = -5$

$-x \quad -x$

$\frac{-y}{-1} = \frac{-x}{-1} - \frac{5}{-1}$

$y = x + 5$

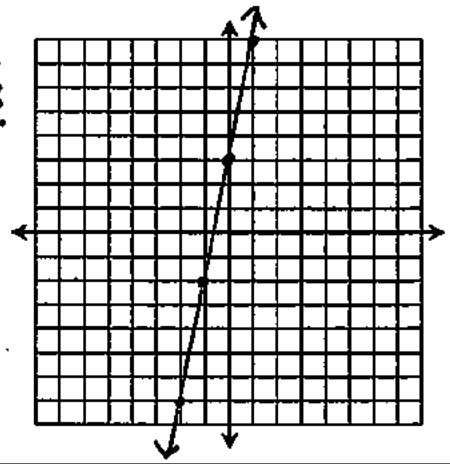


12. $5x - y = -3$

$-5x \quad -5x$

$\frac{-y}{-1} = \frac{-5x}{-1} - \frac{3}{-1}$

$y = 5x + 3$

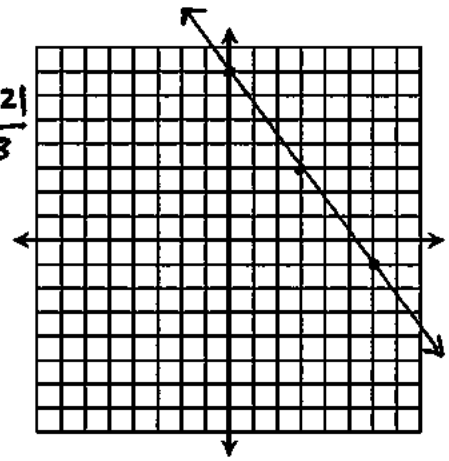


13. $4x + 3y = 21$

$-4x \quad -4x$

$\frac{3y}{3} = \frac{-4x}{3} + \frac{21}{3}$

$y = -\frac{4}{3}x + 7$

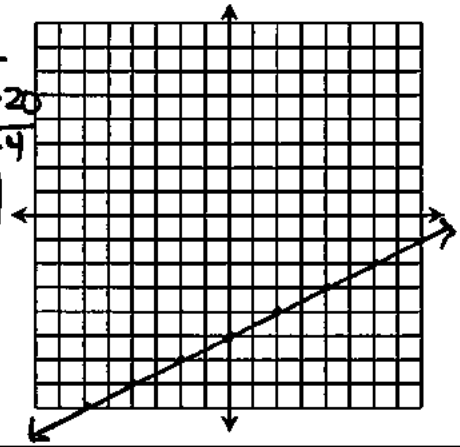


14. $2x - 4y = 20$

$-2x \quad -2x$

$\frac{-4y}{-4} = \frac{-2x}{-4} + \frac{20}{-4}$

$y = \frac{1}{2}x - 5$

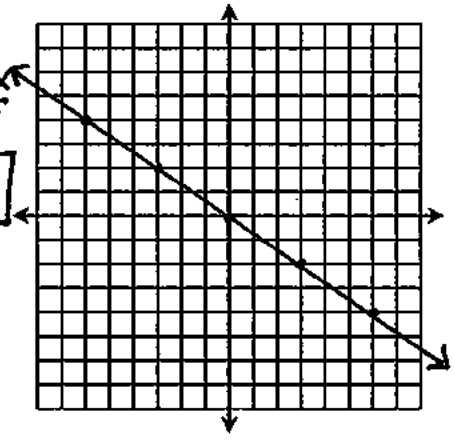


15. $2x + 3y = 0$

$-2x \quad -2x$

$\frac{3y}{3} = \frac{-2x}{3}$

$y = -\frac{2}{3}x$

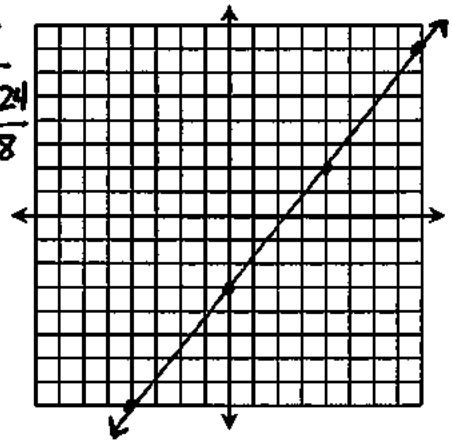


16. $10x - 8y = 24$

$-10x \quad -10x$

$\frac{-8y}{-8} = \frac{-10x}{-8} + \frac{24}{-8}$

$y = \frac{5}{4}x - 3$

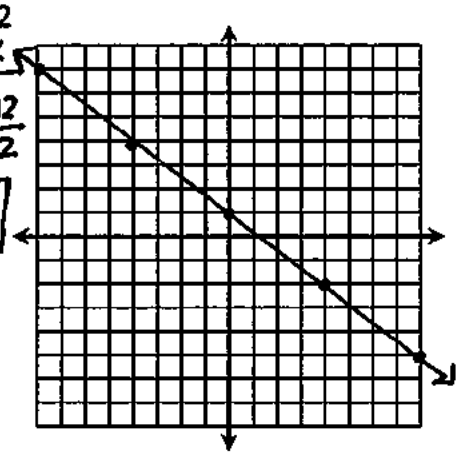


17. $9x + 12y = 12$

$-9x \quad -9x$

$\frac{12y}{12} = \frac{-9x}{12} + \frac{12}{12}$

$y = -\frac{3}{4}x + 1$



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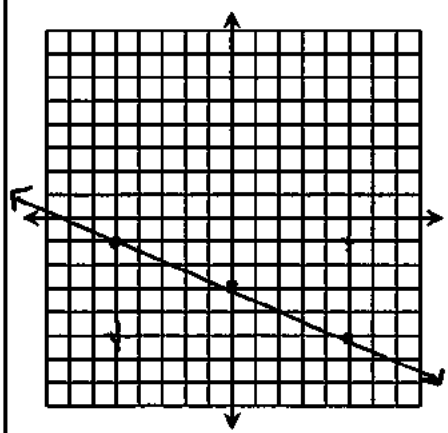
Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 3:** Graphing Linear Equations by Slope-Intercept Form

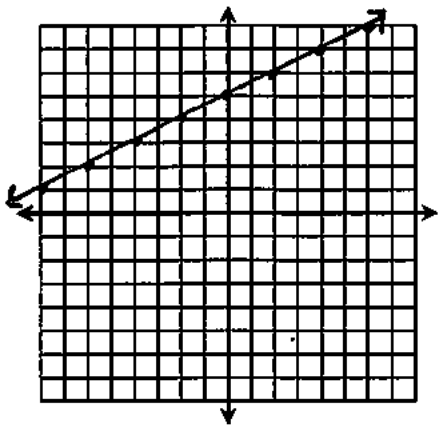
**** This is a 2-page document! ****

Directions: Graph the following linear equations. Convert to slope-intercept form when necessary.

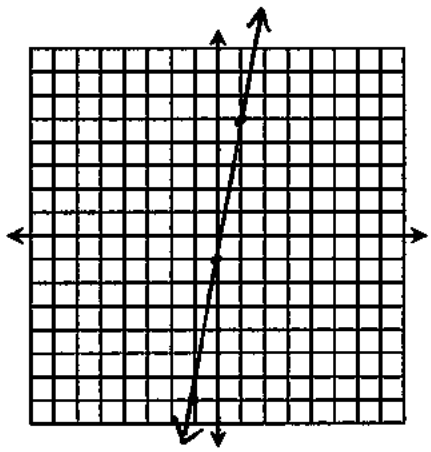
1. $y = -\frac{2}{5}x - 3$



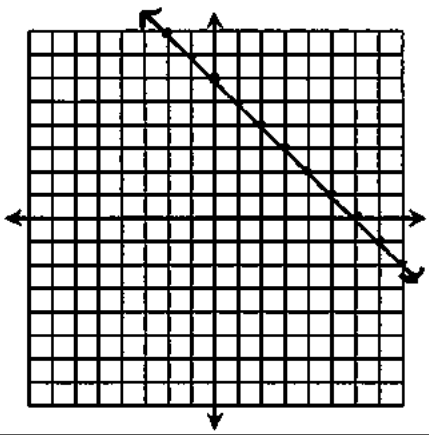
2. $y = \frac{1}{2}x + 5$



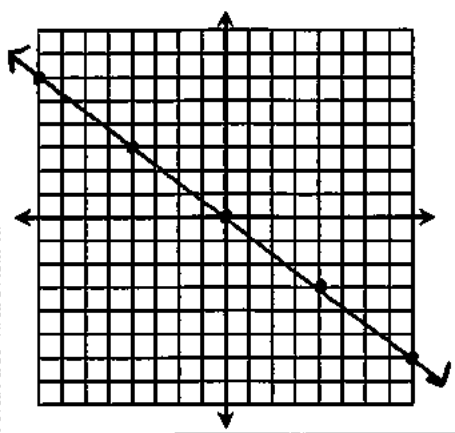
3. $y = 6x - 1$



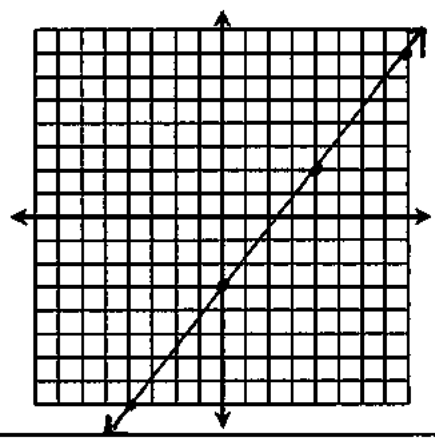
4. $y = -x + 6$



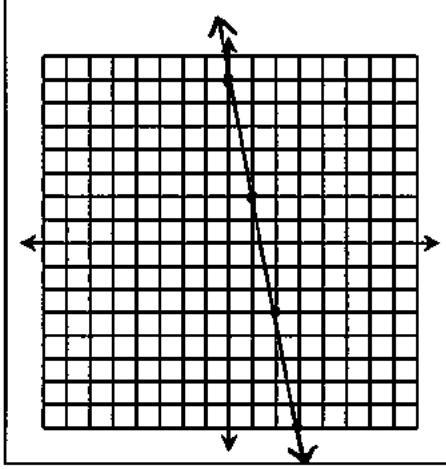
5. $y = -\frac{3}{4}x$



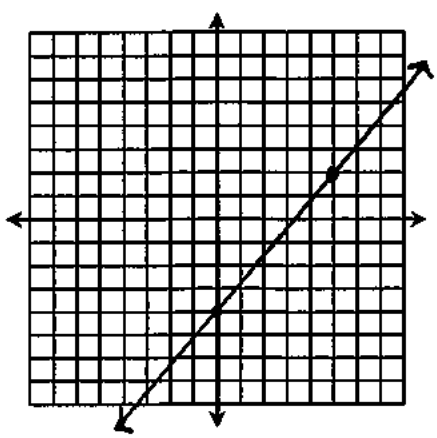
6. $y = \frac{5}{4}x - 3$



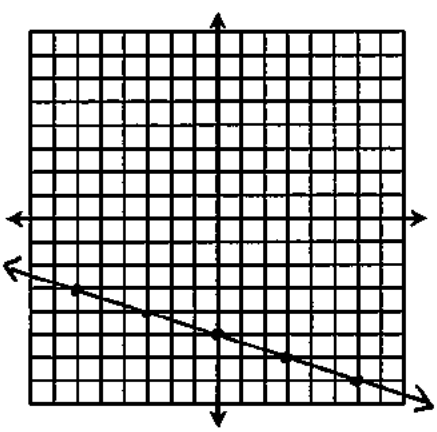
7. $y = 7 - 5x$



8. $y = -4 + \frac{6}{5}x$



9. $y = -5 - \frac{1}{3}x$



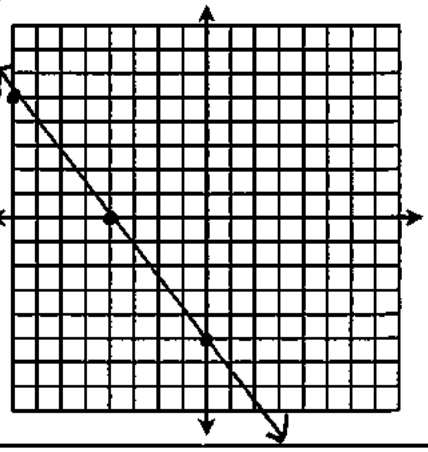
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10. $5x + 4y = -20$

$-5x \quad -5x$

$\frac{4y}{4} = \frac{-5x}{4} - \frac{20}{4}$

$y = -\frac{5}{4}x - 5$

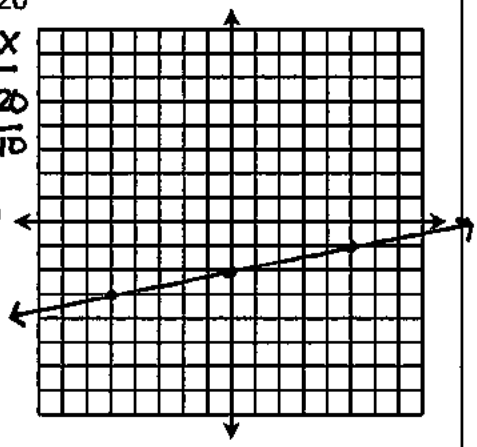


11. $2x - 10y = 20$

$-2x \quad -2x$

$\frac{-10y}{-10} = \frac{-2x}{-10} + \frac{20}{-10}$

$y = \frac{1}{5}x - 2$

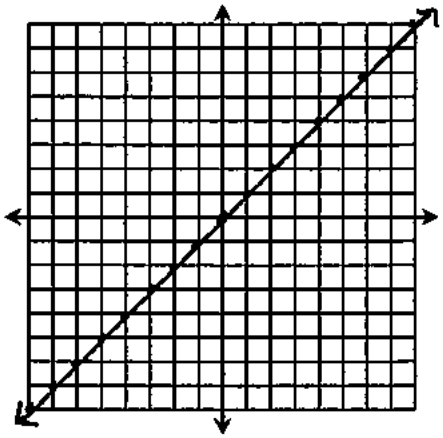


12. $x - y = 0$

$-x \quad -x$

$\frac{-y}{-1} = \frac{-x}{-1}$

$y = x$

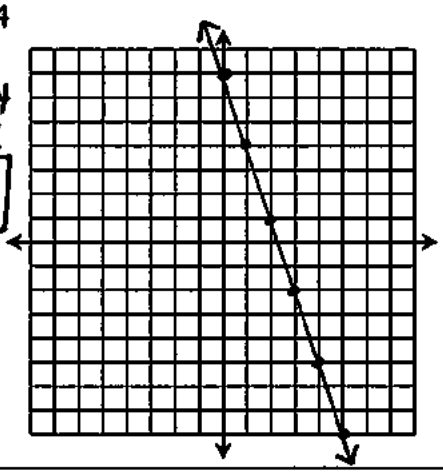


13. $6x + 2y = 14$

$-6x \quad -6x$

$\frac{2y}{2} = \frac{-6x}{2} + \frac{14}{2}$

$y = -3x + 7$

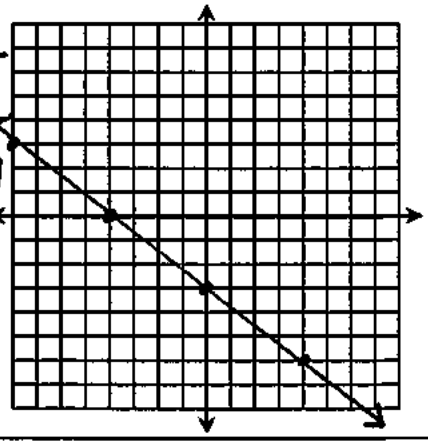


14. $3x + 4y = -12$

$-3x \quad -3x$

$\frac{4y}{4} = \frac{-3x}{4} - \frac{12}{4}$

$y = -\frac{3}{4}x - 3$

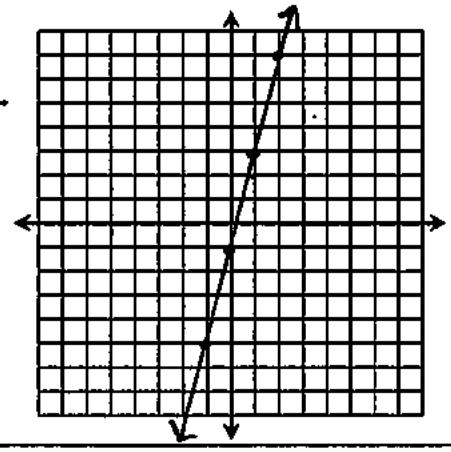


15. $4x - y = 1$

$-4x \quad -4x$

$\frac{-y}{-1} = \frac{-4x}{-1} + \frac{1}{-1}$

$y = 4x - 1$

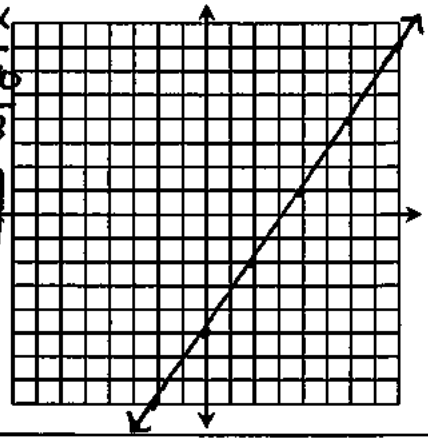


16. $12x - 8y = 40$

$-12x \quad -12x$

$\frac{-8y}{-8} = \frac{-12x}{-8} + \frac{40}{-8}$

$y = \frac{3}{2}x - 5$

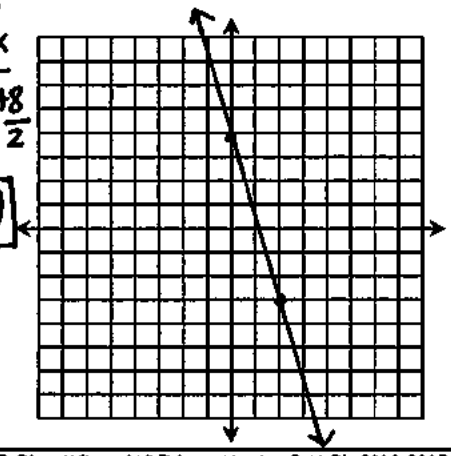


17. $7x + 2y = 8$

$-7x \quad -7x$

$\frac{2y}{2} = \frac{-7x}{2} + \frac{8}{2}$

$y = -\frac{7}{2}x + 4$



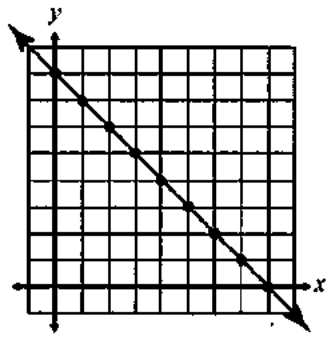
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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

X- and Y-Intercepts

- The point at which the line intersects the x-axis is called the **x-intercept.**
- The point at which the line intersects the y-axis is called the **y-intercept.**
- **Example:** Identify the x- and y-intercept of the graph shown to the right.



x-int: (8,0) y-int: (0,8)

Finding Intercepts Algebraically

- To find the x-intercept of an equation:

set y equal to 0 and solve for x.

- To find the y-intercept of an equation:

set x equal to 0 and solve for y

- **Example:** Find the x- and y-intercept of the equation $y = 3x + 6$.

x-int:
$0 = 3x + 6$
$-6 = 3x$
$-2 = x$
(-2,0)

y-int:
$y = 3(0) + 6$
$y = 6$
(0,6)

Examples

Directions: Find the x- and y-intercept of each equation.

1. $y = -x + 5$

x-int:
$0 = -x + 5$
$-5 = -x$
$5 = x$

y-int:
$y = -0 + 5$
$y = 5$

x-int: (5,0)
y-int: (0,5)

2. $y = \frac{1}{2}x - 8$

x-int:
$0 = \frac{1}{2}x - 8$
$8 = \frac{1}{2}x$
$16 = x$

y-int:
$y = \frac{1}{2}(0) - 8$
$y = -8$

x-int: (16,0)
y-int: (0,-8)

3. $y = -\frac{4}{3}x + 2$

x-int:
$0 = -\frac{4}{3}x + 2$
$-2 = -\frac{4}{3}x$
$x = \frac{3}{2}$

y-int:
$y = -\frac{4}{3}(0) + 2$
$y = 2$

x-int: $(\frac{3}{2}, 0)$
y-int: (0,2)

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4. $x - y = 2$

x-int:
$x - 0 = 2$
$x = 2$

y-int:
$0 - y = 2$
$-y = 2$
$y = -2$

x-int: (2, 0)
y-int: (0, -2)

5. $3x - 2y = 12$

x-int:
$3x - 2(0) = 12$
$3x = 12$
$x = 4$

y-int:
$3(0) - 2y = 12$
$-2y = 12$
$y = -6$

x-int: (4, 0)
y-int: (0, -6)

6. $8x + 10y = -10$

x-int:
$8x + 10(0) = -10$
$8x = -10$
$x = -\frac{5}{4}$

y-int:
$8(0) + 10y = -10$
$10y = -10$
$y = -1$

x-int: $(-\frac{5}{4}, 0)$
y-int: (0, -1)

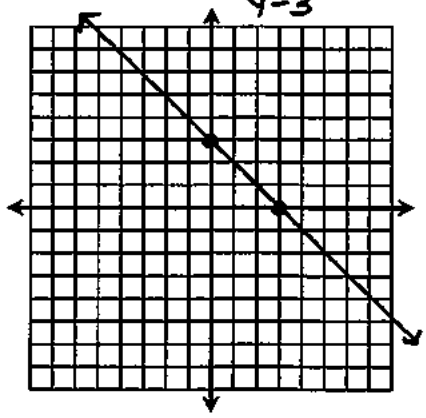
Graphing by Intercepts

Directions: Find the x- and y-intercept of each equation. Graph the equation using its intercepts.

7. $x + y = 3$

x-int:
$x + 0 = 3$
$x = 3$

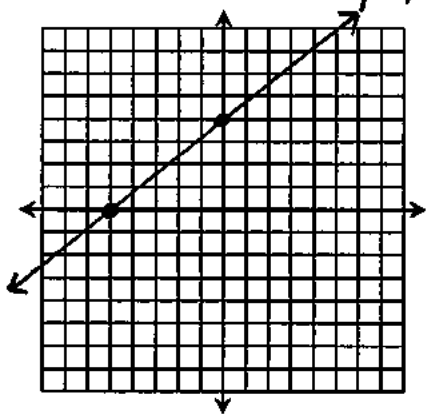
y-int:
$0 + y = 3$
$y = 3$



8. $-4x + 5y = 20$

x-int: $-4x = 20$
$x = -5$

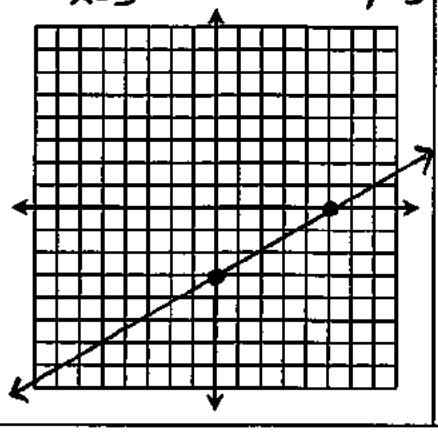
y-int: $5y = 20$
$y = 4$



9. $9x - 15y = 45$

x-int: $9x = 45$
$x = 5$

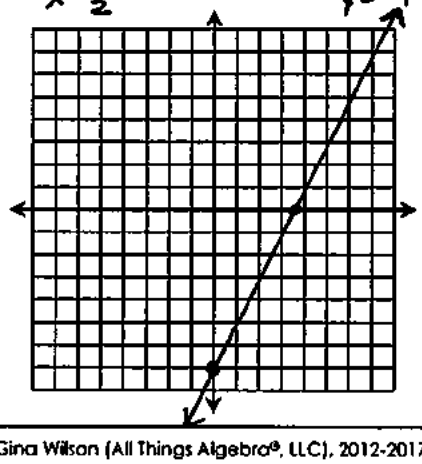
y-int: $-15y = 45$
$y = -3$



10. $2x - y = 7$

x-int: $2x = 7$
$x = \frac{7}{2}$

y-int: $-y = 7$
$y = -7$



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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 4:** x- and y-Intercepts

**** This is a 2-page document! ****

Directions: Find the x-intercept and y-intercept of each equation algebraically.

1. $y = -2x + 6$

x-int:
$0 = -2x + 6$
$-6 = -2x$
$3 = x$

y-int:
$y = -2(0) + 6$
$y = 6$

x-int: $(3, 0)$

y-int: $(0, 6)$

2. $y = x - 7$

x-int:
$0 = x - 7$
$7 = x$

y-int:
$y = 0 - 7$
$y = -7$

x-int: $(7, 0)$

y-int: $(0, -7)$

3. $y = \frac{3}{2}x - 9$

x-int:
$0 = \frac{3}{2}x - 9$
$9 = \frac{3}{2}x$
$6 = x$

y-int:
$y = \frac{3}{2}(0) - 9$
$y = -9$

x-int: $(6, 0)$

y-int: $(0, -9)$

4. $y = -\frac{1}{4}x + 1$

x-int:
$0 = -\frac{1}{4}x + 1$
$-1 = -\frac{1}{4}x$
$4 = x$

y-int:
$y = -\frac{1}{4}(0) + 1$
$y = 1$

x-int: $(4, 0)$

y-int: $(0, 1)$

5. $-x + y = -5$

x-int:
$-x + 0 = -5$
$-x = -5$
$x = 5$

y-int:
$0 + y = -5$
$y = -5$

x-int: $(5, 0)$

y-int: $(0, -5)$

6. $x + 2y = -14$

x-int:
$x + 2(0) = -14$
$x = -14$

y-int:
$0 + 2y = -14$
$2y = -14$
$y = -7$

x-int: $(-14, 0)$

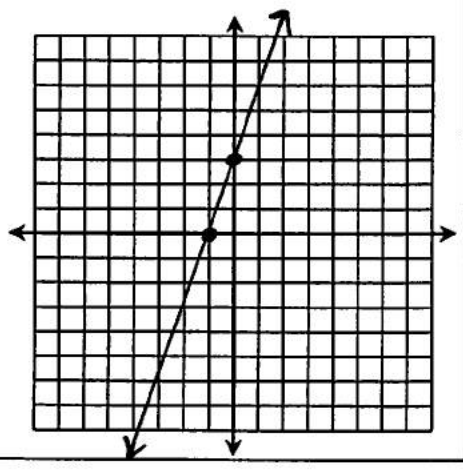
y-int: $(0, -7)$

Directions: Graph each equation by finding its x-intercept and y-intercept.

7. $3x - y = -3$

x-int:
$3x = -3$
$x = -1$

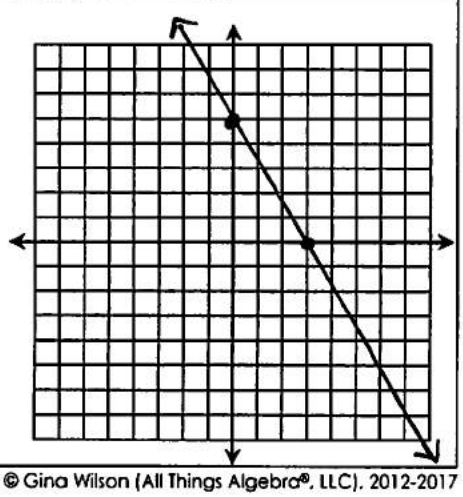
y-int:
$-y = -3$
$y = 3$



8. $5x + 3y = 15$

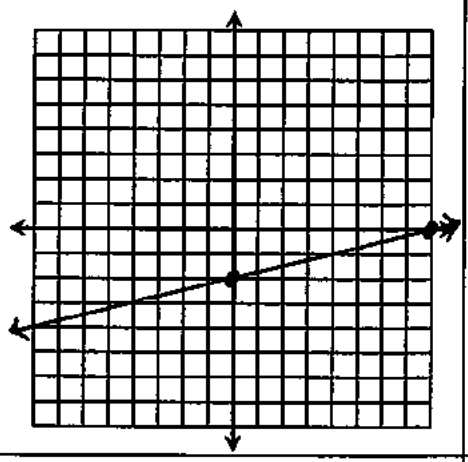
x-int:
$5x = 15$
$x = 3$

y-int:
$3y = 15$
$y = 5$



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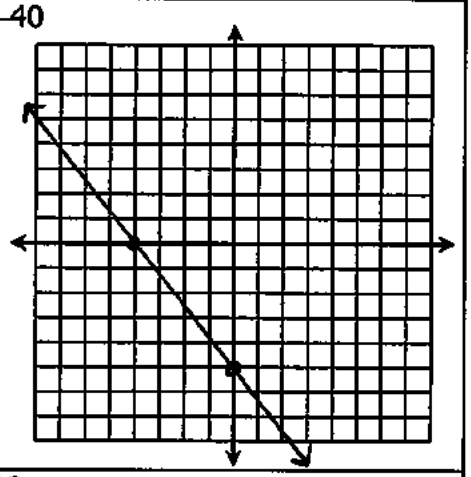
9. $x - 4y = 8$



x-int:
$x = 8$

y-int:
$-4y = 8$
$y = -2$

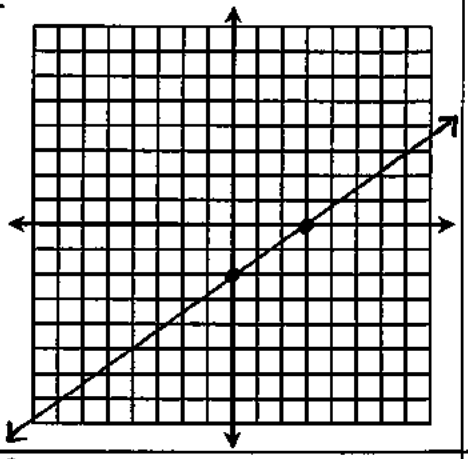
10. $10x + 8y = -40$



x-int:
$10x = -40$
$x = -4$

y-int:
$8y = -40$
$y = -5$

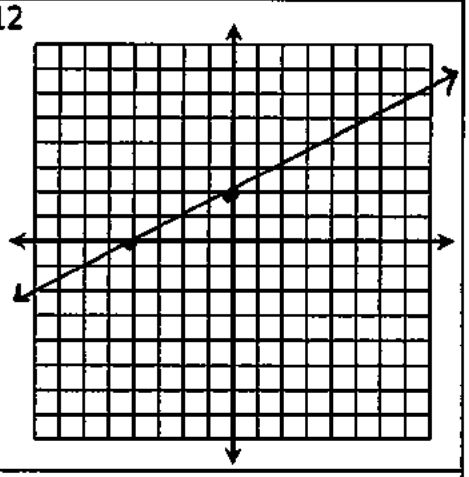
11. $4x - 6y = 12$



x-int:
$4x = 12$
$x = 3$

y-int:
$-6y = 12$
$y = -2$

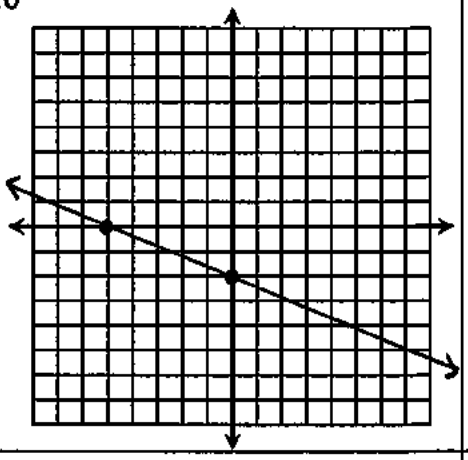
12. $-3x + 6y = 12$



x-int:
$-3x = 12$
$x = -4$

y-int:
$6y = 12$
$y = 2$

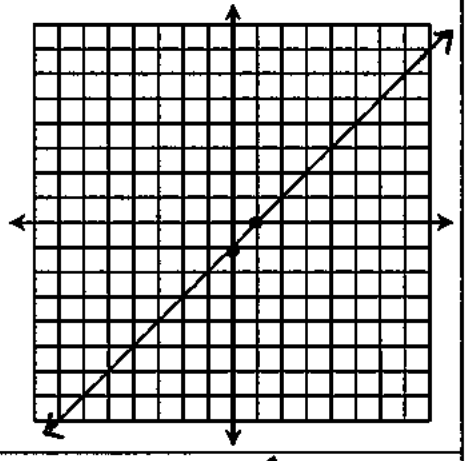
13. $-2x - 5y = 10$



x-int:
$-2x = 10$
$x = -5$

y-int:
$-5y = 10$
$y = -2$

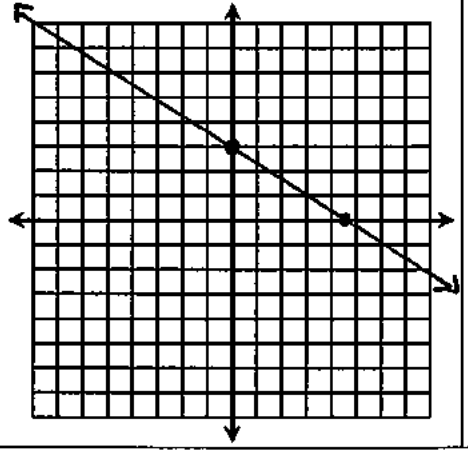
14. $x - y = 1$



x-int:
$x = 1$

y-int:
$-y = 1$
$y = -1$

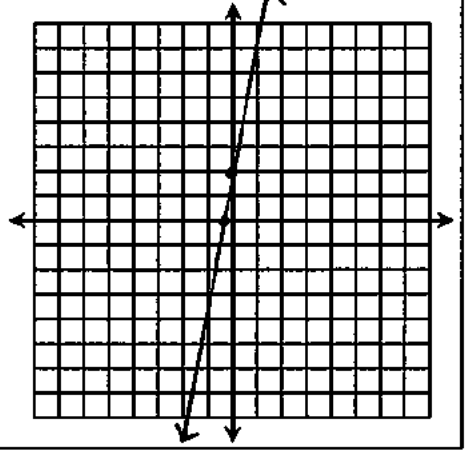
15. $2x + 3y = 9$



x-int:
$2x = 9$
$x = \frac{9}{2}$

y-int:
$3y = 9$
$y = 3$

16. $5x - y = -2$



x-int:
$5x = -2$
$x = \frac{-2}{5}$

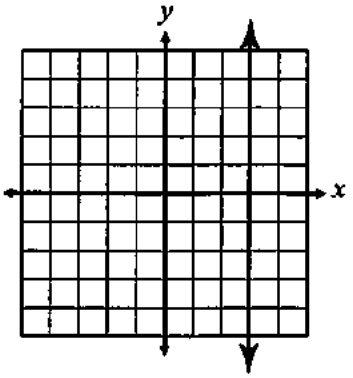
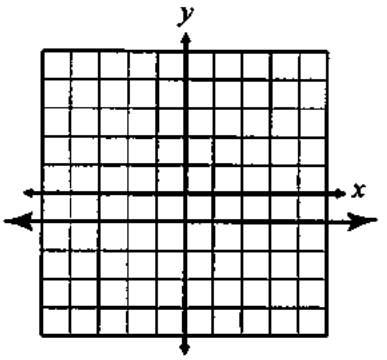
y-int:
$-y = -2$
$y = 2$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

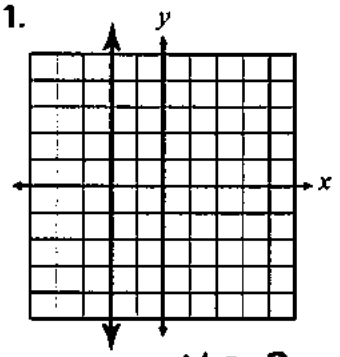
Vertical & Horizontal Lines

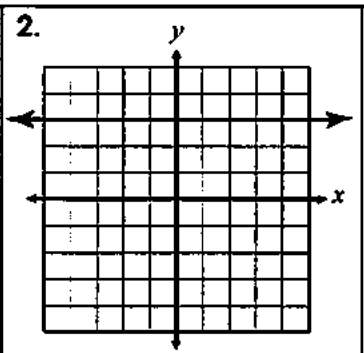
Vertical Lines	Horizontal Lines
A **vertical line** is written in the form $x = a$, where a represents the line's x-intercept.	A **horizontal line** is written in the form $y = a$, where a represents the line's y-intercept.
	
The equation of the vertical line graphed above is $x = 3$	The equation of the **horizontal** line graphed above is $y = -1$

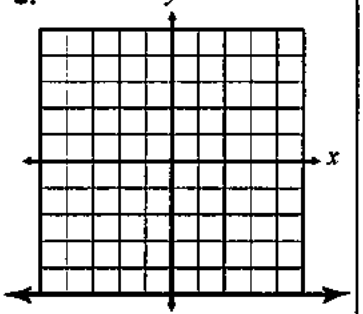
Remember, if the line intersects the **x-axis, it's $x = a$, if a line intersects the **y**-axis, it's $y = a$.**

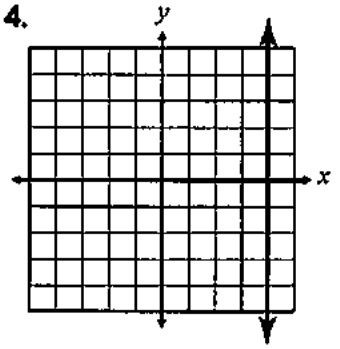
Examples

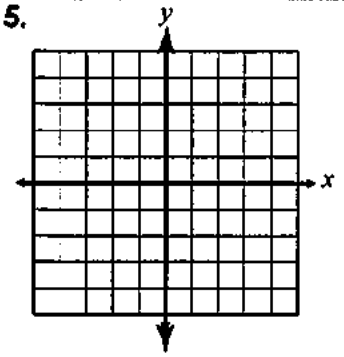
Directions: Write the equation of the line shown on the graph.

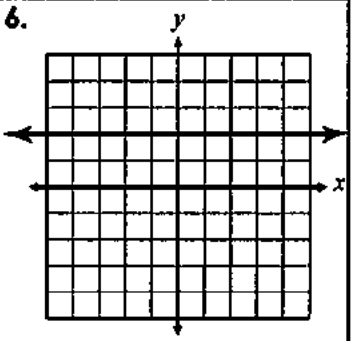
1. 
$x = -2$

2. 
$y = 3$

3. 
$y = -5$

4. 
$x = 4$

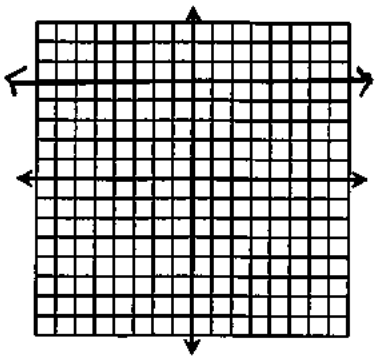
5. 
$x = 0$

6. 
$y = 2$

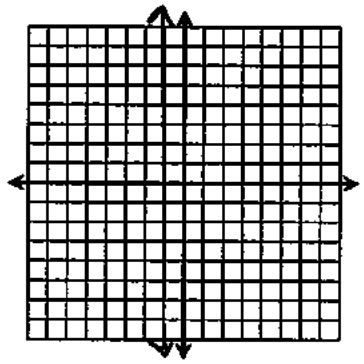
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Directions: Graph each equation.

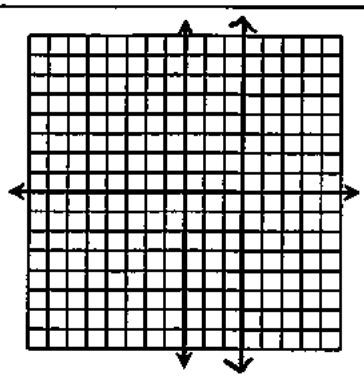
7. $y = 5$



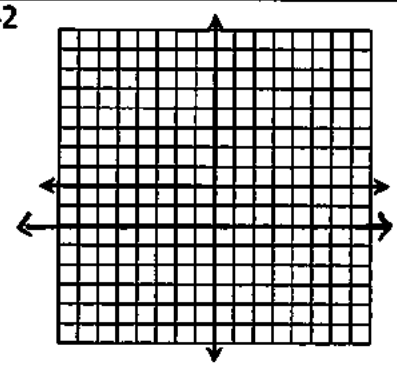
8. $x = -1$



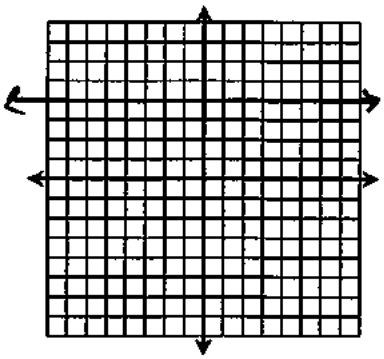
9. $x = 3$



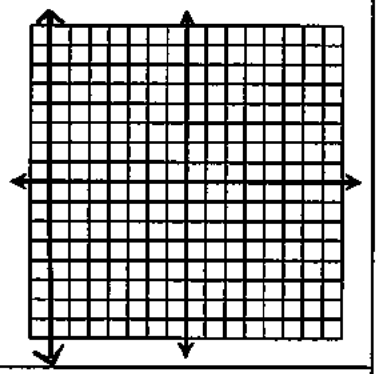
10. $y = -2$



11. $y = 4$



12. $x = -7$



QUESTIONS

13. What is the slope of the line $y = 4$? zero

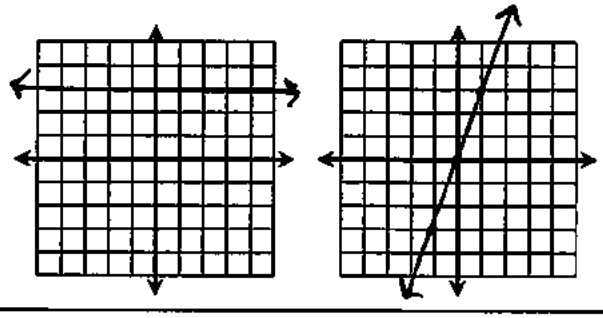
14. What is the slope of the line $x = -2$? undefined

15. What is the slope of the line $x = 0$? undefined

16. Which axis is $y = -1$ parallel to? x-axis

17. Which axis is $x = 4$ parallel to? y-axis

18. How does the graph of $y = 3$ differ from $y = 3x$? Graph both and explain.



y=3 has a slope of 0 + y-int of 3, where y=3x has a slope of 3 + y-int of 0.

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Name: _____ **Unit 4:** Linear Equations

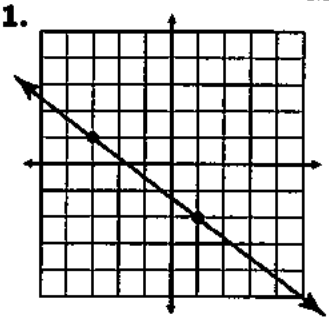
Date: _____ **Bell:** _____ **Homework 5:** Slope & Graphing Lines Review

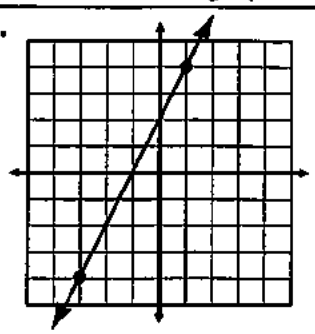
**** This is a 2-page document! ****

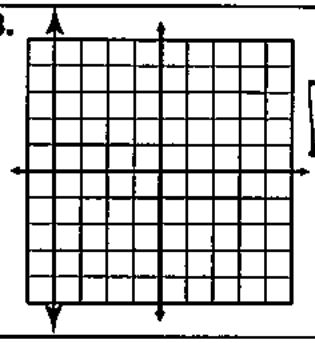
Directions: Identify each of the following below.

SLOPE FORMULA	SLOPE-INTERCEPT FORM	STANDARD FORM
$m = \frac{y_2 - y_1}{x_2 - x_1}$	$y = mx + b$	$Ax + By = C$

Directions: Find the slope of the line shown on the graph.

1.  $\boxed{-\frac{3}{4}}$

2.  $\boxed{2}$

3.  $\boxed{\text{undef.}}$

Directions: Find the slope of the line that passes through the given two points.

4. (1, -4) and (-4, -6)

$m = \frac{-6+4}{-4-1} = \frac{-2}{-5} = \boxed{\frac{2}{5}}$

5. (-6, 2) and (-5, -2)

$m = \frac{-2-2}{-5+6} = \frac{-4}{1} = \boxed{-4}$

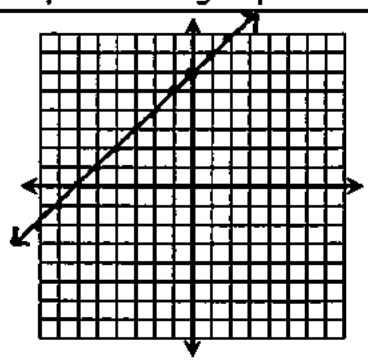
6. (-3, 3) and (-9, 5)

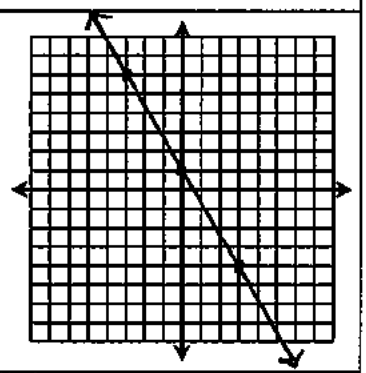
$m = \frac{5-3}{-9+3} = \frac{2}{-6} = \boxed{-\frac{1}{3}}$

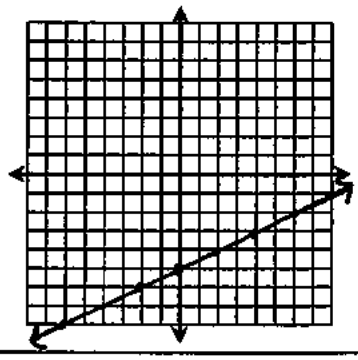
7. (4, -7) and (2, -7)

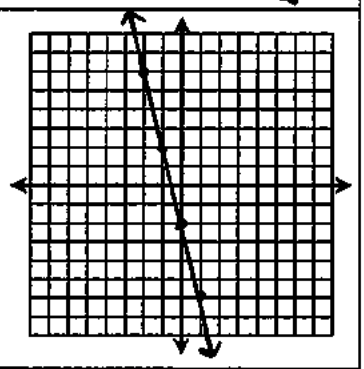
$m = \frac{-7+7}{2-4} = \frac{0}{-2} = \boxed{0}$

Directions: Graph each equation using slope-intercept form.

8. $y = x + 6$ 

9. $y = -\frac{5}{3}x + 1$ 

10. $y = \frac{1}{2}x - 5$ 

11. $y = -2 - 4x$ 

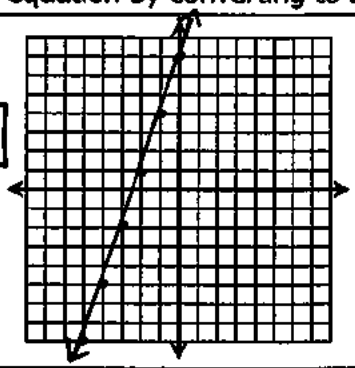
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Directions: Graph each equation by converting to slope-intercept form. **SHOW ALL WORK!**

12. $6x - 2y = -14$

$-2y = -6x - 14$

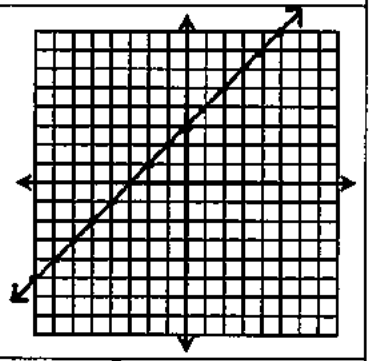
$y = 3x + 7$



13. $x - y = -3$

$-y = -x - 3$

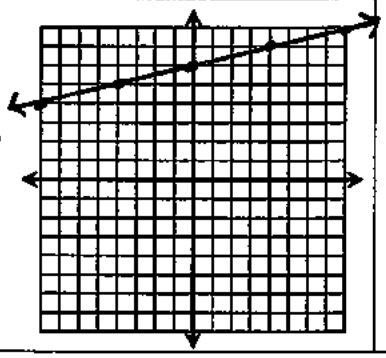
$y = x + 3$



14. $x - 4y = -24$

$-4y = -x - 24$

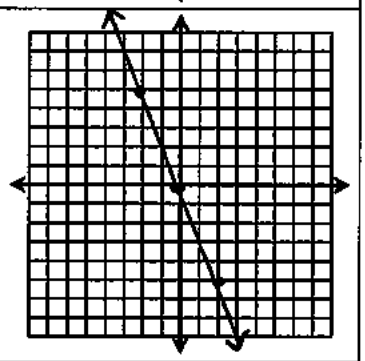
$y = \frac{1}{4}x + 6$



15. $10x + 4y = 0$

$4y = -10x$

$y = \frac{-5}{2}x$



Directions: Graph each equation by finding its x- and y-intercept. **SHOW ALL WORK!**

16. $x - y = 5$

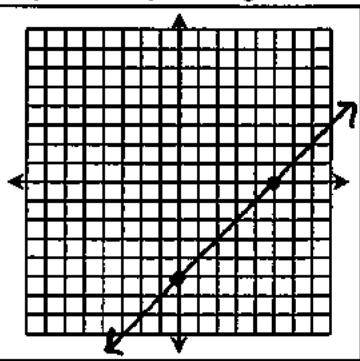
x-int:

$x = 5$

y-int:

$-y = 5$

$y = -5$



17. $x + 5y = -5$

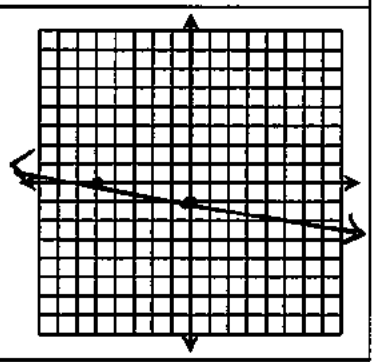
x-int:

$x = -5$

y-int:

$5y = -5$

$y = -1$



18. $9x + 12y = -36$

x-int:

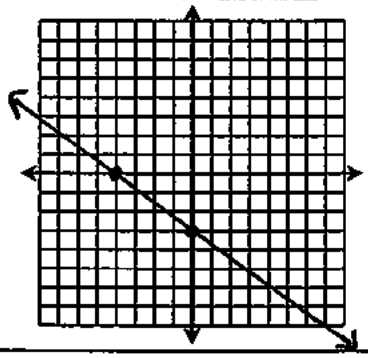
$9x = -36$

$x = -4$

y-int:

$12y = -36$

$y = -3$



19. $16x - 10y = 20$

x-int:

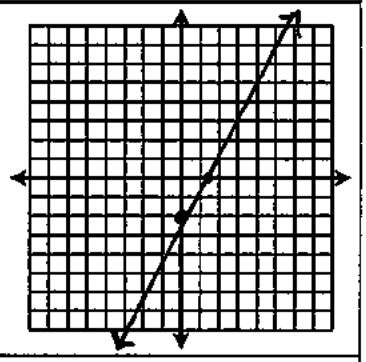
$16x = 20$

$x = \frac{5}{4}$

y-int:

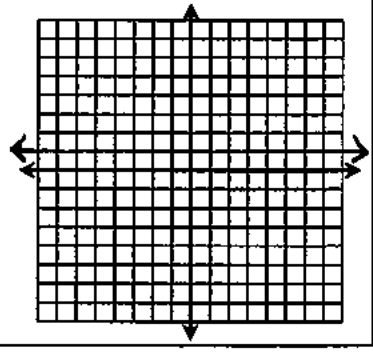
$-10y = 20$

$y = -2$

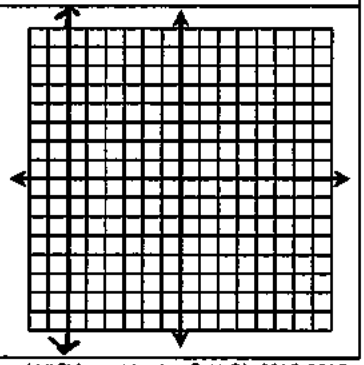


Directions: Graph each equation using slope-intercept form.

20. $y = 1$



21. $x = -6$



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Name: _____ **Algebra I**

Date: _____ **Per:** _____ **Unit 4:** Linear Equations

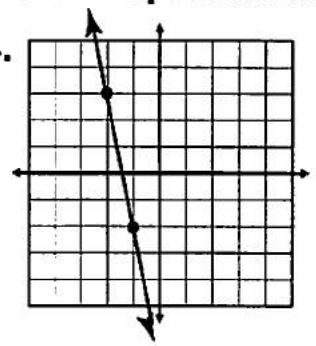
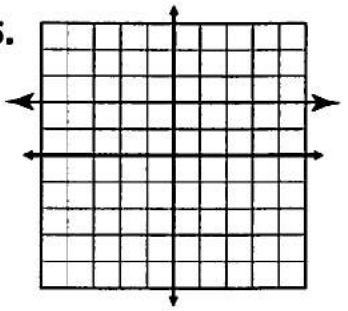
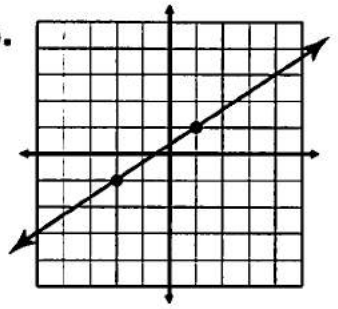
Quiz 4-1: Slope & Graphing Linear Equations

Identify the following formulas:

1. Slope Formula
2. Slope-Intercept Form
3. Standard Form

1. $m = \frac{y_2 - y_1}{x_2 - x_1}$
2. $y = mx + b$
3. $Ax + By = C$

Find the slope of the line shown on each graph below.

4. 
5. 
6. 

4. -5
5. 0
6. $\frac{2}{3}$

Find the slope of the line that passes through the two given points.

7. (-2, -1) and (-4, -7)

$m = \frac{-7+1}{-4+2} = \frac{-6}{-2}$

8. (2, -3) and (6, -13)

$m = \frac{-13+3}{6-2} = \frac{-10}{4}$

9. (-3, 4) and (-3, -8)

$m = \frac{-8-4}{-3+3} = \frac{-12}{0}$

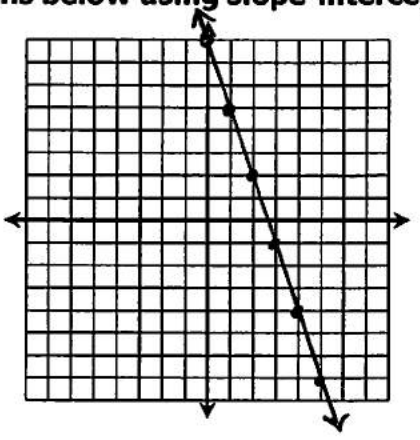
10. (-5, 7) and (3, -1)

$m = \frac{-1-7}{3+5} = \frac{-8}{8}$

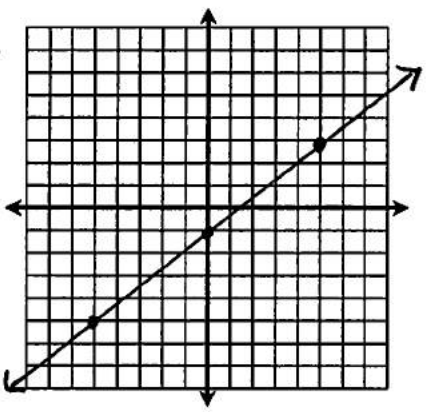
7. 3
8. $-\frac{5}{2}$
9. undefined
10. -1

Graph the equations below using slope-intercept form.

11. $y = -3x + 8$



12. $y = \frac{4}{5}x - 1$



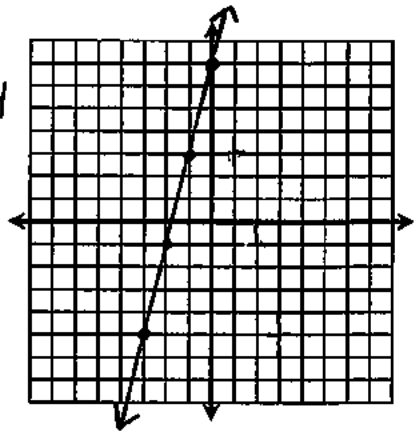
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Convert each equation below to slope-intercept form, then graph. SHOW ALL WORK!

13. $8x - 2y = -14$

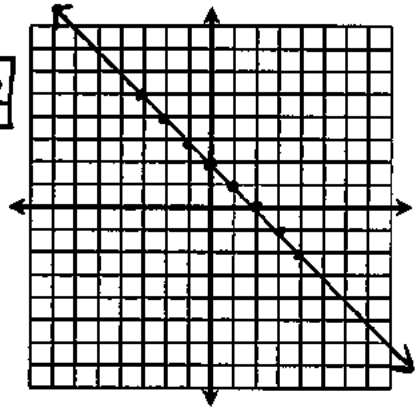
$-2y = -8x - 14$

$y = 4x + 7$



14. $x + y = 2$

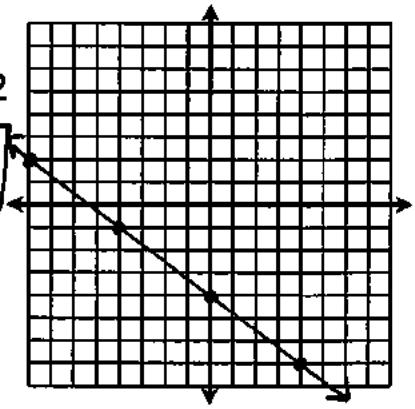
$y = -x + 2$



15. $6x + 8y = -32$

$8y = -6x - 32$

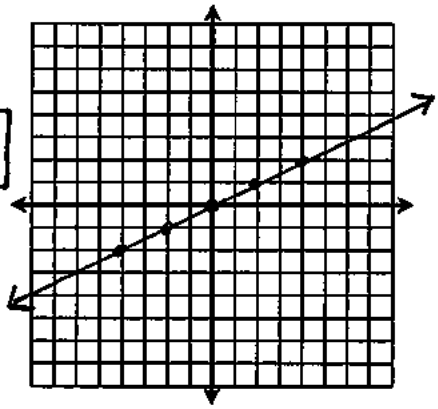
$y = \frac{-3}{4}x - 4$



16. $x - 2y = 0$

$-2y = -x$

$y = \frac{1}{2}x$



Graph each equation below by finding its x- and y-intercept. SHOW ALL WORK!

17. $8x - 4y = 16$

x-int:

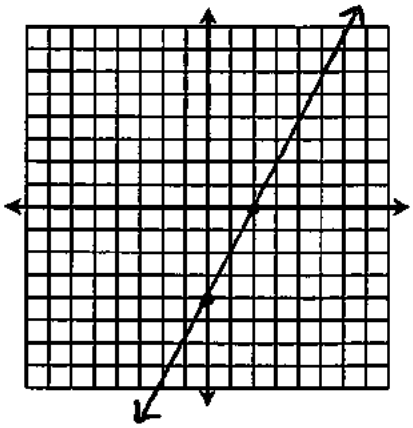
$8x = 16$

$x = 2$

y-int:

$-4y = 16$

$y = -4$



18. $3x - y = 3$

x-int:

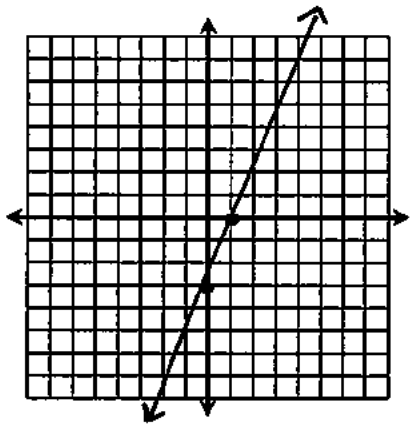
$3x = 3$

$x = 1$

y-int:

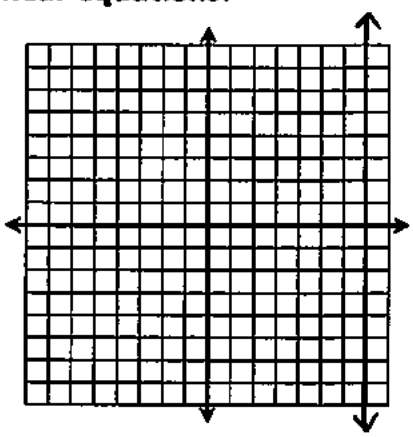
$-y = 3$

$y = -3$

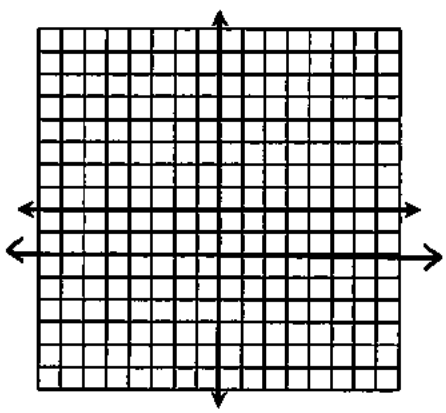


Graph the following linear equations.

19. $x = 7$



20. $y = -2$



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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING LINEAR EQUATIONS (Given a Point and Slope)	To write the equation of the line passing through point (x_1, y_1) with slope (m), you can use the point-slope formula: **Point-Slope Formula:** $y - y_1 = m(x - x_1)$ *Be sure to distribute and solve for y!*
EXAMPLES	**Directions:** Write the equation of the line that passes through the given point and has the given slope. Write all final answers in slope-intercept form.

1. (4, 1); slope = 2

$y - 1 = 2(x - 4)$

$y - 1 = 2x - 8$

$+1 \quad +1$

$y = 2x - 7$

2. (2, 4); slope = $\frac{1}{2}$

$y - 4 = \frac{1}{2}(x - 2)$

$y - 4 = \frac{1}{2}x - 1$

$+4 \quad +4$

$y = \frac{1}{2}x + 3$

3. (-6, 0); slope = $\frac{2}{3}$

$y - 0 = \frac{2}{3}(x + 6)$

$y = \frac{2}{3}x + 4$

4. (-8, -1); slope = $-\frac{3}{4}$

$y + 1 = -\frac{3}{4}(x + 8)$

$y + 1 = -\frac{3}{4}x - 6$

$-1 \quad -1$

$y = -\frac{3}{4}x - 7$

5. (4, -3); slope = -1

$y + 3 = -1(x - 4)$

$y + 3 = -x + 4$

$-3 \quad -3$

$y = -x + 1$

6. (0, -9); slope = 4

$y + 9 = 4(x - 0)$

$y + 9 = 4x$

$-9 \quad -9$

$y = 4x - 9$

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7. (-12, -5); slope = $\frac{1}{3}$

$y+5=\frac{1}{3}(x+12)$

$y+5=\frac{1}{3}x+4$

$-5 \quad -5$

$y=\frac{1}{3}x-1$

8. (3, -8); slope = -3

$y+8=-3(x-3)$

$y+8=-3x+9$

$-8 \quad -8$

$y=-3x+1$

9. (5, -9); slope = $-\frac{2}{5}$

$y+9=-\frac{2}{5}(x-5)$

$y+9=-\frac{2}{5}x+2$

$-9 \quad -9$

$y=-\frac{2}{5}x-7$

10. (-4, -7); slope = 1

$y+7=1(x+4)$

$y+7=x+4$

$-7 \quad -7$

$y=x-3$

11. (-6, -1); slope = $-\frac{4}{3}$

$y+1=-\frac{4}{3}(x+6)$

$y+1=-\frac{4}{3}x-8$

$-1 \quad -1$

$y=-\frac{4}{3}x-9$

12. (4, 9); slope = 5

$y-9=5(x-4)$

$y-9=5x-20$

$+9 \quad +9$

$y=5x-11$

13. (3, -4); slope = 0

$y+4=0(x-3)$

$y+4=0$

$-4 \quad -4$

$y=-4$

14. (-6, -8); slope = $\frac{5}{6}$

$y+8=\frac{5}{6}(x+6)$

$y+8=\frac{5}{6}x+5$

$-8 \quad -8$

$y=\frac{5}{6}x-3$

15. (-4, 1); slope = $-\frac{1}{4}$

$y-1=-\frac{1}{4}(x+4)$

$y-1=-\frac{1}{4}x-1$

$+1 \quad +1$

$y=-\frac{1}{4}x$

16. (7, -8); slope = -2

$y+8=-2(x-7)$

$y+8=-2x+14$

$-8 \quad -8$

$y=-2x+6$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 6:** Writing Linear Equations (Given Point & Slope)

Directions: Given the point and slope, write the equation of the line.

1. (4, 2); slope = 3

$y-2=3(x-4)$

$y-2=3x-12$

$+2 \quad +2$

$y=3x-10$

2. (0, 3); slope = -2

$y-3=-2(x-0)$

$y-3=-2x$

$+3 \quad +3$

$y=-2x+3$

3. (1, -7); slope = -1

$y+7=-1(x-1)$

$y+7=-x+1$

$-7 \quad -7$

$y=-x-6$

4. (-5, -3); slope = $-\frac{3}{5}$

$y+3=\frac{-3}{5}(x+5)$

$y+3=\frac{-3}{5}x-3$

$-3 \quad -3$

$y=\frac{-3}{5}x-6$

5. (-8, 6); slope = $\frac{1}{4}$

$y-6=\frac{1}{4}(x+8)$

$y-6=\frac{1}{4}x+2$

$+6 \quad +6$

$y=\frac{1}{4}x+8$

6. (9, -4); slope = $-\frac{2}{3}$

$y+4=\frac{-2}{3}(x-9)$

$y+4=\frac{-2}{3}x+6$

$-4 \quad -4$

$y=\frac{-2}{3}x+2$

7. (6, -6); slope = $\frac{5}{6}$

$y+6=\frac{5}{6}(x-6)$

$y+6=\frac{5}{6}x-5$

$-6 \quad -6$

$y=\frac{5}{6}x-11$

8. (-8, 9); slope = $-\frac{3}{2}$

$y-9=\frac{-3}{2}(x+8)$

$y-9=\frac{-3}{2}x-12$

$+9 \quad +9$

$y=\frac{-3}{2}x-3$

9. (-2, -11); slope = 4

$y+11=4(x+2)$

$y+11=4x+8$

$-11 \quad -11$

$y=4x-3$

10. (-4, 0); slope = $\frac{1}{2}$

$y-0=\frac{1}{2}(x+4)$

$y=\frac{1}{2}x+2$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING LINEAR EQUATIONS (Given Two Points)	To write a linear equation that passes through two points, (x_1, y_1) and (x_2, y_2), use the slope formula followed by the point-slope formula: **Slope Formula** $m = \frac{y_2 - y_1}{x_2 - x_1}$ ⇒ **Point-Slope Formula** $y - y_1 = m(x_2 - x_1)$
EXAMPLES	**Directions:** Write the equation of the line that passes through the given two points. Write all final answers in slope-intercept form.

1. (-3, 7) and (1, -1)

$m = \frac{-1-7}{1+3} = \frac{-8}{4} = -2$

$y - 7 = -2(x+3)$

$y - 7 = -2x - 6$

$+7 \quad +7$

$\boxed{y = -2x + 1}$

2. (-6, -7) and (3, -4)

$m = \frac{-4+7}{3+6} = \frac{3}{9} = \frac{1}{3}$

$y + 7 = \frac{1}{3}(x+6)$

$y + 7 = \frac{1}{3}x + 2$

$-7 \quad -7$

$\boxed{y = \frac{1}{3}x - 5}$

3. (2, -1) and (4, -6)

$m = \frac{-6+1}{4-2} = \frac{-5}{2}$

$y + 1 = -\frac{5}{2}(x-2)$

$y + 1 = -\frac{5}{2}x + 5$

$-1 \quad -1$

$\boxed{y = -\frac{5}{2}x + 4}$

4. (1, 6) and (2, 5)

$m = \frac{5-6}{2-1} = \frac{-1}{1} = -1$

$y - 6 = -1(x-1)$

$y - 6 = -x + 1$

$+6 \quad +6$

$\boxed{y = -x + 7}$

5. (-8, 1) and (0, -5)

$m = \frac{-5-1}{0+8} = \frac{-6}{8} = -\frac{3}{4}$

$y - 1 = -\frac{3}{4}(x+8)$

$y - 1 = -\frac{3}{4}x - 6$

$+1 \quad +1$

$\boxed{y = -\frac{3}{4}x - 5}$

6. (-3, -8) and (2, 7)

$m = \frac{7+8}{2+3} = \frac{15}{5} = 3$

$y + 8 = 3(x+3)$

$y + 8 = 3x + 9$

$-8 \quad -8$

$\boxed{y = 3x + 1}$

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7. (-9, -14) and (3, 2)

$m = \frac{2+14}{3+9} = \frac{16}{12} = \frac{4}{3}$

$y + 14 = \frac{4}{3}(x + 9)$

$y + 14 = \frac{4}{3}x + 12$

$-14 \qquad -14$

$\boxed{y = \frac{4}{3}x - 2}$

8. (-6, -3) and (-4, -1)

$m = \frac{-1+3}{-4+6} = \frac{2}{2} = 1$

$y + 3 = 1(x + 6)$

$y + 3 = x + 6$

$-3 \qquad -3$

$\boxed{y = x + 3}$

9. (-4, 7) and (6, 2)

$m = \frac{2-7}{6+4} = \frac{-5}{10} = -\frac{1}{2}$

$y - 7 = -\frac{1}{2}(x + 4)$

$y - 7 = -\frac{1}{2}x - 2$

$+7 \qquad +7$

$\boxed{y = -\frac{1}{2}x + 5}$

10. (1, 1) and (-1, -9)

$m = \frac{-9-1}{-1-1} = \frac{-10}{-2} = 5$

$y - 1 = 5(x - 1)$

$y - 1 = 5x - 5$

$+1 \qquad +1$

$\boxed{y = 5x - 4}$

11. (4, -1) and (0, 2)

$m = \frac{2+1}{0-4} = -\frac{3}{4}$

$y + 1 = -\frac{3}{4}(x - 4)$

$y + 1 = -\frac{3}{4}x + 3$

$-1 \qquad -1$

$\boxed{y = -\frac{3}{4}x + 2}$

12. (-2, -2) and (1, -5)

$m = \frac{-5+2}{1+2} = \frac{-3}{3} = -1$

$y + 2 = -1(x + 2)$

$y + 2 = -x - 2$

$-2 \qquad -2$

$\boxed{y = -x - 4}$

13. (3, -5) and (2, -1)

$m = \frac{-1+5}{2-3} = \frac{4}{-1} = -4$

$y + 5 = -4(x - 3)$

$y + 5 = -4x + 12$

$-5 \qquad -5$

$\boxed{y = -4x + 7}$

14. (-3, 14) and (2, -1)

$m = \frac{-1-14}{2+3} = \frac{-15}{5} = -3$

$y - 14 = -3(x + 3)$

$y - 14 = -3x - 9$

$+14 \qquad +14$

$\boxed{y = -3x + 5}$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 7:** Writing Linear Equations (Given Two Points)

Directions: Write a linear equation that passes through the given two points.

1. (-4, -2) and (4, 0)

$m = \frac{0+2}{4+4} = \frac{2}{8} = \frac{1}{4}$

$y+2 = \frac{1}{4}(x+4)$

$y+2 = \frac{1}{4}x+1$

$-2 \quad -2$

$y = \frac{1}{4}x - 1$

2. (3, 4) and (0, 5)

$m = \frac{5-4}{0-3} = -\frac{1}{3}$

$y-4 = -\frac{1}{3}(x-3)$

$y-4 = -\frac{1}{3}x+1$

$+4 \quad +4$

$y = -\frac{1}{3}x+5$

3. (0, 1) and (5, 3)

$m = \frac{3-1}{5-0} = \frac{2}{5}$

$y-1 = \frac{2}{5}(x-0)$

$y-1 = \frac{2}{5}x$

$+1 \quad +1$

$y = \frac{2}{5}x+1$

4. (2, -2) and (0, -1)

$m = \frac{-1+2}{0-2} = -\frac{1}{2}$

$y+2 = -\frac{1}{2}(x-2)$

$y+2 = -\frac{1}{2}x+1$

$-2 \quad -2$

$y = -\frac{1}{2}x-1$

5. (0, 5) and (-5, 1)

$m = \frac{1-5}{-5-0} = \frac{-4}{-5} = \frac{4}{5}$

$y-5 = \frac{4}{5}(x-0)$

$y-5 = \frac{4}{5}x$

$+5 \quad +5$

$y = \frac{4}{5}x+5$

6. (1, 3) and (-3, -5)

$m = \frac{-5-3}{-3-1} = \frac{-8}{-4} = 2$

$y-3 = 2(x-1)$

$y-3 = 2x-2$

$+3 \quad +3$

$y = 2x+1$

7. (1, 4) and (6, -1)

$m = \frac{-1-4}{6-1} = \frac{-5}{5} = -1$

$y-4 = -1(x-1)$

$y-4 = -x+1$

$+4 \quad +4$

$y = -x+5$

8. (3, 3) and (1, -3)

$m = \frac{-3-3}{1-3} = \frac{-6}{-2} = 3$

$y-3 = 3(x-3)$

$y-3 = 3x-9$

$+3 \quad +3$

$y = 3x-6$

9. (-12, 14) and (6, -1)

$m = \frac{-1-14}{6+12} = \frac{-15}{18} = -\frac{5}{6}$

$y-14 = -\frac{5}{6}(x+12)$

$y-14 = -\frac{5}{6}x-10$

$+14 \quad +14$

$y = -\frac{5}{6}x+4$

10. (-1, 4) and (0, -1)

$m = \frac{-1-4}{0+1} = \frac{-5}{1} = -5$

$y-4 = -5(x+1)$

$y-4 = -5x-5$

$+4 \quad +4$

$y = -5x-1$

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Partner A: _____ Partner B: _____

WHY WAS THE CAT KICKED OUT OF SCHOOL?

Either given a point and a slope, or two points, write each equation in slope-intercept form. Partner A should do the left side and Partner B should do the right side. One will have a letter and the other a number. Write the number in the matching numbered box at the bottom of the page.

SET 1

E. (-3, 0); slope = $\frac{2}{3}$	$y = \frac{2}{3}x + 2$	**8.** (2, -2); slope = 1	$y = x - 4$
H. (6, 2) and (-3, -7)	$y = x - 4$	**3.** (4, 1); slope = $\frac{3}{2}$	$y = \frac{3}{2}x - 5$
S. (-4, -6) and (3, 8)	$y = 2x + 2$	**10.** (3, 4) and (-6, -2)	$y = \frac{2}{3}x + 2$
W. (-2, -8) and (6, 4)	$y = \frac{3}{2}x - 5$	**5.** (-3, -4); slope = 2	$y = 2x + 2$

SET 2

A. (-9, 17); slope = $-\frac{4}{3}$	$y = -\frac{4}{3}x + 5$	**12.** (-4, -5); slope = $\frac{1}{2}$	$y = \frac{1}{2}x - 3$
C. (-3, 9) and (0, 1)	$y = -\frac{8}{3}x + 1$	**6.** (3, 1) and (9, -7)	$y = -\frac{4}{3}x + 5$
E. (1, -8); slope = -1	$y = -x - 7$	**7.** (3, -7); slope = $-\frac{8}{3}$	$y = -\frac{8}{3}x + 1$
A. (2, -2) and (8, 1)	$y = \frac{1}{2}x - 3$	**2.** (5, -12) and (-3, -4)	$y = -x - 7$

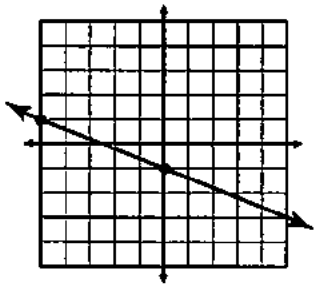
SET 3

H. (-12, 4) and (0, 2)	$y = -\frac{1}{6}x + 2$	**9.** (-6, -4) and (12, 11)	$y = \frac{5}{6}x + 1$
E. (6, 6); slope = $\frac{5}{6}$	$y = \frac{5}{6}x + 1$	**4.** (2, -8); slope = $-\frac{7}{2}$	$y = -\frac{7}{2}x - 1$
H. (-8, -4) and (4, -7)	$y = -\frac{1}{4}x - 6$	**11.** (5, 1) and (10, 5)	$y = \frac{4}{5}x - 3$
T. (-5, -7); slope = $\frac{4}{5}$	$y = \frac{4}{5}x - 3$	**13.** (6, 1); slope = $-\frac{1}{6}$	$y = -\frac{1}{6}x + 2$
A. (-4, 13) and (-2, 6)	$y = -\frac{7}{2}x - 1$	**1.** (-4, -5); slope = $-\frac{1}{4}$	$y = -\frac{1}{4}x - 6$

1	2		3	4	5		6		7	8	9	10	11	12	13	
H	E		W	A	S		A		C	H	E	E	T	A	H	!

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I can write Linear Equations in SLOPE-INTERCEPT FORM GIVEN:

The Slope and y-Intercept slope = $\frac{1}{3}$; y-intercept = -5	$y = \frac{1}{3}x - 5$
A Graph 	$m = -2/5$; $b = -1$ $y = \frac{-2}{5}x - 1$
An Equation in Standard Form $4x - 2y = 14$	$\frac{-2y}{-2} = \frac{-4x}{-2} + \frac{14}{-2}$ $y = 2x - 7$
A Point and Slope (-1, 3); slope = -3	$y - 3 = -3(x + 1)$ $y - 3 = -3x - 3$ $+3 \quad +3$ $y = -3x$
Two Points (-4, -7) and (8, -13)	$m = \frac{-13+7}{8+4} = \frac{-6}{12} = -\frac{1}{2}$ $y + 7 = -\frac{1}{2}(x + 4)$ $y + 7 = -\frac{1}{2}x - 2$ $-7 \quad -7$ $y = -\frac{1}{2}x - 9$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 8:** Writing Linear Equations REVIEW

Directions: Write the linear equation in slope-intercept form given the following:

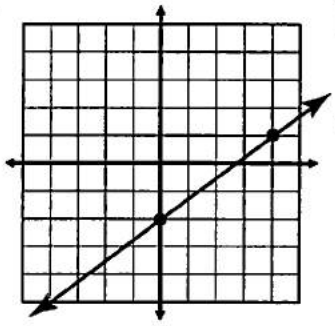
1. slope = $\frac{2}{7}$; y-intercept = 0

$y = \frac{2}{7}x$

2. slope = 1; y-intercept = -4

$y = x - 4$

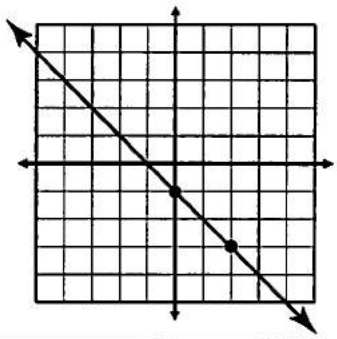
3.



$m = 3/4$; $b = -2$

$y = \frac{3}{4}x - 2$

4.



$m = -1$; $b = -1$

$y = -x - 1$

5. $2x + 5y = 35$

$-2x \quad -2x$

$\frac{5y}{5} = \frac{-2x}{5} + \frac{35}{5}$

$y = -\frac{2}{5}x + 7$

6. $2x - y = 4$

$-2x \quad -2x$

$\frac{-y}{-1} = \frac{-2x}{-1} + \frac{4}{-1}$

$y = 2x - 4$

7. $x - 6y = -30$

$-x \quad -x$

$\frac{-6y}{-6} = \frac{-x}{-6} - \frac{30}{-6}$

$y = \frac{1}{6}x + 5$

8. $5x - 4y = -12$

$-5x \quad -5x$

$\frac{-4y}{-4} = \frac{-5x}{-4} - \frac{12}{-4}$

$y = \frac{5}{4}x + 3$

9. (-2, -3); slope = $\frac{5}{2}$

$y + 3 = \frac{5}{2}(x + 2)$

$y + 3 = \frac{5}{2}x + 5$

$-3 \quad -3$

$y = \frac{5}{2}x + 2$

10. (2, -4); slope = -3

$y + 4 = -3(x - 2)$

$y + 4 = -3x + 6$

$-4 \quad -4$

$y = -3x + 2$

11. (0, 2) and (4, -4)

$m = \frac{-4-2}{4-0} = \frac{-6}{4} = -\frac{3}{2}$

$y - 2 = -\frac{3}{2}(x - 0)$

$y - 2 = -\frac{3}{2}x$

$+2 \quad +2$

$y = -\frac{3}{2}x + 2$

12. (-2, -4) and (-3, -3)

$m = \frac{-3+4}{-3+2} = \frac{1}{-1} = -1$

$y + 4 = -1(x + 2)$

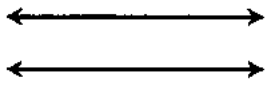
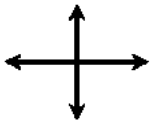
$y + 4 = -x - 2$

$-4 \quad -4$

$y = -x - 6$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Parallel Lines 	**Definition:** lines that never intersect **Algebraically, how do we know if two lines are parallel?** They have the same slope! $y = 2x+1$, $y = 2x-7$
Perpendicular Lines 	**Definition:** lines that intersect at 90° angles. **Algebraically, how do we know if two lines are perpendicular?** They have negative (opposite) reciprocal slopes! $y = 2x+1$, $y = -\frac{1}{2}x+5$
What are Negative Reciprocals?	A fraction "flipped" and the signs are "switched!" **Some examples:** 1) $\frac{3}{4}$ & $-\frac{4}{3}$ 2) 2 & $-\frac{1}{2}$ 3) $-\frac{7}{8}$ & $\frac{8}{7}$ 4) 1 & -1 5) 0 & undef.
Given Ordered Pairs	**Determine if segments AB and CD are parallel, perpendicular, or neither:**

1. $\overline{AB}$ formed by (-2, 3) and (2, 6)
$\overline{CD}$ formed by (-1, 0) and (3, 3)

$\overline{AB}$: $m = \frac{6-3}{2+2} = \frac{3}{4}$

$\overline{CD}$: $m = \frac{3-0}{3+1} = \frac{3}{4}$

Parallel

2. $\overline{AB}$ formed by (0, 2) and (5, 4)
$\overline{CD}$ formed by (1, 8) and (3, 3)

$\overline{AB}$: $m = \frac{4-2}{5-0} = \frac{2}{5}$

$\overline{CD}$: $m = \frac{3-8}{3-1} = -\frac{5}{2}$

Perpendicular

3. $\overline{AB}$ formed by (-1, 8) and (2, 6)
$\overline{CD}$ formed by (-1, 2) and (3, 3)

$\overline{AB}$: $m = \frac{6-8}{2+1} = -\frac{2}{3}$

$\overline{CD}$: $m = \frac{3-2}{3+1} = \frac{1}{4}$

Neither

4. $\overline{AB}$ formed by (2, 3) and (-1, 4)
$\overline{CD}$ formed by (-5, 3) and (-4, 6)

$\overline{AB}$: $m = \frac{4-3}{-1-2} = \frac{1}{-3}$

$\overline{CD}$ $m = \frac{6-3}{-4+5} = 3$

Perpendicular

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5. $\overline{AB}$ formed by (0, -2) and (0, 7)
$\overline{CD}$ formed by (3, -5) and (6, -5)

$\overline{AB}$: $m = \frac{7+2}{0-0} = \frac{9}{0}$ = undef.

$\overline{CD}$: $m = \frac{-5+5}{6-3} = \frac{0}{3} = 0$

Perp.

6. $\overline{AB}$ formed by (-4, 7) and (-2, 6)
$\overline{CD}$ formed by (2, -2) and (-8, 3)

$\overline{AB}$: $m = \frac{6-7}{-2+4} = \frac{-1}{2}$

$\overline{CD}$: $m = \frac{3+2}{-8-2} = \frac{5}{-10} = -\frac{1}{2}$

Parallel

7. $\overline{AB}$ formed by (3, 1) and (3, -4)
$\overline{CD}$ formed by (-4, 1) and (-4, 5)

$\overline{AB}$: $m = \frac{-4-1}{3-3} = \frac{-5}{0}$ = undef.

$\overline{CD}$: $m = \frac{5-1}{-4+4} = \frac{4}{0}$ = undef.

Parallel

8. $\overline{AB}$ formed by (-3, 8) and (3, 2)
$\overline{CD}$ formed by (7, 1) and (5, -1)

$\overline{AB}$: $m = \frac{2-8}{3+3} = \frac{-6}{6} = -1$

$\overline{CD}$: $m = \frac{-1-1}{5-7} = \frac{-2}{-2} = 1$

Perp.

Given Equations

9. $y = 7x + 2$ and $y = 7x - 1$

$m=7$ $m=7$

Parallel

10. $y = \frac{4}{5}x - 8$ and $y = -\frac{5}{4}x + 3$

$m = 4/5$ $m = -5/4$

Perp.

11. $y = -\frac{1}{3}x + 2$ and $y = \frac{1}{3}x$

$m = -1/3$ $m = 1/3$

Neither

12. $x + 6y = 30$ and $3y = 18x - 6$

$6y = -x + 30$ $y = 6x - 2$

$y = -\frac{1}{6}x + 5$ $m = 6$

$m = -1/6$

Perp.

13. $5x - y = 4$ and $y = -\frac{1}{5}x + 7$

$-y = -5x + 4$ $m = -1/5$

$y = 5x - 4$

$m = 5$

Perp.

14. $3x - y = 2$ and $12x - 4y = 4$

$-y = -3x + 2$ $-4y = -12x + 4$

$y = 3x - 2$ $y = 3x - 1$

$m = 3$ $m = 3$

Parallel

15. $y = x + 3$ and $y = -x - 5$

$m = 1$ $m = -1$

Perp.

16. $y = 6$ and $x = -1$

$m = 0$ $m =$ undef.

Perp.

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PARALLEL, PERPENDICULAR, OR NEITHER?

Directions: Determine whether the lines given in each box are parallel, perpendicular, or neither. Color the boxes as follows:

Parallel Lines: Yellow **Perpendicular Lines:** Light Blue **Neither:** Uncolored

$y = 3x - 7$ $y = 3x + 1$ ∥	$y = -\frac{2}{5}x + 3$ $y = \frac{2}{5}x + 8$ N	$y = -\frac{1}{4}x$ $y = 4x - 5$ ⊥
$2x + 7y = 28$ $7x - 2y = 4$ ⊥	$y = -5x + 1$ $x - 5y = 30$ ⊥	$3x + 2y = 8$ $2x + 3y = -12$ N
$y = -4x - 1$ $8x + 2y = 14$ ∥	$x + y = 7$ $x - y = 9$ ⊥	$y = \frac{1}{3}x + 9$ $x - 3y = 3$ ∥
$4x + 9y = 18$ $y = 4x + 9$ N	$5x - 10y = 20$ $y = -2x + 6$ ⊥	$-9x + 12y = 24$ $y = \frac{3}{4}x - 5$ ∥
$y = x - 3$ $x - y = 8$ ∥	$10x + 8y = 16$ $5y = 4x - 15$ ⊥	$y = \frac{5}{3}x + 7$ $6x - 10y = 10$ N
$x - 2y = 18$ $2x + y = 6$ ⊥	$x = 4$ $x = -6$ ∥	$x = 1$ $y = -8$ ⊥

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 9:** Parallel & Perpendicular Lines (Day 1)

Directions: Determine if segments AB and CD are parallel, perpendicular, or neither.

1. $\overline{AB}$ formed by (-2, 13) and (0, 3)
$\overline{CD}$ formed by (-5, 0) and (10, 3)

$\overline{AB}$: $m = \frac{3-13}{0+2} = -5$

$\overline{CD}$: $m = \frac{3-0}{10+5} = \frac{1}{5}$

Perp.

2. $\overline{AB}$ formed by (3, 7) and (-6, 1)
$\overline{CD}$ formed by (-6, -5) and (0, -1)

$\overline{AB}$: $m = \frac{1-7}{-6-3} = \frac{2}{3}$

$\overline{CD}$: $m = \frac{-1+5}{0+6} = \frac{2}{3}$

Parallel

3. $\overline{AB}$ formed by (-6, 2) and (-2, 4)
$\overline{CD}$ formed by (-1, 11) and (5, -7)

$\overline{AB}$: $m = \frac{4-2}{-2+6} = \frac{1}{2}$

$\overline{CD}$: $m = \frac{-7-11}{5+1} = -3$

Neither

4. $\overline{AB}$ formed by (-3, 8) and (2, 3)
$\overline{CD}$ formed by (-4, 6) and (-8, 2)

$\overline{AB}$: $m = \frac{3-8}{2+3} = -1$

$\overline{CD}$: $m = \frac{2-6}{-8+4} = 1$

Perp.

5. $\overline{AB}$ formed by (-8, -1) and (-4, 2)
$\overline{CD}$ formed by (0, -3) and (12, 6)

$\overline{AB}$: $m = \frac{2+1}{-4+8} = \frac{3}{4}$

$\overline{CD}$: $m = \frac{6+3}{12-0} = \frac{3}{4}$

Parallel

6. $\overline{AB}$ formed by (6, 5) and (3, -1)
$\overline{CD}$ formed by (2, -5) and (-4, 7)

$\overline{AB}$: $m = \frac{-1-5}{3-6} = 2$

$\overline{CD}$: $m = \frac{7+5}{-4-2} = -2$

Neither

Directions: Determine if the given equations are parallel, perpendicular, or neither.

7. $3x + 2y = 6$ and $y = -\frac{3}{2}x + 5$

$2y = -3x + 6$ $\quad m = -3/2$

$y = \frac{-3}{2}x + 3$

$m = -3/2$

Parallel

8. $3y = 4x + 15$ and $9x + 12y = 12$

$y = \frac{4}{3}x + 5$ $\quad 12y = -9x + 12$

$y = \frac{-3}{4}x + 1$

$m = 4/3$ $\quad m = -3/4$

Perp.

9. $8x - 2y = 4$ and $x + 4y = -12$

$-2y = -8x + 4$ $\quad 4y = -x - 12$

$y = 4x - 2$ $\quad y = -\frac{1}{4}x - 3$

$m = 4$ $\quad m = -1/4$

Perp.

10. $3x + 2y = 10$ and $2x + 3y = -3$

$2y = -3x + 10$ $\quad 3y = -2x - 3$

$y = \frac{-3}{2}x + 5$ $\quad y = \frac{-2}{3}x - 1$

$m = -3/2$ $\quad m = -2/3$

Neither

11. $-4y = -2x + 8$ and $3x - 6y = 6$

$y = \frac{1}{2}x - 2$ $\quad -6y = -3x + 6$

$m = 1/2$ $\quad y = \frac{1}{2}x - 1$

$m = 1/2$

Parallel

12. $y = 8$ and $x = -1$

$m = 0$ $\quad m =$ undef.

Perp.

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
WRITING Parallel & Perpendicular **EQUATIONS**	• **What are we doing?** Given a linear equation and a certain point, you can create ANOTHER equation that passes through this point and is either parallel or perpendicular to the given line. • **Keep in mind the following points:** ➢ Parallel lines have Same slopes. ➢ Perpendicular lines have negative reciprocal slopes.
PARALLEL EXAMPLES	**1.** Write the equation of the line that passes through the point **(-2, 7)** and is **PARALLEL** to the line $y = -4x + 1$. $m = -4$ $y - 7 = -4(x + 2)$ $y - 7 = -4x - 8$ $+7 \quad +7$ $y = -4x - 1$
	2. Write the equation of the line that passes through the point **(3, -1)** and is **PARALLEL** to the line $x - 3y = 9$. $-3y = -x + 9$ $y = \frac{1}{3}x - 3$ $m = \frac{1}{3}$ $y + 1 = \frac{1}{3}(x - 3)$ $y + 1 = \frac{1}{3}x - 1$ $-1 \quad -1$ $y = \frac{1}{3}x - 2$
PERPENDICULAR EXAMPLES	**3.** Write the equation of the line that passes through the point **(4, 3)** and is **PERPENDICULAR** to the line $y = 2x - 4$. $m = -1/2$ $y - 3 = -1/2(x - 4)$ $y - 3 = -\frac{1}{2}x + 2$ $+3 \quad +3$ $y = -\frac{1}{2}x + 5$
	4. Write the equation of the line that passes through the point **(-5, 1)** and is **PERPENDICULAR** to the line $5x + 3y = -21$. $y - 1 = \frac{-3}{5}(x + 5)$ $3y = 5x - 21$ $y = \frac{5}{3}x - 7$ $m = -\frac{3}{5}$ $y - 1 = \frac{-3}{5}x - 3$ $+1 \quad +1$ $y = -\frac{3}{5}x - 2$

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MORE PRACTICE

Directions: Write an equation passing through the point that is **PARALLEL** to the given equation.

5. (-4, -1); $y = 2x + 4$ $m = 2$

$y + 1 = 2(x + 4)$

$y + 1 = 2x + 8$

$-1 \quad -1$

$\boxed{y = 2x + 7}$

6. (8, 3); $y = -\frac{1}{4}x + 7$ $m = -1/4$

$y - 3 = -\frac{1}{4}(x - 8)$

$y - 3 = -\frac{1}{4}x + 2$

$+3 \quad +3$

$\boxed{y = -\frac{1}{4}x + 5}$

7. (4, 5); $x - 2y = 14$

$-2y = -x + 14$

$y = \frac{1}{2}x - 7$

$y - 5 = \frac{1}{2}(x - 4)$

$y - 5 = \frac{1}{2}x - 2$

$+5 \quad +5$

$\boxed{y = \frac{1}{2}x + 3}$

8. (-6, 7); $5x + 2y = 10$

$2y = -5x + 10$

$y = -\frac{5}{2}x + 5$

$y - 7 = -\frac{5}{2}(x + 6)$

$y - 7 = -\frac{5}{2}x - 15$

$+7 \quad +7$

$\boxed{y = -\frac{5}{2}x - 8}$

Directions: Write an equation passing through the point that is **PERPENDICULAR** to the given equation.

9. (3, -3); $y = \frac{3}{4}x + 5$ $m = -\frac{4}{3}$

$y + 3 = -\frac{4}{3}(x - 3)$

$y + 3 = -\frac{4}{3}x + 4$

$-3 \quad -3$

$\boxed{y = -\frac{4}{3}x + 1}$

10. (-3, -2); $y = x - 2$ $m = -1$

$y + 2 = -1(x + 3)$

$y + 2 = -x - 3$

$-2 \quad -2$

$\boxed{y = -x - 5}$

11. (2, 3); $2x + 10y = 20$

$10y = -2x + 20$

$y = -\frac{1}{5}x + 2$

$m = 5$

$y - 3 = 5(x - 2)$

$y - 3 = 5x - 10$

$+3 \quad +3$

$\boxed{y = 5x - 7}$

12. (-1, -6); $x + 3y = 6$

$3y = -x + 6$

$y = -\frac{1}{3}x + 2$

$m = 3$

$y + 6 = 3(x + 1)$

$y + 6 = 3x + 3$

$-6 \quad -6$

$\boxed{y = 3x - 3}$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 10:** Parallel & Perpendicular Lines (Day 2)

Directions: Write an equation passing through the point and **PARALLEL** to the given line.

1. (4, 7); $y = 3x + 6$ $m=3$

$y-7=3(x-4)$

$y-7=3x-12$

$\boxed{y=3x-5}$

2. (-2, 3); $y = -x + 4$ $m=-1$

$y-3=-1(x+2)$

$y-3=-x-2$

$\boxed{y=-x+1}$

3. (-4, -5); $y=\frac{1}{2}x-6$ $m=1/2$

$y+5=\frac{1}{2}(x+4)$

$y+5=\frac{1}{2}x+2$

$\boxed{y=\frac{1}{2}x-3}$

4. (-8, 2); $5x - 4y = 4$ $-4y=-5x+4$ $y=\frac{5}{4}x-1$ $m=5/4$

$y-2=\frac{5}{4}(x+8)$

$y-2=\frac{5}{4}x+10$

$\boxed{y=\frac{5}{4}x+12}$

5. (-10, 1); $2x + 5y = 15$ $5y=-2x+15$ $y=\frac{-2}{5}x+3$ $m=-2/5$

$y-1=\frac{-2}{5}(x+10)$

$y-1=\frac{-2}{5}x-4$

$\boxed{y=\frac{-2}{5}x-3}$

6. (-5, -1); $2y = 2x - 4$ $y=x-2$

$y+1=1(x+5)$

$y+1=x+5$

$\boxed{y=x+4}$

Directions: Write an equation passing through the point and **PERPENDICULAR** to the given line.

7. (-2, -2); $y=-\frac{1}{5}x+9$ $m=5$

$y+2=5(x+2)$

$y+2=5x+10$

$\boxed{y=5x+8}$

8. (4, -1); $y = 2x - 4$ $m=-1/2$

$y+1=\frac{-1}{2}(x-4)$

$y+1=\frac{-1}{2}x+2$

$\boxed{y=-\frac{1}{2}x+1}$

9. (-3, 5); $y=\frac{3}{4}x-4$ $m=-4/3$

$y-5=\frac{-4}{3}(x+3)$

$y-5=\frac{-4}{3}x-4$

$\boxed{y=\frac{-4}{3}x+1}$

10. (2, -5); $x - 4y = 20$ $-4y=-x+20$ $y=\frac{1}{4}x-5$ $m=-4$

$y+5=-4(x-2)$

$y+5=-4x+8$

$\boxed{y=-4x+3}$

11. (10, 7); $5x - 6y = 18$ $-6y=-5x+18$ $y=\frac{5}{6}x-3$ $m=-\frac{6}{5}$

$y-7=-\frac{6}{5}(x-10)$

$y-7=-\frac{6}{5}x+12$

$\boxed{y=-\frac{6}{5}x+19}$

12. (-2, 2); $6x + 3y = -9$ $3y=-6x-9$ $y=-2x-3$ $m=\frac{1}{2}$

$y-2=\frac{1}{2}(x+2)$

$y-2=\frac{1}{2}x+1$

$\boxed{y=\frac{1}{2}x+3}$

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Name: _____ **Algebra I**

Date: _____ **Per:** _____ **Unit 4:** Linear Equations

Quiz 4-2: Writing Linear Equations; Parallel & Perpendicular Lines

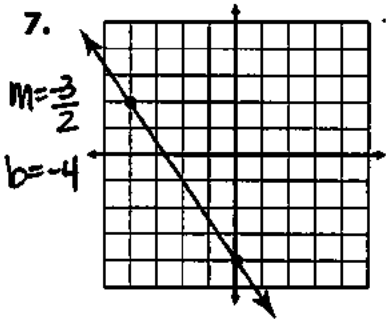
Identify the following formulas:

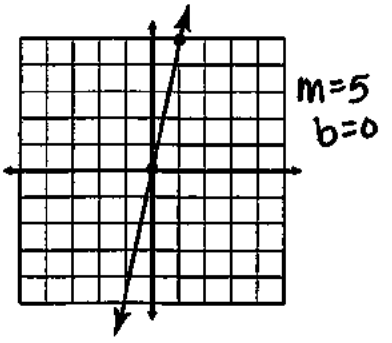
1. Slope Formula
2. Slope-Intercept Form
3. Standard Form
4. Point-Slope Formula

1. $m = \frac{y_2 - y_1}{x_2 - x_1}$
2. $y = mx + b$
3. $Ax + By = C$
4. $y - y_1 = m(x - x_1)$

Write the linear equation in slope-intercept form given the following:

5. slope = $\frac{2}{3}$; y-intercept = -1
6. slope = -1; y-intercept = 8

7. 

8. 

9. $2x + 8y = 16$

$\frac{8y}{8} = \frac{-2x}{8} + \frac{16}{8}$

10. $3x - y = 5$

$\frac{-y}{-1} = \frac{-3x}{-1} + \frac{5}{-1}$

11. (1, 4); slope = -2

$y - 4 = -2(x - 1)$

$y - 4 = -2x + 2$

12. (-4, 6); slope = $-\frac{3}{4}$

$y - 6 = -\frac{3}{4}(x + 4)$

$y - 6 = -\frac{3}{4}x - 3$

13. (-5, -11) and (-2, 1)

$m = \frac{1+11}{-2+5} = \frac{12}{3} = 4$

$y - 1 = 4(x + 2)$

$y - 1 = 4x + 8$

14. (-3, 7) and (6, -8)

$m = \frac{-8-7}{6+3} = \frac{-15}{9} = -\frac{5}{3}$

$y - 7 = -\frac{5}{3}(x + 3)$

$y - 7 = -\frac{5}{3}x - 5$

5. $y = \frac{2}{3}x - 1$
6. $y = -x + 8$
7. $y = -\frac{3}{2}x - 4$
8. $y = 5x$
9. $y = -\frac{1}{4}x + 2$
10. $y = 3x - 5$
11. $y = -2x + 6$
12. $y = -\frac{3}{4}x + 3$
13. $y = 4x + 9$
14. $y = -\frac{5}{3}x + 2$

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Determine if segments AB and CD are parallel (||), perpendicular ($\perp$), or neither.

15. $\overline{AB}$ formed by $A(-1, 3)$ and $B(1, 5)$; $\overline{CD}$ formed by $C(2, 3)$ and $D(-2, 7)$

$\overline{AB}$: $m = \frac{5-3}{1+1} = \frac{2}{2} = 1$ $\overline{CD}$: $m = \frac{7-3}{-2-2} = \frac{4}{-4} = -1$

16. $\overline{AB}$ formed by $A(3, 8)$ and $B(-6, 5)$; $\overline{CD}$ formed by $C(-6, 7)$ and $D(-3, 8)$

$\overline{AB}$: $m = \frac{5-8}{-6-3} = \frac{-3}{-9} = \frac{1}{3}$ $\overline{CD}$: $m = \frac{8-7}{-3+6} = \frac{1}{3}$

17. $\overline{AB}$ formed by $A(2, -7)$ and $B(2, 3)$; $\overline{CD}$ formed by $C(-4, 0)$ and $D(-4, -5)$

$\overline{AB}$: $m = \frac{3+7}{2-2} = \frac{10}{0}$ = undef. $\overline{CD}$: $m = \frac{-5-0}{-4+4} = \frac{-5}{0}$ = undef

15. -1
16. $||$
17. $||$
18. C

18. Given the following four lines, pick the true statement.

Line 1:	**Line 2:**	**Line 3:**	**Line 4:**
$3y = 4x + 3$	$4y = 3x - 4$	$3x + 4y = 8$	$4x + 3y = -6$
$y = \frac{4}{3}x + 1$	$y = \frac{3}{4}x - 1$	$4y = -3x + 8$	$3y = -4x - 6$
		$y = \frac{-3}{4}x + 2$	$y = \frac{-4}{3}x - 2$

A. Lines 1 and 4 are parallel

B. Lines 2 and 3 are parallel

C. Lines 2 and 4 are perpendicular

D. Lines 1 and 2 are perpendicular

For questions 19-22, write a linear equation with the given information.

19. Passing through (2, 3) and **PARALLEL** to $y = x - 4$ $m = 1$

$y - 3 = 1(x - 2)$

$y - 3 = x - 2$

20. Passing through (-6, -1) and **PARALLEL** $2x + 3y = 3$

$y + 1 = \frac{-2}{3}(x + 6)$ $3y = -2x + 3$

$y + 1 = -\frac{2}{3}x - 4$ $y = \frac{-2}{3}x + 1$ $m = \frac{-2}{3}$

21. Passing through (-2, 8) and **PERPENDICULAR** to $y = 2x + 5$ $m = -\frac{1}{2}$

$y - 8 = -\frac{1}{2}(x + 2)$

$y - 8 = -\frac{1}{2}x - 1$

22. Passing through (-2, -11) and **PERPENDICULAR** to $x + 4y = 8$

$y + 11 = 4(x + 2)$ $4y = -x + 8$ $m = 4$

$y + 11 = 4x + 8$ $y = -\frac{1}{4}x + 2$

19. $y = x + 1$
20. $y = \frac{-2}{3}x - 5$
21. $y = -\frac{1}{2}x + 7$
22. $y = 4x - 3$

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LINEAR EQUATIONS WORD PROBLEMS

Type 1 — SLOPE-INTERCEPT

Notes: Use when given a rate of change (m) and a starting point (b).

1. Evan is going to the county fair this weekend. The admission to the fair is $5 and the cost per ride is 50¢. If his parents gave him $20, write and solve a linear equation to find how many rides he can go on.

$m = 0.50$
$b = 5.00$

$y = .50x + 5.00$

$20 = .50x + 5$
$\frac{15}{.5} = \frac{.50x}{.5}$

$x = 30$ rides

2. While visiting Crimson Lake, Sally decided to go kayaking. The rangers charge $8.50 per hour in addition to a $25.00 deposit to rent the kayak. If she rented the kayak from 11:30 a.m. to 2:30 p.m., write and solve a linear equation to find the total cost to rent the kayak.

$m = 8.50$
$b = 25.00$

$y = 8.50x + 25.00$

$y = 8.50(3) + 25$
$y = \$50.50$

3. Alex bought a new truck for $42,935. According to the dealer, the truck will depreciate approximately $4,200 per year. Write and solve a linear equation to find how many years until the car is worth $5,135.

$m = -4200$
$b = 42935$

$y = -4200 + 42935$

$5135 = -4200x + 42935$
$-37800 = -4200x$

$x = 9$ years

4. If you buy a car wash at the gas station for $6.00, the cost per gallon is $2.35. If you have $40, write and solve a linear equation to find the number of gallons of gas you can afford.

$m = 2.35$
$b = 6$

$y = 2.35x + 6.00$

$40 = 2.35x + 6$
$34 = 2.35x$

$x = 14.47$ gal

Type 2 — STANDARD FORM

Notes: Use when the problem relates 2 different objects.

5. Sam ordered two hamburgers and three hotdogs from the concession stand at the baseball game. His bill came to $19.05. If hamburgers cost $5.25 each, write and solve a linear equation to find the cost of each hot dog.

x = hamburgers
y = hotdogs

$2x + 3y = 19.05$

$2(5.25) + 3y = 19.05$
$10.50 + 3y = 19.05$
$3y = 8.55$

$y = \$2.85$

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6. Tickets at a school play cost $4 in advance or $5 at the door. Total ticket sales for an evening production were $440. If no tickets were sold in advance, write and solve a linear equation to find the how many were sold at the door.

X = advance
y = door

$4x + 5y = 440$

$4(0) + 5y = 440$
$5y = 440$
$y = 88$ tickets

7. Karen's piggy bank has all dimes and quarters. The total value of the money in her piggy bank is $11.40. If Karen has 36 quarters, write and solve a linear equation to find the number of dimes she has.

X = dimes
y = quarters

$.10x + .25y = 11.40$

$.10x + .25(36) = 11.40$
$.10x + 9 = 11.40$
$.10x = 2.40$
$x = 24$ dimes

8. The Oakville High School Marching Band sold gift wrap to earn money for a band trip. Solid color gift wrap sells for $3.50 per roll and printed gift wrap sells for $5 per roll. The band needs to earn $2,500 for an upcoming competition. If the band sells 325 rolls of printed paper, write and solve a linear equation to find the number of solid paper they must sell.

X = solid
y = printed

$3.50x + 5y = 2500$

$3.50x + 5(325) = 2500$
$3.50x + 1625 = 2500$
$3.50x = 875$
$x = 250$ rolls

Type 3

POINT & SLOPE

Notes:
Use when the problem gives a sample point (x, y) and rate of change (m).

9. At Eagle Bay, it costs $12 per hour to rent a canoe. Nate and his friends rented a canoe for 4 hours and paid $68. Write and solve a linear equation to find the cost to rent the canoe for 7 hours.

$m = 12$; $(4, 68)$

$y - 68 = 12(x - 4)$
$y - 68 = 12x - 48$
$+68 \quad +68$
$y = 12x + 20$

$y = 12(7) + 20$
$y = \$104$

10. A construction company charges $18 per hour for debris removal, plus a one-time fee for the use of the trash dumpster. The total fee for 8 hours of service was $219. Write and solve a linear equation to find the one-time fee for the trash dumpster.

$m = 18$; $(8, 219)$

$y - 219 = 18(x - 8)$
$y - 219 = 18x - 144$
$+219 \quad +219$
$y = 18x + 75$

$y = 18(0) + 75$
$y = \$75$

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11. A company offers premium cable for $39.95 plus a one-time setup fee. The total cost for setup and 6 months of service is $264.70. Write and solve a linear equation to find the total cost for 2 years of service.

$m = 39.95$; $(6, 264.70)$

$y - 264.70 = 39.95(x - 6)$

$y - 264.70 = 39.95x - 239.70$

$+264.70 \quad +264.70$

$\boxed{y = 39.95x + 25}$

$y = 39.95(24) + 25$

$\boxed{y = \$983.80}$

12. A home security company provides security systems for $5 per week, plus an installation fee. The total fee for 12 weeks of service is $210. Write and solve a linear equation to find the cost of the installation fee.

$m = 5$; $(12, 210)$

$y - 210 = 5(x - 12)$

$y - 210 = 5x - 60$

$+210 \quad +210$

$\boxed{y = 5x + 150}$

$y = 5(0) + 150$

$\boxed{y = \$150}$

Type 4

TWO POINTS

Notes:
Use when the problem gives 2 sample points (x_1, y_1) and (x_2, y_2).

13. To surf the internet for 15 minutes at an airport, it costs $4.05. For 40 minutes, it costs $5.80. Write and solve a linear equation to find the cost for surfing the web for one hour.

$(15, 4.05)$ & $(40, 5.80)$

$m = \frac{5.80 - 4.05}{40 - 15} = \frac{1.75}{25} = 0.07$

$y - 4.05 = .07(x - 15)$

$y - 4.05 = .07x - 1.05$

$\boxed{y = .07x + 3}$

$y = .07(60) + 3$

$\boxed{y = \$7.20}$

14. Water boils at 100° Celsius or 212° Fahrenheit. Water freezes at 0° Celsius or 32° Fahrenheit. If the weather forecaster says it will be 25° Celsius today, write and solve a linear equation to find what Fahrenheit temperature this is.

$(100, 212)$ & $(0, 32)$

$m = \frac{32 - 212}{0 - 100} = \frac{-180}{-100} = 1.8$

$y - 32 = 1.8(x - 0)$

$y - 32 = 1.8x$

$\boxed{y = 1.8x + 32}$

$y = 1.8(25) + 32$

$\boxed{y = 77^{\circ}F}$

15. The value of a car decreases at a constant rate. After 3 years, the value of the car is $15,000. After 2 more years, the value of the car is $11,000. Write and solve a linear equation to find the value of the car after 8 years.

$(3, 15000)$ & $(5, 11000)$

$m = \frac{11000 - 15000}{5 - 3} = \frac{-4000}{2} = -2000$

$y - 11000 = -2000(x - 5)$

$y - 11000 = -2000x + 10000$

$\boxed{y = -2000x + 21000}$

$y = -2000(8) + 21000$

$\boxed{y = \$5000}$

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16. Customers of a satellite dish company must pay for installation of the dish, and a monthly fee for service. After 6 months, Ryan had paid $594. After 11 months, he had paid $989. Write and solve a linear equation to find the cost for installation.

$(6,594)$ & $(11,989)$

$m = \frac{989-594}{11-6} = \frac{395}{5} = 79$

$y - 594 = 79(x-6)$

$y - 594 = 79x - 474$

$\boxed{y = 79x + 120}$

$y = 79(0) + 120$

$\boxed{y = \$120}$

NOW... YOU DETERMINE THE TYPE!

17. Kate bought $23.40 worth of two types of bird seed. Thistle bird seed sells for $1.60 per pound and wild bird seed sells for $0.95 per pound. If she bought 12 pounds of wild bird seed, write and solve a linear equation to find the amount of thistle bird she purchased. seed purchased.

x = thistle

y = wild

$\boxed{1.60x + .95y = 23.40}$

$1.60x + .95(12) = 23.40$

$1.60x + 11.4 = 23.40$

$1.6x = 12$

$\boxed{x = 7.5 \text{ pounds}}$

18. Jeff is keeping track of his weight over several weeks. After 2 weeks, he weighs 194 pounds. After 6 weeks, he weighs 186 pounds. Write and solve a linear equation to find Jeff's weight after 12 weeks.

$(2,194)$ & $(6,186)$

$m = \frac{186-194}{6-2} = \frac{-8}{4} = -2$

$y - 194 = -2(x-2)$

$y - 194 = -2x + 4$

$\boxed{y = -2x + 198}$

$y = -2(12) + 198$

$\boxed{y = 174 \text{ pounds}}$

19. A landscape supply business charges $35 to deliver mulch. The cost of the mulch is $29 per cubic yard. Write and solve a linear equation to find the cost of having 6 cubic yards of mulch delivered to a site.

$m = 29$; $b = 35$

$\boxed{y = 29x + 35}$

$y = 29(6) + 35$

$\boxed{y = \$209}$

20. Hallie used a birthday check her mother gave her to start a savings account. She plans to save $25 per month as well. After 9 months, Hallie has $300. Write and solve a linear equation to find the original check amount from Hallie's mother.

$m = 25$; $(9,300)$

$y - 300 = 25(x-9)$

$y - 300 = 25x - 225$

$\boxed{y = 25x + 75}$

$y = 25(0) + 75$

$\boxed{y = \$75}$

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 11:** Linear Equation Word Problems (Day 1)

**** This is a 2-page document! ****

Slope-Intercept Applications ($y = mx + b$)

1. A music app charges $2 to download the app plus $1.29 per song downloaded. Write and solve a linear equation to find the total cost to download 30 songs.

$m = 1.29$
$b = 2$

$y = 1.29x + 2$

$y = 1.29(30) + 2$
$y = \$40.70$

2. Carly has already written 35 pages of a novel. She plans to write 12 additional pages per month until she is finished. Write and solve a linear equation to find the total number of pages written at 5 months.

$m = 12$
$b = 35$

$y = 12x + 35$

$y = 12(5) + 35$
$y = 95$ pages

3. The late fee for library books is $2.00 plus 15¢ each day for a book that is late. If Maria's fee for a late book was $3.20, write and solve a linear equation to find many days late the book was.

$m = .15$
$b = 2$

$y = .15x + 2$

$3.20 = .15x + 2$
$-2 \quad -2$
$1.20 = .15x$
$\frac{1.20}{.15} = \frac{.15x}{.15}$
$x = 8$ days

4. A group of mountain climbers begin an expedition with 250 pounds of food. They plan to eat a total of 12 pounds of food per day. Write and solve a linear equation to find the number of days their food will last.

$m = -12$
$b = 250$

$y = -12x + 250$

$0 = -12x + 250$
$-250 \quad -250$
$\frac{-250}{-12} = \frac{-12x}{-12}$
$x = 20.8\overline{3}$ days

5. An airplane at 30,100 feet above the ground begins descending at the rate of 2,150 feet per minute. Write and solve a linear equation to find how many minutes it will take the plane to reach the ground.

$m = -2150$
$b = 30100$

$y = -2150x + 30100$

$0 = -2150x + 30100$
$-30100 \quad -30100$
$\frac{-30100}{-2150} = \frac{-2150x}{-2150}$
$x = 14$ min

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Standard Form Applications ($Ax + By = C$)

6. The admission fee at a small fair is \$1.50 for children and \$4.00 for adults. On a certain day, \$5,050 is collected. If 1,000 adults attended the fair, write and solve a linear equation to find the number of children that attended.

X= children
y= adult

$1.50x + 4.00y = 5050$

$1.50x + 4(1000) = 5050$
$1.50x + 4000 = 5050$
$1.50x = 1050$
$x = 700$ children

7. A landscaping company ordered 17 plants and 8 trees for a total of \$964. If plants are \$12 each, write and solve a linear equation to find the cost for each tree.

X= plants
y= trees

$17x + 8y = 964$

$17(12) + 8y = 964$
$204 + 8y = 964$
$8y = 760$
$y = 95$ trees

8. A road side vegetable stand sells pumpkins for \$5 each and tomatoes for \$3 each. The total sales on a certain day was \$98. If they sold 10 pumpkins, write and solve a linear equation to find the number of tomatoes sold.

X= pumpkins
y= tomatoes

$5x + 3y = 98$

$5(10) + 3y = 98$
$50 + 3y = 98$
$3y = 48$
$y = 16$ tomatoes

9. A tire dealer sells all-season tires for \$68 each and all-terrain tires for \$125 each. In one day, the sales both tires totaled \$3,926. If 14 all-terrain tires were sold, write and solve a linear equation to find the number of all-season tires sold.

X= all-season tires

y= all-terrain tires

$68x + 125y = 3926$

$68x + 125(14) = 3926$
$68x + 1750 = 3926$
$68x = 2176$
$x = 32$ tires

10. You are buying the hamburgers and chicken for a party. The hamburgers cost \$2.75 per pound and the chicken is \$3.50 per pound. If you have \$75 to spend, write and solve a linear equation to find how many pounds of chicken you can afford if you buy 12 pounds of hamburger.

X= hamburgers
y= chicken

$2.75x + 3.50y = 75$

$2.75(12) + 3.50y = 75$
$33 + 3.50y = 75$
$3.50y = 42$
$y = 12$ lb

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Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 12:** Linear Equation Word Problems (Day 2)

**** This is a 2-page document! ****

Point & Slope Applications

1. ChitChat Wireless charges $0.30 per minute plus a monthly base fee. In September, Maggie used 120 minutes and her bill was $51. Write and solve an equation to find the cost if Maggie uses 165 minutes in October.

$m = 0.30$
$(120, 51)$

$y - 51 = .3(x - 120)$
$y - 51 = .3x - 36$
$\boxed{y = .3x + 15}$

$y = .3(165) + 15$
$\boxed{y = \$64.50}$

2. There is a daily fee for renting a moving truck, plus a charge of $0.50 per mile driven. If driven 48 miles, it costs $64 to rent a truck. Write and solve an equation to find the daily fee.

$m = 0.50$
$(48, 64)$

$y - 64 = .5(x - 48)$
$y - 64 = .5x - 24$
$\boxed{y = .5x + 40}$

$y = .5(0) + 40$
$\boxed{y = \$40}$

3. To lease a new car, you must make a down payment when you sign the lease, then pay $199 per month. Six months after signing his lease, Mr. Scott had paid $2,694. Write and solve an equation to find the total amount he will have paid after 3 years.

$m = 199$
$(6, 2694)$

$y - 2694 = 199(x - 6)$
$y - 2694 = 199x - 1194$
$\boxed{y = 199x + 1500}$

$y = 199(36) + 1500$
$\boxed{y = \$8664}$

4. Since Beth was born, the population of her town has increased at a rate of 850 people per year. On Beth's 9th birthday, the total population was nearly 307,650. Write and solve a linear equation to find the population on Beth's 16th birthday.

$m = 850$
$(9, 307650)$

$y - 307650 = 850(x - 9)$
$y - 307650 = 850x - 7650$
$\boxed{y = 850x + 300000}$

$y = 850(16) + 300000$
$\boxed{y = 313,600 \text{ people}}$

5. A candlestick burns at a rate of 0.2 inches per hour. After eight straight hours of burning, the candlestick is 13.4 inches tall. Write and solve a linear equation to find the original height of the candle.

$m = -0.2$
$(8, 13.4)$

$y - 13.4 = -0.2(x - 8)$
$y - 13.4 = -.2x + 1.6$
$\boxed{y = -.2x + 15}$

$y = -.2(0) + 15$
$\boxed{y = 15 \text{ in}}$

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Two Points Applications

6. To serve 1 person at a local restaurant, it takes 10 minutes. To serve 5 people, it takes 18 minutes. Write and solve an equation to find the number of minutes it will take to serve a party of 8.

(1, 10)
(5, 18)

$m = \frac{18-10}{5-1} = \frac{8}{4} = 2$

$y - 10 = 2(x-1)$
$y - 10 = 2x - 2$
$\boxed{y = 2x + 8}$

$y = 2(8) + 8$
$\boxed{y = 24 \text{ min}}$

7. The cost for 7 dance lessons is $82. The cost for 11 lessons is $122. Write and solve an equation to find the cost of 4 lessons.

(7, 82)
(11, 122)

$m = \frac{122-82}{11-7} = \frac{40}{4} = 10$

$y - 82 = 10(x-7)$
$y - 82 = 10x - 70$
$\boxed{y = 10x + 12}$

$y = 10(4) + 12$
$\boxed{y = \$52}$

8. To order tickets online to a hockey game, there is a processing fee plus the cost per ticket. The cost for 5 tickets is $174.45. The cost for 8 tickets is $271.95. Write and solve a linear equation to find the total cost to order 12 tickets.

(5, 174.45)
(8, 271.95)

$m = \frac{97.5}{3} = 32.5$

$y - 174.45 = 32.5(x-5)$
$y - 174.45 = 32.5x - 162.5$
$\boxed{y = 32.5x + 11.95}$

$y = 32.5(12) + 11.95$
$\boxed{y = \$401.95}$

9. An international calling plan charges a rate per minute plus a flat fee. A 10-minute call to France costs $3.19. A 15-minute call to France costs $4.29. Write and solve a linear equation to find the cost of a 12-minute call to France

(10, 3.19)
(15, 4.29)

$m = \frac{1.1}{5} = 0.22$

$y - 3.19 = 0.22(x-10)$
$y - 3.19 = .22x - 2.2$
$\boxed{y = .22x + 0.99}$

$y = .22(12) + .99$
$\boxed{y = \$3.63}$

10. Jake spent a three-day weekend on the lake. On Saturday, he rented a pontoon boat for three hours and paid $435. On Sunday, he rented the same boat for five hours and it cost him $190 more than it did the day before. If he rented the same boat on Monday for just two hours, find the the total cost to rent the boat for the three days.

(3, 435)
(5, 625)

$m = \frac{190}{2} = 95$

$y - 435 = 95(x-3)$
$y - 435 = 95x - 285$
$\boxed{y = 95x + 150}$

$y = 95(2) + 150$
$y = 340$

$\boxed{\$1400 \text{ total}}$

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entrance ticket

Name: _____

1. The start-up cost to join eTunes music is $7.95 plus $0.95 per song downloaded. If Will spent $26, write and solve a linear equation to find how many songs he downloaded.

$m = .95$

$b = 7.95$

$\boxed{y = .95x + 7.95}$

$26 = .95x + 7.95$

$18.05 = .95x$

$19 = x$

$\boxed{19 \text{ songs}}$

2. At a pet store, it costs $5 to wash a cat and $7.50 to wash a dog. Last week, the store made $60 from washing cats and dogs. If 6 cats were washed, write and solve a linear equation to find how many dogs were washed.

$x = cat$

$y = dog$

$\boxed{5x + 7.50y = 60}$

$5(6) + 7.50y = 60$

$7.50y = 30$

$y = 4$

$\boxed{4 \text{ dogs}}$

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entrance ticket

Name: _____

1. The start-up cost to join eTunes music is $7.95 plus $0.95 per song downloaded. If Will spent $26, write and solve a linear equation to find how many songs he downloaded.

2. At a pet store, it costs $5 to wash a cat and $7.50 to wash a dog. Last week, the store made $60 from washing cats and dogs. If 6 cats were washed, write and solve a linear equation to find how many dogs were washed.

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entrance ticket

Name: _____

1. The cost for new carpet is $8.55 per square yard, plus an installation fee. If 700 square yards of carpet cost $6164, write and solve a linear equation to find the cost of 1000 square yards.

$m = 8.55$

$(700, 6164)$

$y - 6164 = 8.55(x - 700)$

$y - 6164 = 8.55x - 5985$

$\boxed{y = 8.55x + 179}$

$y = 8.55(1000) + 179$

$\boxed{y = \$8729}$

2. According to the problem above, how much is the installation fee?

$y = 8.55(0) + 179$

$\boxed{y = \$179}$

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entrance ticket

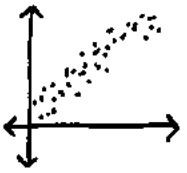
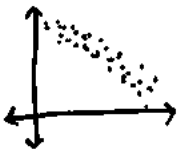
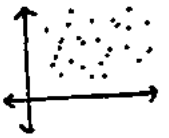
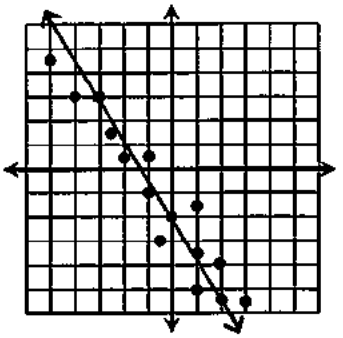
Name: _____

1. The cost for new carpet is $8.55 per square yard, plus an installation fee. If 700 square yards of carpet cost $6164, write and solve a linear equation to find the cost of 1000 square yards.

2. According to the problem above, how much is the installation fee?

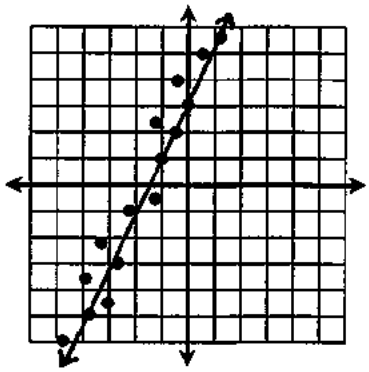
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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Scatter Plot	Shows the relationship between two variables.
Types of Relationships	**Positive**  **Negative**  **None** 
Line of Best Fit	Given a positive or negative correlation, it's possible to write a prediction equation. **Example:** The data in the graph below shows the resting heart rates and ages of a group of 20 people at the gym. Which line best represents the data? Graph each line to justify your answer.  **Line A:** $y = -\frac{2}{3}x + 80$ **Line B:** $y = -\frac{1}{3}x + 75$ **Line C:** $y = -\frac{1}{3}x + 80$ **Line D:** $y = 50$
More Examples	**Directions:** Determine which line best fits the data shown on the graph. 1.  A. $y = \frac{3}{5}x - 2$ C. $y = \frac{5}{3}x - 2$ B. $y = -\frac{3}{5}x - 2$ (D.) $y = -\frac{5}{3}x - 2$

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2.



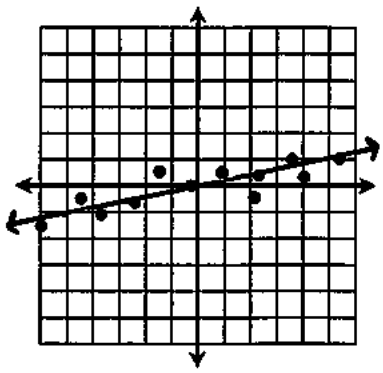
(A) $y = 2x + 3$

C. $y = -2x + 3$

B. $y = \frac{1}{2}x + 3$

D. $y = -\frac{1}{2}x + 3$

3.



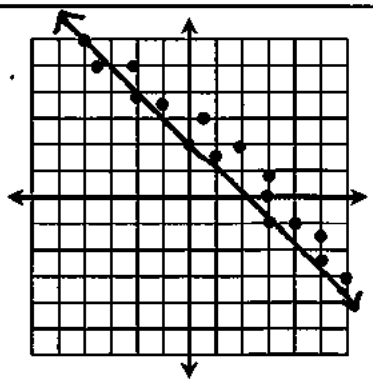
A. $y = 4x$

(C) $y = \frac{1}{4}x$

B. $y = -4x$

D. $y = -\frac{1}{4}x$

4.



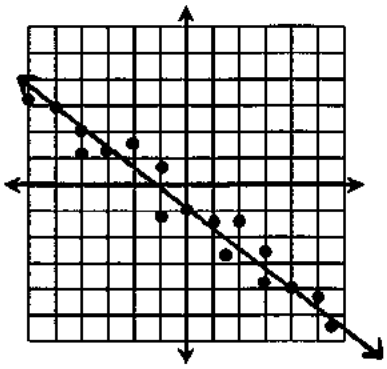
A. $y = x + 2$

C. $y = -x - 2$

(B) $y = -x + 2$

D. $y = -x - 2$

5.



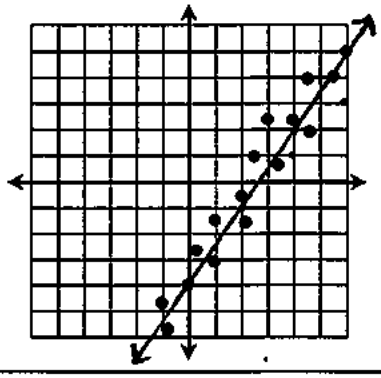
A. $y = \frac{3}{4}x - 1$

(C) $y = -\frac{3}{4}x - 1$

B. $y = \frac{4}{3}x - 1$

D. $y = -\frac{4}{3}x - 1$

6.



$-2y = -3x + 8$

$y = \frac{3}{2}x - 4$

A. $2x + 3y = -12$

C. $2x - 3y = 12$

B. $3x + 2y = -8$

(D) $3x - 2y = 8$

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Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples

Linear Regression

Linear regression is a method used to find the line of best fit given a set of data. This can be done through your graphing calculator and will create the most precise equation possible.



Calculator Steps

Enter your data:	Find the line of best fit:
• Hit **STAT, ENTER** • Enter **x-values** in L_1 • Enter **y-values** in L_2	• Hit **STAT** • Arrow over to **CALC** • Choose **#4-LinReg(ax+b)** • Hit **ENTER**

Examples

1. The data in the table below shows the average temperature in Northern Latitudes:

Latitude (°N)	0	10	20	30	40	50	60	70	80
Temp (°F)	79	81	79	68	58	43	28	13	1

a) Find the line of best fit: $y = -1.07x + 92.87$

b) Estimate the average temperature for a city with a latitude of 48°:

$y = -1.07(48) + 92.87$

$y = 41.51°F$

2. The data in the table below shows the number of passengers and number of suitcases on various airplanes.

Passengers	75	92	115	128	143	154	178	200
Suitcases	159	180	239	272	290	310	357	405

a) Find the line of best fit: $y = 1.98x + 7.97$

b) Estimate the number of suitcases on a flight carrying 250 people.

$y = 1.98(250) + 7.97$

$y = 502.97$

503 bags

3. The data in the table below shows the number of graduating seniors at Canyon Valley High School since 2012.

	0	1	2	3	4	5
Year	~~2012~~	~~2013~~	~~2014~~	~~2015~~	~~2016~~	~~2017~~
Graduates	340	348	356	361	375	387

a) Find the line of best fit: $y = 9.17x + 338.24$

b) Estimate the number of graduating seniors in ~~2025~~. $x = 13$

$y = 9.17(13) + 338.24$

$y = 457.45$

457 seniors

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	Year	Time (s)
0	1980	422
4	1984	432
8	1988	404
12	1992	420
14	1994	395
18	1998	382

4. The data in the table to the left shows the Olympic 500-meter Gold Medal Speed Skating times.

a) Find the line of best fit: $y = -2.29x + 430.56$

b) Estimate the 500-meter time for the 2020 Olympics. $(x = 40)$

$y = -2.29(40) + 430.56$

$y = 338.96$ sec

	Year	Sales
0	1994	216
1	1995	235
2	1996	252
3	1997	267
4	1998	282
5	1999	300

5. The data in the table to the left shows sales for a certain department store (in billions of dollars).

a) Find the line of best fit: $y = 16.46x + 217.52$

b) Estimate the stores sales in 2018. $(x = 24)$

$y = 16.46(24) + 217.52$

$y = \$612.56$

Weight	BMI
150	20.3
155	21.0
160	21.7
165	22.4
170	23.1
175	23.7

6. The data in the table to the left gives the weight, in pounds, and BMI (body mass index) for six people of the same height.

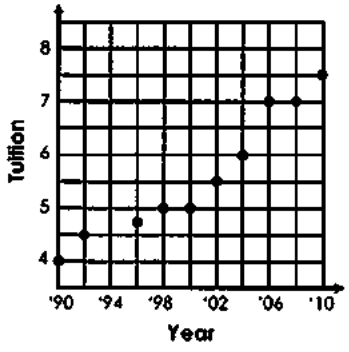
a) Find the line of best fit: $y = 0.14x - 0.25$

b) Sam is the same height as the peope in this group. If he has a BMI of 25, estimate his weight.

$25 = 0.14x - 0.25$

$\frac{25.25}{.14} = \frac{0.14x}{.14}$

$x = 180.3616$



7. The graph to the left shows the annual tuition at a public four-year college from 1990 to 2010.

$(x = 0)$ $(x = 20)$

a) Find the line of best fit: $y = 0.17x + 3.79$

b) Estimate the year in which the annual tuition will reach \$10,000.

$10 = 0.17x + 3.79$

$\frac{6.21}{0.17} = \frac{0.17x}{0.17}$

$x = 36.53 \rightarrow 37$ yrs. 2027

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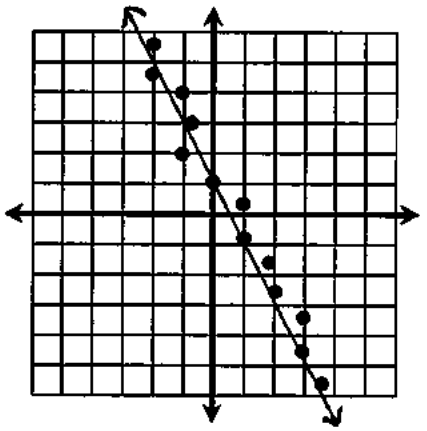
Name: _____ **Unit 4:** Linear Equations

Date: _____ **Bell:** _____ **Homework 13:** Line of Best Fit & Linear Regression

**** This is a 2-page document! ****

Directions: Choose the linear equation that best fits the data shown on the graph.

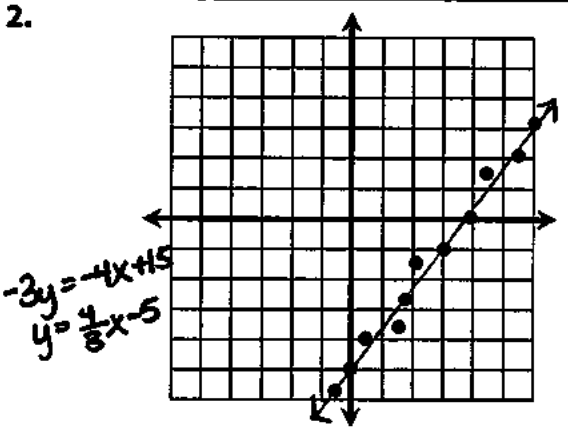
1.



A. $y = 2x + 1$ **C.** $y = -2x + 1$

B. $y = \frac{1}{2}x + 1$ **D.** $y = -\frac{1}{2}x + 1$

2.



A. $3x - 4y = 20$ **C.** $3x + 4y = -20$

B. $4x - 3y = 15$ **D.** $4x + 3y = -15$

Directions: Use your graphing calculator for questions 3-8.

3. The table below shows the value of a car based on the number of miles displayed on its odometer.

Miles	Value
5,000	$25,500
12,000	$24,800
18,000	$24,200
24,000	$23,600
30,000	$23,000
50,000	$21,000

a) Write an equation for the line of best fit.

$y = -0.1x + 26000$

b) Estimate the value of the car if its odometer reads 95,000 miles.

$y = -0.1(95000) + 26000$

$y = \$16500$

4. The table below shows the number of rabbits a petting zoo each year since 2008.

	Year	Rabbits
0	2008	14
1	2009	24
2	2010	30
3	2011	39
4	2012	45
5	2013	57

a) Write an equation for the line of best fit.

$y = 8.2x + 14.33$

b) Estimate the number of rabbits at the farm in 2016. $(x = 8)$

$y = 8.2(8) + 14.33$

$y = 79.93$

≈ 80 rabbits

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5. The data below shows the population of a city, in millions of people, since 1950.

	Year	People
0	1950	1.75
10	1960	1.68
20	1970	1.52
30	1980	1.16
40	1990	1.04
50	2000	0.98

a) Write an equation for the line of best fit.

$y = -0.02x + 1.79$

b) Estimate the population of the city in 2020. $(x=70)$

$y = -0.02(70) + 1.79$

$\boxed{y = 0.39}$

6. The table below shows the average price of a movie ticket at the theater.

	Year	Price
0	1990	$4.22
5	1995	$4.35
10	2000	$5.39
15	2005	$6.41
20	2010	$7.89
25	2015	$8.43

a) Write an equation for the line of best fit.

$y = 0.19x + 3.78$

b) In what year will the average price of a movie ticket reach $15?

$15 = 0.19x + 3.78$

$11.22 = 0.19x$

$59.1 = x$

$\boxed{2050}$

7. The table below shows the number of people, in millions, who voted in U.S. Presidential Elections.

	Year	Price
0	1980	86.5
4	1984	92.7
8	1988	91.6
12	1992	104.4
16	1996	96.3
20	2000	122.3

a) Write an equation for the line of best fit.

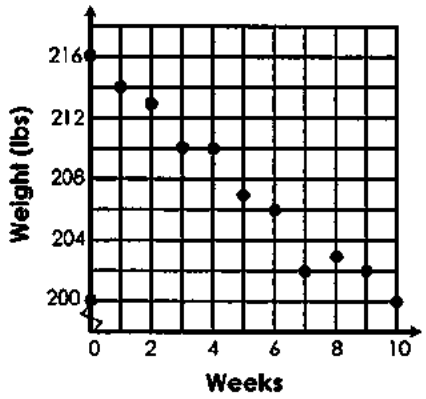
$y = 1.45x + 84.50$

b) Estimate the number of voters in the 2020 election. $(x=40)$

$y = 1.45(40) + 84.50$

$\boxed{y = 142.5 \text{ million people}}$

8. The graph below shows Carter's weight each week since he started his diet.



a) Write an equation for the line of best fit.

$y = -1.62x + 215.64$

b) Estimate Carter's weight 15 weeks into his diet.

$y = -1.62(15) + 215.64$

$\boxed{y = 191.34 \text{ lb}}$

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Name: _____ **Algebra I**

Date: _____ **Per:** _____ **Unit 4:** Linear Equations

Quiz 4-3: Linear Word Problems & Line of Best Fit

1. It costs $350 to rent a room for a party. You also plan to hire a caterer that charges $9.25 per person. Write and solve a linear equation to find the total cost of 60 people attend the party.

$m = 9.25$

$b = 350$

$y = 9.25(60) + 350$

Equation	Solution
$y = 9.25x + 350$	$905

2. As the student council treasurer, you prepare the budget for your class rafting trip. Each large raft costs $100 to rent and each small raft costs $40 to rent. You have $1,600 to spend. Write and solve a linear equation to find the number of small rafts you can rent if you rent 12 large rafts.

$x =$ large

$y =$ small

$100(12) + 40y = 1600$

$40y = 400$

$y = 10$

Equation	Solution
$100x + 40y = 1600$	10 rafts

3. Alex is on a diet to lose some weight. He is losing weight at a rate of 2 pounds per week. After 6 weeks, he weighs 205 pounds. Write and solve a linear equation to find how many weeks it will take Alex to reach his target weight of 175 pounds.

$m = -2$

$(6, 205)$

$y - 205 = -2(x - 6)$

$y - 205 = -2x + 12$

$y = -2x + 217$

$175 = -2x + 217$

$-42 = -2x$

$x = 21$

Equation	Solution
$y = -2x + 217$	21 weeks

4. Jamie ordered 200 business cards and paid $23. She ordered 500 business cards a few months later and paid $35. Write and solve a linear equation to find the cost to order 700 business cards.

$(200, 23)$

$(500, 35)$

$m = \frac{12}{300} = 0.04$

$y - 23 = 0.04(x - 200)$

$y - 23 = 0.04x - 8$

$y = 0.04x + 15$

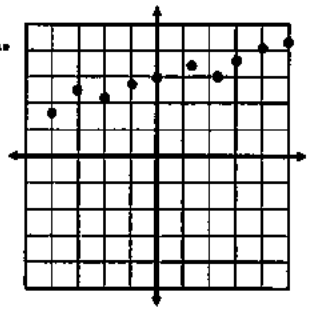
$y = 0.04(700) + 15$

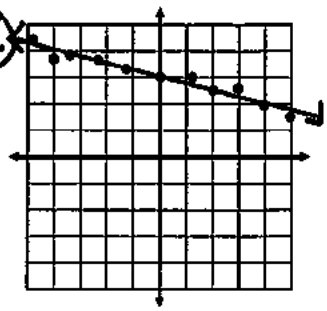
$y = 43$

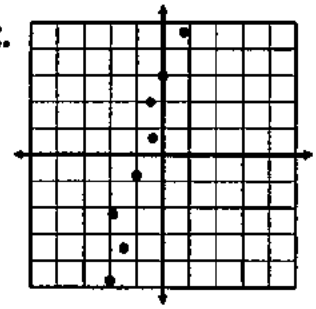
Equation	Solution
$y = 0.04x + 15$	$43

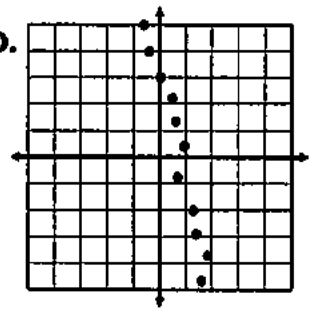
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5. Which scatter plot is most likely to have a line of best fit represented by $y = -\frac{1}{4}x + 3$?

A. 

B. 

C. 

D. 

6. Hannah weighed 7.2 pounds at birth. Her weight was recorded at each of her infant doctor visits. The records from her appointments from birth to six months are shown in the table below.

Months	Weight (lbs)
0	7.2
1	9.4
2	10.5
4	13.2
6	15.9

a) Find the line of best fit: $y = 1.4x + 7.6$

b) Estimate Hannah's weight at her one-year visit: 24.4 lb

$(x = 12)$

$y = 1.4(12) + 7.6$

$y = 24.4$

7. The table below shows the average life expectancy (in years) based on certain birth years.

	Birth Year	Life Expectancy
0	1900	47.3
10	1910	50.0
20	1920	54.1
30	1930	59.7
40	1940	62.9
50	1950	68.2
60	1960	69.7

a) Find the line of best fit: $y = 0.4x + 46.8$

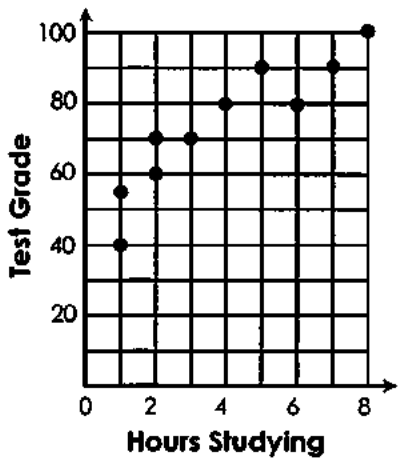
b) Estimate the life expectancy for someone born in 2005: 88.8 yrs

$(x = 105)$

$y = 0.4(105) + 46.$

$y = 88.8$

8. The table below shows the relationship between the number of hours a student studied and their grade on a certain test.



a) Find the line of best fit: $y = 6.65x + 47.56$

b) Estimate the grade for a student that did not study at all: ≈48%

$y = 6.65(0) + 47.56$

$y = 47.56$

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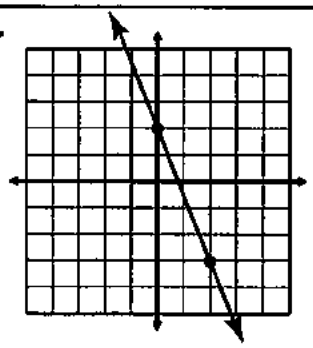
Unit 4 Test Study Guide
(Linear Equations)

Name: _____

Date: _____ Block: _____

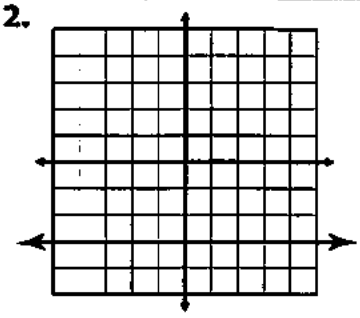
Topic 1: Slope (Given a Graph or Ordered Pairs)

1.



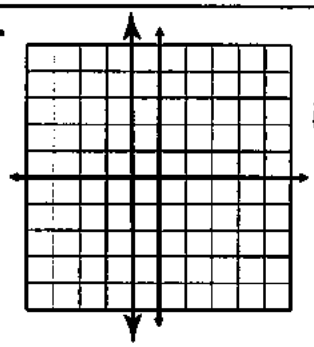
$\frac{-5}{2}$

2.



0

3.



undef

4. (-12, -1) and (-3, -4)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$m = \frac{-4+1}{-3+12} = \frac{-3}{9} = \boxed{-\frac{1}{3}}$

5. (-11, 7) and (-11, -2)

$m = \frac{-2-7}{-11+11} = \frac{-9}{0} = \boxed{\text{undefined}}$

Topic 2: Slope-Intercept Form vs. Standard Form

6. Write a linear equation with a slope of 1 and the y-intercept of -7

$y = mx + b$

$\boxed{y = x - 7}$

7. Identify the slope and y-intercept of the equation $y = -3x$

$\boxed{m = -3;\ b = 0}$

8. Identify the slope and y-intercept of the equation $2x + 5y = 20$

$-2x \quad -2x$

$\frac{5y}{5} = \frac{-2x}{5} + \frac{20}{5}$

$y = -\frac{2}{5}x + 4$

$\boxed{m = -\frac{2}{5};\ b = 4}$

9. Identify the slope and y-intercept of the equation $x - 2y = 2$

$-x \quad -x$

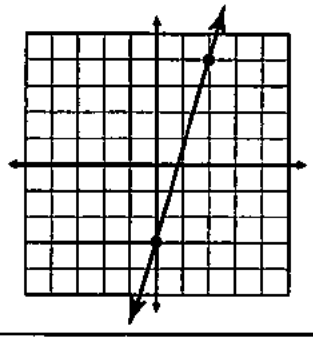
$\frac{-2y}{-2} = \frac{-x}{-2} + \frac{2}{-2}$

$y = \frac{1}{2}x - 1$

$\boxed{m = \frac{1}{2};\ b = -1}$

Topic 3: Writing Linear Equations (Given a Graph)

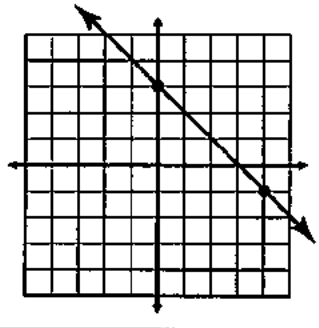
10. Write a linear equation given the graph.



$m = \frac{7}{2};\ b = -3$

$\boxed{y = \frac{7}{2}x - 3}$

11. Write a linear equation given the graph.

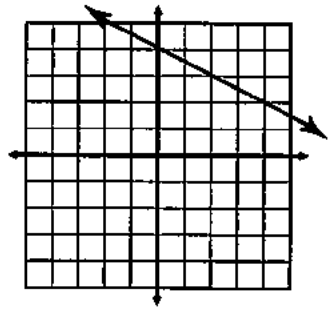


$m = -1;\ b = 3$

$\boxed{y = -x + 3}$

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12. Identify the equation of the line graphed below. Prove by converting!



A. $2x + y = 4$

B. $2x - y = 4$

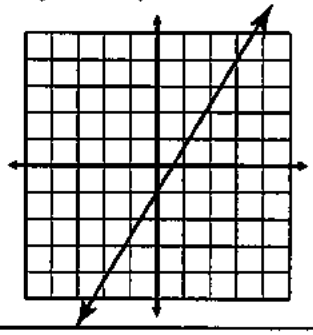
C. $x + 2y = 8$

D. $x - 2y = 8$

$\frac{2y}{2} = \frac{-x}{2} + \frac{8}{2}$

$y = -\frac{1}{2}x + 4$

13. Identify the equation of the line graphed below. Prove by converting!



A. $3x + 5y = 5$

B. $3x - 5y = 5$

C. $5x + 3y = 3$

D. $5x - 3y = 3$

$\frac{-3y}{-3} = \frac{-5x}{-3} + \frac{3}{-3}$

$y = \frac{5}{3}x - 1$

Topic 4: x- and y-Intercepts

14. $4x - y = 4$

x-int: $(y=0)$

$4x - 0 = 4$

$4x = 4$

$x = 1$ (1,0)

y-int: $(x=0)$

$4(0) - y = 4$

$-y = 4$

$y = -4$ (0,-4)

15. $12x - 8y = 24$

x-int:

$12x = 24$

$x = 2$

(2,0)

y-int:

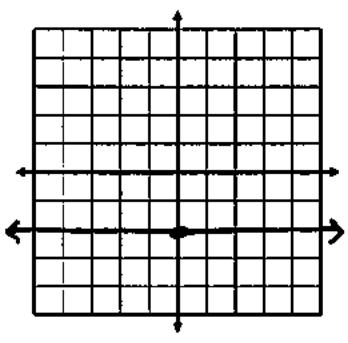
$-8y = 24$

$y = -3$

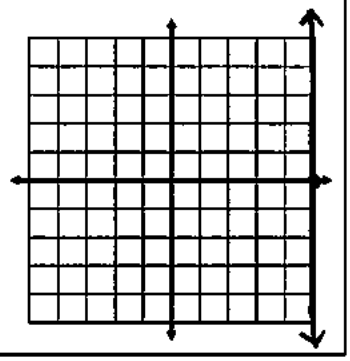
(0,-3)

Topic 5: Vertical & Horizontal Lines

16. $y = -2$



17. $x = 5$



Topic 6: Writing Linear Equations (Given a Point and Slope, or Two Points)

18. (-4, -7); slope = 3

$y + 7 = 3(x + 4)$

$y + 7 = 3x + 12$

$-7 \quad -7$

$y = 3x + 5$

19. (8, -9); slope = $-\frac{5}{4}$

$y + 9 = -\frac{5}{4}(x - 8)$

$y + 9 = -\frac{5}{4}x + 10$

$-9 \quad -9$

$y = -\frac{5}{4}x + 1$

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20. (-3, -9) and (4, -2)

$m = \frac{-2+9}{4+3} = \frac{7}{7} = 1$

$y+9 = 1(x+3)$

$y+9 = x+3$

$-9 \quad -9$

$\boxed{y = x-6}$

21. (-2, 7) and (6, -5)

$m = \frac{-5-7}{6+2} = \frac{-12}{8} = \frac{-3}{2}$

$y-7 = \frac{-3}{2}(x+2)$

$y-7 = \frac{-3}{2}x-3$

$+7 \quad +7$

$\boxed{y = \frac{-3}{2}x+4}$

Topic 7: Parallel & Perpendicular Lines

22. Describe the slopes of parallel lines.

Same slopes

23. Describe the slopes of perpendicular lines.

negative reciprocal slopes (opposite sign and flipped fraction)

24. Which line is parallel to the line $5x - 2y = 18$?

$\frac{-2y}{-2} = \frac{-5x}{-2} + \frac{18}{-2}$

$y = \frac{5}{2}x - 9$

A. $y = \frac{2}{5}x + 1$ **B.** $y = -\frac{2}{5}x - 7$

(C.) $y = \frac{5}{2}x - 4$ **D.** $y = -\frac{5}{2}x + 2$

25. Which line is perpendicular to the line $y = -3x + 2$?

A. $3x + y = 4$

B. $x + 3y = 24$

C. $3x - y = 5$

(D.) $x - 3y = 3$

$\frac{-3y}{-3} = \frac{-x}{-3} + \frac{3}{-3}$

$y = \frac{1}{3}x - 1$

26. Write an equation that is **parallel** to $y = -4x + 5$ and passes through the point (-2, -1).

$m = -4$

$y+1 = -4(x+2)$

$y+1 = -4x-8$

$-1 \quad -1$

$\boxed{y = -4x-9}$

27. Write an equation that is **perpendicular** to $3x - 5y = 5$ and passes through the point (9, -14).

$-5y = -3x + 5$

$y = \frac{3}{5}x - 1$

$m = \frac{-5}{3}$

$y+14 = \frac{-5}{3}(x-9)$

$y+14 = \frac{-5}{3}x+15$

$-14 \quad -14$

$\boxed{y = \frac{-5}{3}x+1}$

Topic 8: Line of Best Fit

28. The number of students enrolled at Beach Manor Preschool during various years is given in the table below. Use the line of best fit to predict the number of students enrolled in 2010. $x = 26$

	0	3	6	9	12	15	18
Year	1984	1987	1990	1993	1996	1999	2002
Entrants	39	45	52	61	72	78	85

$\boxed{y = 2.67x + 37.71}$

$y = 2.67(26) + 37.71$

$\boxed{y = 107 \text{ students}}$

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29. The table below shows Luke's golf score each week he participated in a golf tournament. Use the line of best fit to estimate Luke's score for week 12.

Week	1	2	3	4	5	6
Score	95	92	89	88	86	84

$y = -2.11x + 96.4$

$y = -2.11(12) + 96.4$

$y = 71$

Topic 9: Word Problems

30. MaryAnn is printing pictures from her recent trip to Europe. An online print shop charges $0.15 per 4x6 inch print, along with a flat rate shipping charge of $3.00. If MaryAnn has $35 to spend, how many prints can she order?

$m = .15$

$b = 3.00$

$y = 0.15x + 3.00$

$35 = .15x + 3$

$\frac{32}{.15} = \frac{.15x}{.15}$

$213.\overline{3} = x$

213 pictures

31. At Smith Mountain Lake Boat Rentals, it cost $25 per hour to rent a pontoon boat, plus a one-time charge for cleaning. The Bengel family rented a boat from 11 am – 6:30 pm and paid $226.50. If the Jones Family rented a boat from 8 am – 1 pm, how much did they pay?

$m = 25$

$(7.5, 226.50)$

hr $

$y - 226.5 = 25(x - 7.5)$

$y - 226.5 = 25x - 187.5$

$y = 25x + 39$

$y = 25(5) + 39$

$y = \$164$

32. Clint is in study hall reading The Hunger Games. After 20 minutes, he is on page 151. After 45 minutes, he is on page 181. How many pages had Clint read prior to study hall?

$(20, 151)$

$(45, 181)$

$m = \frac{181 - 151}{45 - 20} = \frac{30}{25} = \frac{6}{5}$

$y - 151 = \frac{6}{5}(x - 20)$

$y - 151 = \frac{6}{5}x - 24$

$y = \frac{6}{5}x + 127$

$y = \frac{6}{5}(0) + 127$

$y = 127$ pages

33. Alaina is selling wreaths and poinsettias for her chorus fundraiser. Wreaths cost $27 each poinsettias cost $20 each. If she sold 15 poinsettias and made $543, how many wreaths did she sell?

x = wreaths

y = poinsettias

$27x + 20y = 543$

$27x + 20(15) = 543$

$27x + 300 = 543$

$27x = 243$

$x = 9$ wreaths

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Name: _____

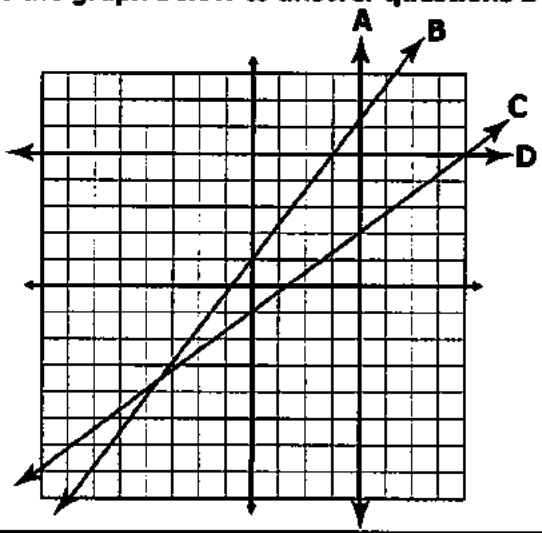
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Algebra I Unit 4 Test

Linear Equations

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION.
PLACE YOUR FINAL ANSWER IN THE BOX. GOOD LUCK! ☺

Use the graph below to answer questions 1 & 2.



1. Which line has a slope of $\frac{3}{4}$?

C

2. Which line has an undefined slope?

A

3. What is the slope of the line that contains the points (-6, 1) and (4, -4)?

$m = \frac{-4-1}{4+6} = \frac{-5}{10}$

A. 2

B. -2

C. $\frac{1}{2}$

D. $-\frac{1}{2}$

D

4. What is the slope of the line that contains the points (13, -2) and (3, -2)?

$m = \frac{-2+2}{3-13} = \frac{0}{-10}$

A. $-\frac{2}{5}$

B. 0

C. $-\frac{5}{2}$

D. undefined

B

5. What is the slope of the line whose equation is $\frac{2y}{2} = \frac{3x}{2} + \frac{4}{2}$?

$y = \frac{3}{2}x + 2$

A. 3

B. 2

C. $\frac{3}{2}$

D. $\frac{2}{3}$

C

6. Which is an equation of the line with a slope of $\frac{1}{4}$ and a y-intercept of -2?

$y = \frac{1}{4}x - 2$

A. $x - 4y = 8$ → $\frac{-4y}{-4} = \frac{-x}{-4} \frac{+8}{-4}$

B. $x + 4y = -8$ $y = \frac{1}{4}x - 2$

C. $4x + y = -2$

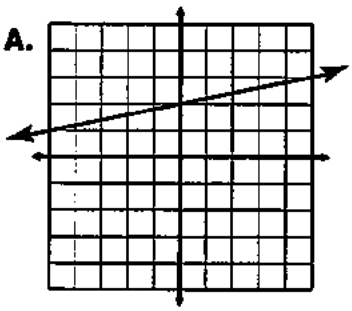
D. $4x - y = 2$

A

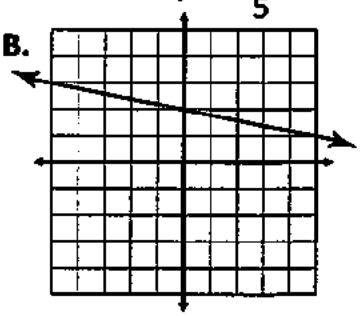
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7. Which graph best represents the line $y = -\frac{1}{5}x + 2$?

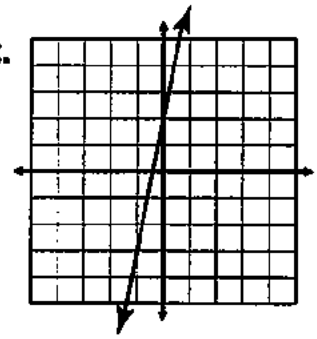
A.



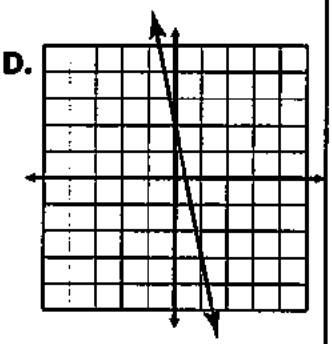
B.



C.



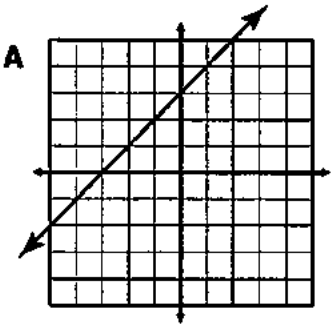
D.



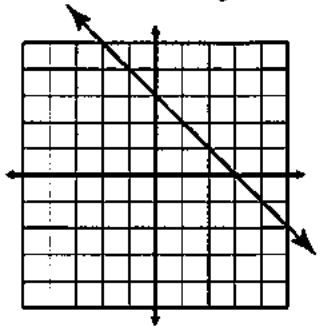
B

8. Which graph best represents the line $x - y = 3$? $-y = -x+3$ $y = x-3$

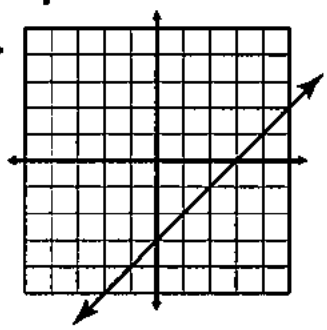
A



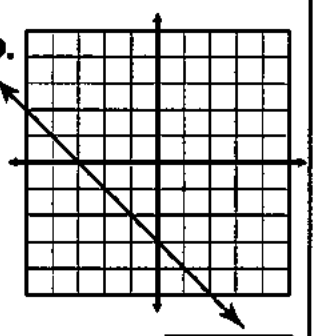
B



C.

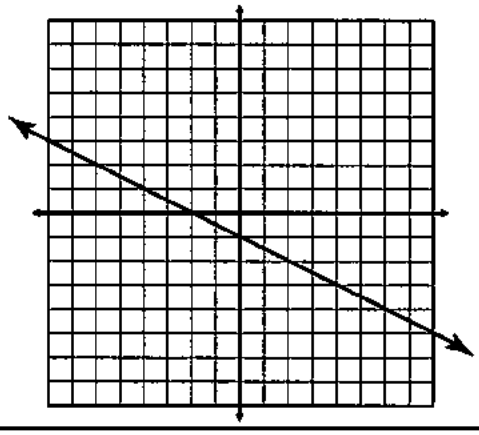


D.



C

9. Identify the equation of the line graphed below. $y = -\frac{1}{2}x - 1$



A. $x + 2y = -2$ → $2y = -x - 2$, $y = -\frac{1}{2}x - 1$

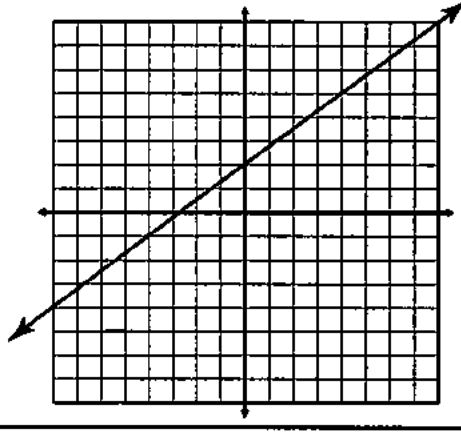
B. $x - 2y = 2$

C. $2x + y = -1$

D. $2x - y = 1$

A

10. Identify the equation of the line graphed below. $y = \frac{3}{4}x + 2$



A. $4x + 3y = 6$

B. $4x - 3y = -6$

C. $3x + 4y = 8$

D. $3x - 4y = -8$ → $-4y = -3x - 8$, $y = \frac{3}{4}x + 2$

D

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11. Identify the x- and y-intercepts:

$$10x + 4y = -20$$

$10(0) + 4y = -20$
$4y = -20$
$y = -5$

$10x + 4(0) = -20$
$10x = -20$
$x = -2$

x-intercept	y-intercept
$(-2, 0)$	$(0, -5)$

12. Which line has an x-intercept of -1 and a y-intercept of 1?

$x = -1$; $-y = -1$
$y = 1$

A. $y = -x$
B. $x - y = 1$
C. $x - y = -1$
D. $x + y = 1$

C

13. Which lines have a y-intercept of 5? Check all that apply.

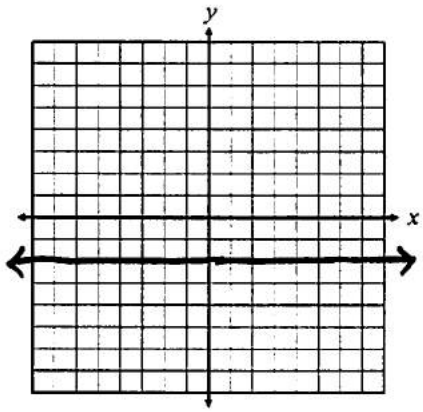
$y = \frac{1}{2}x + 5$ $\quad$ $y = x - 5$

- ☐ $y = 5x - 1$
- ☑ $4x - 8y = -40$
- ☐ $x - y = 5$
- ☐ $x - 3y = 15$
- ☐ $2x - y = 10$
- ☑ $6x + 4y = 20$

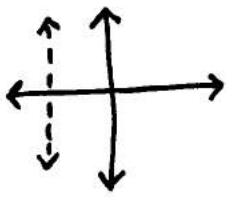
$y = \frac{1}{3}x - 5$ $\quad$ $y = 2x - 10$ $\quad$ $y = -\frac{3}{2}x + 5$

14. Graph the linear equation below:

$y = -2$



15. What is the slope of the line $x = -5$?



A. -5
B. 5
C. 0
D. undefined

D

16. Write the equation of the line that passes through the point (2, -1) and has a slope of -3.

$y + 1 = -3(x - 2)$
$y + 1 = -3x + 6$

A. $y = -3x - 1$
B. $y = -3x - 5$
C. $y = -3x + 5$
D. $y = -3x + 1$

C

17. Write the equation of the line that passes through the point (-4, -9) and has a slope of $\frac{1}{2}$.

$y + 9 = \frac{1}{2}(x + 4)$
$y + 9 = \frac{1}{2}x + 2$

A. $y = \frac{1}{2}x - 7$
B. $y = \frac{1}{2}x + 7$
C. $y = \frac{1}{2}x + 11$
D. $y = \frac{1}{2}x - 11$

A

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18. Which is the equation of the line that passes through the points (-4, 3) and (2, -6)?

$m = \frac{-6-3}{2+4} = \frac{-9}{6} = -\frac{3}{2}$

A. $y = -\frac{3}{2}x + 3$

B. $y = -\frac{3}{2}x - 3$

C. $y = \frac{3}{2}x + 3$

D. $y = \frac{3}{2}x - 3$

$y - 3 = -\frac{3}{2}(x+4)$

$y - 3 = -\frac{3}{2}x - 6$

B

19. Which is the equation of the line that passes through the points (-4, 8) and (1, 3)?

$m = \frac{3-8}{1+4} = \frac{-5}{5} = -1$

A. $y = -x + 4$

B. $y = -x + 12$

C. $y = x + 4$

D. $y = x + 12$

$y - 3 = -1(x-1)$

$y - 3 = -x + 1$

A

20. Which of the following lines is parallel to the line $4x + y = 5$?

$y = -4x - 5$

A. $y = 4x - 2$

B. $y = -4x + 9$

C. $y = \frac{1}{4}x + 8$

D. $y = -\frac{1}{4}x - 1$

B

21. Which equation represents a line parallel to the y-axis?

A. $y = x$

B. $x = 4$

C. $y = 4$

D. $y = x + 4$

B

22. Which of the following lines is perpendicular to the line $y = -\frac{3}{5}x + 1$?

A. $3x + 5y = 10$

B. $3x - 5y = 20$

C. $5x + 3y = 21$

D. $5x - 3y = 27$

$-3y = -5x + 27$

$y = \frac{5}{3}x - 9$

D

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23. What is the equation of the line that passes through the point (3, -1) and is perpendicular to the equation $y = -3x + 2$?

$y+1 = \frac{1}{3}(x-3)$

$y+1 = \frac{1}{3}x - 1$

$y = \frac{1}{3}x - 2$

A. $y = -3x + 8$

B. $y = -3x$

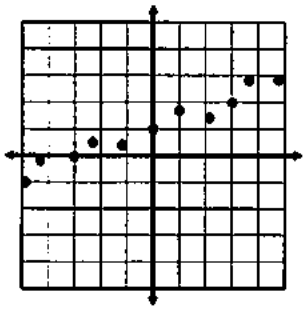
C. $y = \frac{1}{3}x$

D. $y = \frac{1}{3}x - 2$

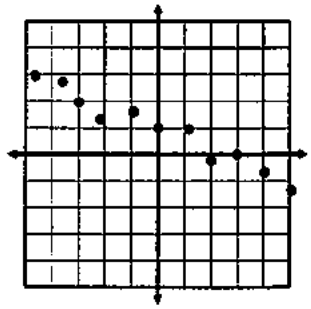
D

24. Which scatter plot most likely has a line of best fit represented by $y = 3x + 1$?

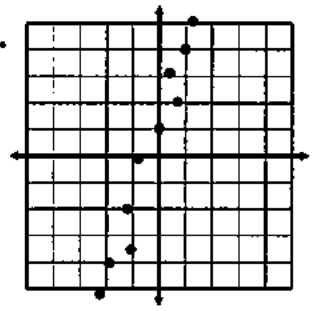
A



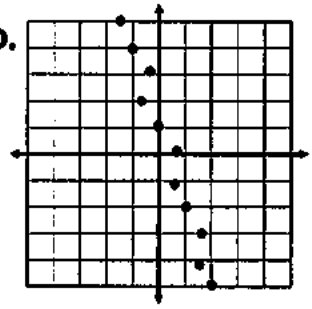
B



C.



D.



C

25. Marsha recorded the time it took six children of different ages to run one lap around the track. Using the line of best fit, which is closest to the length of time it should take Marsha's 8 year old son to run one lap?

$y = -12.55x + 276.52$

$y = -12.55(8) + 276.52 = 176.12$

Age of Child	Time (Seconds)
4	230
6	205
10	132
13	119
14	108
17	62

A. 168 seconds

B. 176 seconds

C. 182 seconds

D. 190 seconds

B

26. The table shows the approximate population of Virginia for five years. If the trend continues, which could be a reasonable prediction if the population in 2018?

$y = 0.09x + 7.10$ $(x = 18)$

$y = 0.09(18) + 7.10$

$y = 8.72$

Year	Population
2000	7.08
2004	7.48
2006	7.64
2007	7.71
2008	7.77

A. 8.24 million people

B. 8.57 million people

C. 8.72 million people

D. 8.95 million people

C

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27. You plan to go snowboarding this weekend in Pennsylvania. The resort charges $15.75 per hour in addition to a $25 deposit to rent a snowboard. Write and solve linear equation to find the total cost to rent the snowboard from 8:30 a.m. to 12:30 p.m.

$m = 15.75$

$b = 25$

$y = 15.75(4) + 25$

$y = 88$

Equation	Solution
$y = 15.75x + 25$	$88

28. When Brandon went bowling, it cost $4.95 per game, plus a one-time fee to rent the shoes. Brandon played 5 games and paid $32. Write and solve a linear equation to find the cost to rent the shoes.

$m = 4.95$

$(5, 32)$

$y - 32 = 4.95(x - 5)$

$y - 32 = 4.95x - 24.75$

$y = 4.95(0) + 7.25$

$y = 7.25$

Equation	Solution
$y = 4.95x + 7.25$	$7.25

29. The monthly rent for an 800 square foot office is $975. For an 1100 square foot office, the rent is $1125. Write and solve a linear equation to find the monthly rent for a 1500 square foot office.

$(800, 975)$

$(1100, 1125)$

$m = \frac{150}{300} = .5$

$y - 975 = .5(x - 800)$

$y - 975 = .5x - 400$

$y = .5x + 575$

$y = .5(1500) + 575$

$y = 1325$

Equation	Solution
$y = .5x + 575$	$1325

30. Brielle's piggy bank has all nickels and dimes. The total value of the money in her piggy bank is $7.80. If Brielle has 66 nickels, write and solve a linear equation to find the number of dimes she has.

x = nickels

y = dimes

$.05x + .10y = 7.80$

$.05(66) + .10y = 7.80$

$.10y = 4.50$

$y = 45$

Equation	Solution
$.05x + .10y = 7.80$	45 dimes

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